

Turning Aminated Polyolefin into a More Sustainable Polymer via Physical and Chemical Approaches

by

Hsiang-En (Shawn) Chen

B.Sc., University of British Columbia, 2023

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF

MASTER OF SCIENCE

in

THE FACULTY OF GRADUATE AND POSTDOCTORAL STUDIES
(Chemistry)

THE UNIVERSITY OF BRITISH COLUMBIA
(Vancouver)

December 2025

© Hsiang-En (Shawn) Chen, 2025

The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

Turning Aminated Polyolefin into a More Sustainable Polymer via Physical and Chemical Approaches

submitted by Hsiang-En (Shawn) Chen in partial fulfilment of the requirements for

the degree of Master of Science

in Chemistry

Examining Committee:

Prof. Laurel L. Schafer, Professor, Department of Chemistry, UBC

Supervisor

Prof. Savvas G. Hatzikiriakos, Professor, Department of Chemical and Biological Engineering, UBC

Supervisory Committee Member

Prof. Zachary M. Hudson, Professor, Department of Chemistry, UBC

Supervisory Committee Member

Abstract

Aminated polyolefins (APOs) represent a unique class of specialty polymers, due to the amine functionalities on polyolefin backbones. The pendant amines enable self-healing and adhesive behaviour, through dynamic non-covalent interactions. The dynamic non-covalent interactions within APOs enhance cohesive strength, while the amines increase the polymer surface energy and improve the adhesion to a wide range of surfaces. APOs exhibit potential as specialized adhesives for niche applications, however, the dynamic non-covalent interactions promote amine aggregation over time. Aggregation results in reduced adhesion and insolubility that shortens product lifetime.

This thesis presents two approaches to improve the sustainability of APOs using process-level and molecular-level strategies, with the goal to extend the functional lifetime of APOs. First, a physical recycling method was developed and optimized to recover APO from a polymer blend. Next, the molecular structure of APO was modified to incorporate cleavable amination linkages to form a recyclable network.

The first approach establishes a process for selective APO extraction from an APO/EVA-40 polymer blend using minimal organic solvents. A solvent/anti-solvent system was identified to recover APO under mild conditions. Solvents were reused to improve the sustainability of the process. The recycled APO maintains its molecular weight, thermal stability, surface topography, and dynamic behaviour, with slight changes in thermal and viscoelastic properties. In contrast, the recycled EVA-40 exhibited reduced thermal stability and crystallinity due to residual APO. Nevertheless, the recycling process was efficient and effective to recover APO with minimal aging, demonstrating the potential for future process scale-up.

The second approach introduces dynamic amination linkages to convert linear thermoplastic APO into recyclable thermosets. The crosslinker, *N,N'*-diphenyl-*N,N'*-bis(4-cyclooctene-1-methyl)methylenediamine (**a3**), was synthesized via CO₂ reductive functionalization. Highly crosslinked APO were rigid and brittle, whereas low crosslinked APO retained dynamic behaviour. A post-polymerization modification of APO with CH₂I₂ crosslinkers afforded *para-para*

methylene linkage with controllable crosslinking density and tunable thermal and rheological properties.

Overall, this thesis demonstrates that physical recycling methods and dynamic covalent networks offer viable pathways for enhancing APO sustainability. These findings lay the groundwork for the development of recyclable aminated thermosets.

Lay Summary

Functionalized plastics are crucial to modern society, as they serve countless niche applications from adhesives to anti-corrosive coatings. Amine-functionalized plastics developed in the Schafer group show potential use as adhesives due to their self-healing and adhesive behaviour. However, functionalized plastics are often single-use or difficult-to-recycle. Common recycling streams cannot recover such materials without compromising their functionality. In this thesis, two approaches were explored to enhance the recyclability and sustainability of amine-functionalized plastics. In the first approach, a plastic blend containing economically valuable amine-functionalized plastic was used as a model system to develop and optimize a physical recycling process. Organic solvents were used to selectively recover the valuable fraction without significantly altering its material properties. In the second approach, the amine-functionalized plastic was modified into a recyclable plastic via CO₂ fixation to extend its product lifetime. Overall, this thesis serves as a foundation to improve the sustainability of amine-functionalized plastics.

Preface

All work in this thesis is original and unpublished content completed by me under the supervision and guidance of Dr. Laurel L. Schafer, except the following portions.

All rheological measurements and data collection in Chapters 2 and 3 were conducted by Dr. Ziyue Zhang from Dr. Savvas Hatzikiriakos's lab in the Department of Chemical and Biological Engineering in UBC. Analyses regarding polymer viscoelastic properties were interpreted with the help of Dr. Ziyue Zhang and written by me.

Data in Chapter 2 has been used as part of the manuscript for publication through a collaboration with Dr. Elmira Mohammadi, Dr. Ziyue Zhang, Dr. Savvas Hatzikiriakos, and Dr. Laurel L. Schafer, and is expected to be submitted in the near future: Universal Hot-melt Adhesives from Miscible Blends of Amine-functionalized Polyolefin with Poly (ethylene-vinyl acetate). *Manuscript in preparation.*

X-ray crystallography measurement of **a1** in Chapter 3 was collected by Dr. Ben Nadeau. The crystal structure was refined and solved by 3rd year PhD candidate Natalie Wright.

Microsoft Copilot installed in Microsoft 365 Word was used for editorial purposes to improve paragraph flow and correct grammar mistakes during the writing process.

Table of Contents

Abstract	iii
Lay Summary	v
Preface	vi
Table of Contents	vii
List of Tables	x
List of Figures	xi
List of Schemes	xiii
List of Abbreviations	xv
Acknowledgements	xvii
Dedication	xix
Chapter 1: Introduction	1
1.1 The Plastic Waste Crisis in Canada	1
1.2 Current End-of-Life Treatments of Plastic Waste - Their Advantages and Limitations	3
1.3 Recycling Specialty Polymers – The Overlooked Role in Plastic Waste	5
1.4 Specialty Polymer Development in Schafer Group	5
1.4.1 Aminated Polyolefins (APOs)	5
1.4.2 Synthetic Strategies	6
1.4.3 Material Properties and Potential Applications	10
1.4.4 Economic Value and Recyclability Challenges	10
1.5 Molecular Design Principles for Recyclable Polymers	11
1.6 Scope of Thesis	14
Chapter 2: Physical Recycling of APO-Containing Blend	16
2.1 Introduction	16
2.1.1 APO/EVA-40 Blend	16
2.1.2 Scope of Chapter	19

2.2 Results and Discussion	20
2.2.1 Solvent Screening	20
2.2.2 Recycling Procedure Development and Optimization.....	24
2.2.3 Characterization of Recovered Polymers.....	25
2.2.3.1 Comparison of Pristine APO and Recycled APO.....	26
2.2.3.2 Comparison of Pristine EVA-40 and recycled EVA-40	30
2.2.4 Discussion	35
2.3 Conclusion	36
2.4 Experimental	36
2.4.1 Materials	36
2.4.2 Instrumentation	37
Chapter 3. Design of Cleavable Crosslinked APO	41
3.1 Introduction.....	41
3.1.1 Dynamic Covalent Networks (DCNs) in Polymers	41
3.1.2 Dual System of DCNs and Non-covalent Interactions in Polymers	43
3.1.3 Incorporating DCNs into APO.....	44
3.1.4 Scope of Chapter.....	47
3.2 Results and Discussion	48
3.2.1 Small-Molecule Reaction with Formaldehyde	48
3.2.2 Small-Molecule Reaction with Diiodomethane.....	51
3.2.3 Small-Molecule Reaction with CO ₂	54
3.2.4 Transamination of Aromatic Amino Species.....	57
3.2.5 Crosslinking APO with Aromatic Amino Linkage.....	58
3.2.6 Crosslinking APO with Diiodomethane	60
3.3 Conclusion	64
3.4 Experimental	66
3.4.1 Materials	66
3.4.2 Instrumentation	70
Chapter 4. Conclusion and Future Work	71

4.1 Thesis Overview	71
4.2 Future Work	73
References	76
Appendices	84

List of Tables

Table 2-1. Solubility of APO and EVA-40 in common organic solvents listed based on ascending Reichardt polarity index at 25 °C for 24 hours.	22
Table 2-2. Comparison of molecular weight and thermal properties of pristine and recycled APO based on GPC, DSC, and TGA measurements.	27
Table 2-3. Comparison of molecular weight and thermal properties of pristine and recycled EVA-40, and APO/EVA-40 blend based on GPC, DSC, and TGA measurements.....	33
Table 3-1. Summary of reactions of aniline derivatives with formaldehyde under different conditions.	49
Table 3-2. Summary of reactions of aniline derivatives with CH ₂ I ₂ under different conditions..	53
Table 3-3. Summary of reactions of ACC-NMA with CO ₂ under different conditions.	56
Table 3-4. Summary of crosslinked APO using CH ₂ I ₂ under different conditions.	61

List of Figures

Figure 1-1. Sankey diagram of the flow of plastic waste in Canada in 2021 by resin in thousand tonnes. ^{1,2}	1
Figure 1-2. Workflow comparison of different End-of-Life Treatments of Plastics Solid Waste (PSW). ¹⁰	3
Figure 1-3. General illustration of backbone entanglement, H-bonding, π - π interactions in APO matrix.	6
Figure 1-4. Venn diagram of thermoplastic, thermoset, and polymer with DCNs.	13
Figure 1-5. Illustration of dynamic covalent networks.	13
Figure 2-1. Illustration of dynamic H-bonding between 2° amines.	16
Figure 2-2. General illustration of backbone entanglement and H-bonding interactions between APO and EVA-40.	18
Figure 2-3. Schematic APO/EVA-40 solvent-based recycling steps using NMP and MeOH.	25
Figure 2-4. ¹ H NMR spectra (400 MHz, 298 K, CDCl ₃) of pristine APO, recycled APO, APO/EVA-40 blend, and pristine EVA-40 (Spectra intensity adjusted based on signal at 5.41 ppm).	26
Figure 2-5. Thermograms of pristine and recycled APO, and 80/20 wt% APO/EVA-40 blend..	27
Figure 2-6. SEM images of a) pristine APO, b) recycled APO, and c) 80/20 wt% APO/EVA-40.	28
Figure 2-7. AFM image of 80/20 wt% APO/EVA-40.	29
Figure 2-8. Master curves of pristine and recycled APO from frequency sweep experiments at T _{ref} = 30 °C.	30
Figure 2-9. FT-IR spectra of pristine APO, 80/20 wt% APO/EVA-40 blend, and recycled EVA-40.	31
Figure 2-10. ¹ H NMR spectra (400 MHz, 298 K, CDCl ₃) of pristine EVA-40, recycled EVA-40, and 80/20 wt% APO/EVA-40 blend (Spectra intensity adjusted based on signal at 4.86 ppm).	32
Figure 2-11. Thermograms of pristine and recycled EVA-40, and the blend.	34
Figure 2-12. SEM images of a) pristine EVA-40 and b) recycled EVA-40.	35
Figure 3-1. Transamination between 2° amines via formation of iminium ion. ⁷⁸	46

Figure 3-2. General illustration of aminated polymer followed by reversible transamination.	47
Figure 3-3. The proposed reaction pathway of reductive functionalization of CO ₂ . ^{83,86}	57
Figure 3-4. Gelled 40 mol% aminated-linked APO.	59
Figure 3-5. Master curves of linear and crosslinked APOs synthesized in DCM at room temperature from frequency sweep experiments at T _{ref} = 30 °C.	61
Figure 3-6. Master curves of linear and crosslinked APOs synthesized in DCM at 40 °C from frequency sweep experiments at T _{ref} = 30 °C.	62
Figure 3-7. Master curves of linear and crosslinked APOs synthesized in THF at room temperature from frequency sweep experiments at T _{ref} = 30 °C.	63
Figure 3-8. Master curves of linear and crosslinked APOs synthesized in THF at 60 °C from frequency sweep experiments at T _{ref} = 30 °C.	64

List of Schemes

Scheme 1-1. Hydroaminoalkylation to form C-C bond at α -carbon to the amine and the alkene. ²⁴	7
Scheme 1-2. General reaction scheme of ring-opening-metathesis polymerization (ROMP) using G2.	8
Scheme 1-3. General synthetic protocol of poly(aminonorbornenes).	8
Scheme 1-4. General synthetic protocol of aminopolyolefines (APOs).	9
Scheme 1-5. General synthetic scheme of HAA of vinyl-terminated polyolefines.	9
Scheme 1-6. General synthetic scheme of HAA of 1,2-polybutadiene.	9
Scheme 2-1. Physical blending of APO and EVA-40.	17
Scheme 3-1. Polymer with DCN using reversible Diels-Alder reaction. ⁶⁰	42
Scheme 3-2. Exchange reaction via transesterification in epoxide carboxylic acid crosslinked networks. ⁴¹	43
Scheme 3-3. Dual-DCN polymer matrix containing disulfide metathesis and carboxylate transesterification. ⁷¹	43
Scheme 3-4. Polymer containing 2° amide H-bonding site and G2-mediated self-healing olefin-containing network. ⁷²	44
Scheme 3-5. Equilibrium between isocyanate, amine, and corresponding urea. ⁷⁵	45
Scheme 3-6. Synthesis of poly(aminal)s using diamine and paraformaldehyde. ⁷⁸	46
Scheme 3-7. Reaction scheme of NMA and formaldehyde.	48
Scheme 3-8. Reaction scheme of 4-MeO-NMA and formaldehyde.	49
Scheme 3-9. General reaction between ACC-NMA and PFA with target product a3.	50
Scheme 3-10. General reaction between aniline derivatives and CH ₂ I ₂ with targeted aminal products and by-products.	52
Scheme 3-11. Single-step reductive functionalization of CO ₂ using TBD organocatalyst to yield formamide. ⁸⁵	54
Scheme 3-12. Single-step reductive functionalization of CO ₂ using TBD organocatalyst to yield aminal a1. ⁸³	54
Scheme 3-13. Single-step reductive functionalization of CO ₂ using TBD organocatalyst to yield aminal a3.	55
Scheme 3-14. Formation of asymmetric aminal species.	58

Scheme 3-15. <i>In-situ</i> reaction of aminated crosslinked APO.....	59
Scheme 3-16. General reaction of crosslinking APO with CH ₂ I ₂	60

List of Abbreviations

4-MeO-NMA	4-methoxy- <i>N</i> -methylaniline
ACC-NMA	Amine-containing cyclooctene <i>N</i> -methyl aniline
AFM	Atomic force microscope
APO	Aminated polyolefin
CHCl ₃	Chloroform
COSY	Correlated spectroscopy
$\bar{D}$	Dispersity
DCM	Dichloromethane
DCN	Dynamic covalent network
DMF	Dimethyl formamide
DMSO	Dimethyl sulfoxide
DSC	Differential scanning calorimetry
ϵ	Dielectric constant
E_T^N	Reichardt scale
EoL	End-of-life
eq.	Equivalent
EtOAc	Ethyl acetate
EVA	poly(ethylene-vinyl acetate)
EVE	Ethyl Vinyl Ether
FT-IR	Fourier-Transform Infrared
G'	Storage modulus
G''	Loss modulus
G2	Grubbs catalyst 2nd generation
GC-MS	Gas chromatography-mass spectrometry
G _e	equilibrium modulus
GPC	Gel permeation chromatography
HAA	hydroaminoalkylation
HSQC	heteronuclear single quantum correlation
KTp	potassium tri(1-pyrazolyl) borohydride
m/z	Mass to charge ratio
MeCN	Acetonitrile
MeOH	Methanol
M _w	Weight average molecular weight
NBA	<i>N</i> -butylaniline
<i>n</i> BuOH	<i>n</i> -butanol
NMA	<i>N</i> -methylaniline
NMP	<i>N</i> -methyl-2-pyrrolidone
NMR	Nuclear magnetic resonance
P'	Snyder polarity index

PCL	Polycaprolactone
PFA	Paraformaldehyde
PLA	Polylactic acid
PPM	Post-polymerization modification
ppm	parts per million
PSW	Plastic solid waste
PTFE	Polytetrafluoroethylene
RBF	Round bottom flask
ROMP	Ring-opening-metathesis polymerization
SEM	Scanning electron microscope
TBD	1,5,7-triazabicyclo[4.4.0]dec-5-ene
T _d , 5%	5% decomposition temperature
T _d , 50%	50% decomposition temperature
TFA	Trifluoroacetic acid
T _g	Glass transition temperature
TGA	Thermogravimetric analysis
THF	Tetrahydrofuran
T _m	Melting temperature
μ	Dipole moment
WtE	Waste-to-Energy
X_d	Gas-liquid partition coefficient using ethanol
X_e	Gas-liquid partition coefficient using dioxane
X_n	Gas-liquid partition coefficient using nitromethane
XRD	X-ray diffraction

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Dr. Laurel L. Schafer. Thank you for your invaluable guidance, mentorship, and support throughout my Master's degree. This work would not have been possible without your knowledge and support.

I would also like to thank Dr. Zachary M. Hudson and Dr. Savvas G. Hatzikiriakos for being on my thesis committee. I truly appreciate the time and insightful feedback you have provided on my work.

I gratefully acknowledge Maria Ezhova from the NMR facility, Ben Herring from SIF, and Derrick Horne from BIF for their technical expertise. Their patient training and troubleshooting were essential to the completion of this work. The data presented in this thesis would not exist without their support.

I am deeply grateful to my mentor and dear friend, Dr. Elmira Mohammadi. Your unconditional support, patience, and encouragement was valuable to me. Your contributions to this research extended far beyond the scientific aspects. No matter how stressful and stubborn I was, you always worked through challenges together with me. Thank you for your scientific and moral support throughout this journey. This is something I will remember for the rest of my life. Knowing you as both a mentor and a friend has truly been an honour.

I would also like to thank the members of the Schafer group for their kindness, help, and collaboration over the past two years. In particular, I would like to acknowledge Natalie Wright, Dr. Victoria S. Pfennig, Keegan Milliken, Hafiza Ruqiya Tasneem, and Matthew Isaac for their constructive feedback and troubleshooting during my graduate school. I have learned so much from you. Completing this thesis would not have been possible without your support and careful revisions. Working with you has been a truly inspiring experience.

I would also like to thank Dr. Ziyue Zhang for her expertise in polymer rheology. Your contribution is crucial in providing a more comprehensive understanding of the polymers investigated in this work.

I would like to express my heartfelt appreciation to my partner, Mia Chen, for her emotional support throughout this challenging and stressful journey. Thank you for always cheering me up, believing in me, and helping me grow into a more resilient and confident person.

Finally, I would like to thank my family for always being there for me. You have been my strongest source of support in my entire life. Being more than 10,000 km away from home has not been easy, but because of your love, I was able to achieve this milestone.

Dedication

To my beloved parents,

陳維經 (Wei-Ching Chen) and 于貞華 (Chen-Hua Yu);

And my sisters,

Vicky and Tilly Chen.

Chapter 1: Introduction

1.1 The Plastic Waste Crisis in Canada

The Canadian plastic industry follows a linear “take-make-waste” model. Non-renewable resources are extracted and transformed into plastic products and discarded at the end of life. For example, Canadians generated nearly 5 million tonnes of plastic waste in 2021. However, only 7% of the plastic waste was diverted into recycling streams and 2% was incinerated for energy recovery. Instead, the majority of plastic waste was landfilled (79%), released into the environment (1%), or exported to regions with less developed waste management systems (10%).^{1,2}

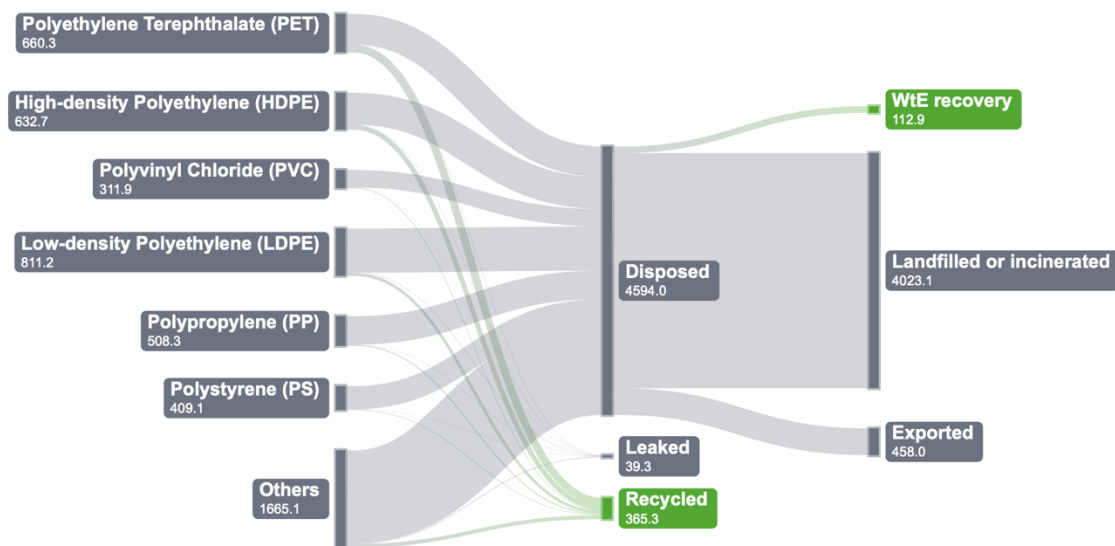


Figure 1-1. Sankey diagram of the flow of plastic waste in Canada in 2021 by resin in thousand tonnes.^{1,2}

The accumulation of plastic waste is both an environmental and economical concern. Plastic waste threatens ecosystems through landfill persistence, microplastic formation, and marine pollution, while generating a large economic loss.³ In 2016, an estimated \$7.8 billion CAD was lost due to unmanaged plastic waste, with this figure projected to rise to \$11 billion CAD by 2030.⁴ Despite the urgency, the recycling rate of plastics are strikingly low due to economic disincentives and technical challenges. The lack of infrastructure also contributes to the low recycling rates.⁵ From an economic point of view, the low cost of producing pristine plastics from cheap petroleum-based feedstocks make recycled plastics uncompetitive on the market.

Among all the plastic products, six types of plastic materials are the most common ones, including polyethylene terephthalate (PET), high-density polyethylene (HDPE), low-

density polyethylene (LDPE), polyvinyl chloride (PVC), polypropylene (PP), and polystyrene (PS). They are categorized as “commodity plastics” or “commodity polymers,” where polymers refer to large molecules connected by repeated molecular units in a long chain. 67% of the plastic waste collected in 2021 was contributed by these six plastic types while the other 33% was everything else. The commodity plastics are commonly used in food packaging, single-use item, textile, electronics, automobile, and construction materials due to their low production cost and excellent material properties. The cheap production of commodity plastics, therefore, discourages investment in new and improved plastics recycling technologies.^{3,6}

Technical challenges like thermal degradation, loss of molecular weight, or reduced mechanical properties are common in recycled plastics.⁷ Additionally, some plastic products contain additives like stabilizers, plasticizers, or pigment dyes to modify the physical or mechanical properties of the plastics to meet certain product needs. Besides additives, contaminants like food residues are also very common in the collected plastic wastes. All of these impurities in plastic waste further degrade the quality of recyclates as these substances can decompose into undesired by-products upon recycling.⁸ Complicated plastic products like multilayer packaging films or polymer blends are not separable by conventional methods.⁹ Purifying these materials and reprocessing them is even more difficult. In terms of infrastructure, collection and sorting facilities are severely lacking in most areas. These factors highlight the inefficiency of the current recycling systems that contributes to the low recycling rate of plastic wastes.

However, transitioning to a circular economy, where materials are kept in use at their highest value for as long as possible, is essential to reducing both carbon emissions and plastic waste. A circular economy approach focuses on extending product lifecycles through reuse, recycling, and sustainable design, thereby conserving resources and minimizing environmental impact in a long-term manner. To address the urgent need of plastic waste crisis, innovative and sustainable strategies need to be developed.

1.2 Current End-of-Life Treatments of Plastic Waste - Their Advantages and Limitations

The limitations in current plastic recycling streams have prompted the development of various end-of-life (EoL) treatments aimed at recovering economic value from plastic waste. The most widely adopted approaches include mechanical recycling, waste-to-energy (WtE) incineration, and chemical recycling. Each method offers unique advantages but also carries significant limitations.¹⁰ The workflow of different EoL treatments of plastic solid waste (PSW) was shown in Figure 1-2.

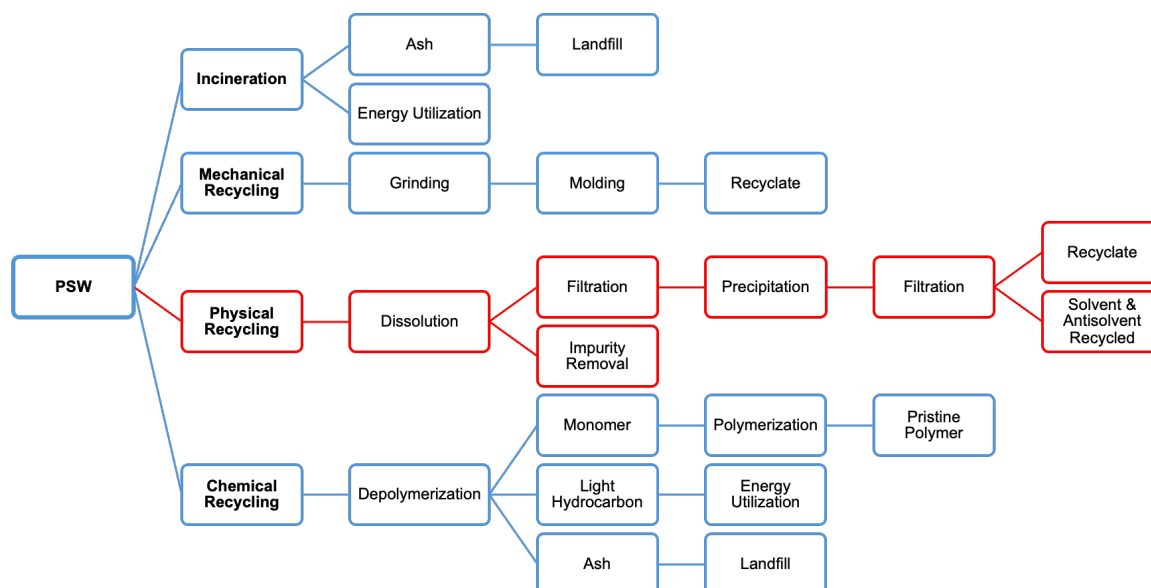


Figure 1-2. Workflow comparison of different End-of-Life Treatments of Plastics Solid Waste (PSW).¹⁰

Mechanical recycling transforms plastic waste into secondary raw materials without significantly altering the chemical structure of the polymers.¹¹ The process involves sorting, washing, shredding, and melting plastic waste to produce granulates that are used to manufacture new plastic products. Mechanical recycling is the most prevalent method for plastic value recovery in Canada, however, its capacity to drive a circular plastic economy is limited. The quality of recycled plastics is dependent on the purity and consistency of the input waste stream.⁸ In practice, feedstocks are often contaminated or composed of mixed plastics, which “downcycles” the products into lower economic value products with compromised material properties. Furthermore, recyclates suffer from thermomechanical degradation during the melting step, resulting in chain scission, loss of molecular weight, and compromised material performance.^{8,12} Consequently, plastics that are classified as recyclable often end up in landfills, undermining the intended benefits of mechanical recycling.

WtE incineration is the second most common method of plastic waste treatment in Canada. In WtE facilities, plastics generate ash, flue gas, and heat through combustion reactions.¹³ The heat is used to produce steam that drives turbine generators to produce electricity, while the ash is discarded in landfills. Plastics are petroleum-derived and possess high calorific value. Therefore, plastics serve as efficient fuels for energy recovery. While this process helps to offset fossil fuel based energy production, it is controversial due to the release of greenhouse gases and other pollutants during incineration.¹⁴

Chemical recycling offers a different approach by breaking down plastics into their monomeric form for reuse in polymer synthesis or alternative products with higher economic values.¹⁵ In theory, processes such as pyrolysis, solvolysis, and depolymerization can regenerate pristine quality monomers or produce valuable hydrocarbon feedstocks.¹⁶ While chemical recycling is promising in its ability to address difficult-to-recycle plastics and reduce reliance on fossil resources, it remains in an early stage of maturity and faces barriers to scalability, energy efficiency, and economic viability.¹⁷

In recent years, solvent extraction, also known as physical recycling, has emerged as a promising alternative, particularly for multi-polymer blends and multilayer plastic wastes.¹⁸ Solvent extraction, or physical recycling, has gained attention as an industrially viable technique for separating mixed plastics and multilayer plastic packages. By using solubility differences, solvent extraction enables the selective dissolution of one plastic component under mild conditions, while leaving the others plastic intact.¹⁵ The recovered plastics retain properties close to their pristine state, avoiding the thermomechanical degradation observed using mechanical recycling. In addition, this method can remove additives and plasticizers, thereby improving recycle quality.^{19,20} For multi-component plastic blends, solvent extraction presents a more suitable solution compared to conventional methods, particularly when high economic value materials are involved.

1.3 Recycling Specialty Polymers – The Overlooked Role in Plastic Waste

While current recycling strategies provide partial solutions for commodity polymers, they are far less effective when applied to specialty polymers. Specialty polymers include adhesives, coatings, composites, and thermosets, which are produced in relatively small volumes compared to commodity polymers. They serve critical roles in high-value or niche applications. Specialty polymers are designed for performance rather than recyclability in most cases. Current EoL treatments cannot effectively recycle specialty polymers with quality close to their pristine state.²¹ Furthermore, specialty polymers are integrated into products, posing challenges to separate. For example, permanent crosslinking in thermosets (a type of polymer that is hardened and cannot be melted) allows excellent mechanical durability but makes depolymerization extremely difficult.²² Heterogeneous formulations with stabilizers, plasticizers, and curing agents in blends and composites further complicate recycling.¹⁰ Coatings and adhesives possess strong interfacial adhesion to substrates, which prevents mechanical separation without damaging the underlying base material.²³

As a result, recycling systems prioritize commodity polymers such as PE, PP, PET, and PS as they are thermoplastics that can be melted, solubilized, or reshaped. This leaves specialty polymers overlooked. Specialty polymers account for a small fraction of global plastic production by weight, however they exert a disproportionate environmental burden.²³ Specialty polymers compromise the recyclability of larger quantities of commodity plastics, which often end up as contaminated residues leading to incineration or discard in the landfill.

1.4 Specialty Polymer Development in Schafer Group

1.4.1 Aminated Polyolefins (APOs)

Aminated polyolefins (APOs) represent a unique class of high-value specialty polymers derived from commodity polyolefins through the incorporation of secondary amine functionalities. In the Schafer group, aniline derivatives have been incorporated into side chains of polymers.²⁴ This structural modification introduces the possibility of dynamic hydrogen bonding through the N–H bond and provides π – π stacking interactions between aromatic rings embedded within the polymer matrix as shown in Figure 1-3. The combination of intermolecular and intramolecular H–

bonding and π - π stacking introduce dynamic non-covalent interactions that is absent in conventional non-polar polyolefins.

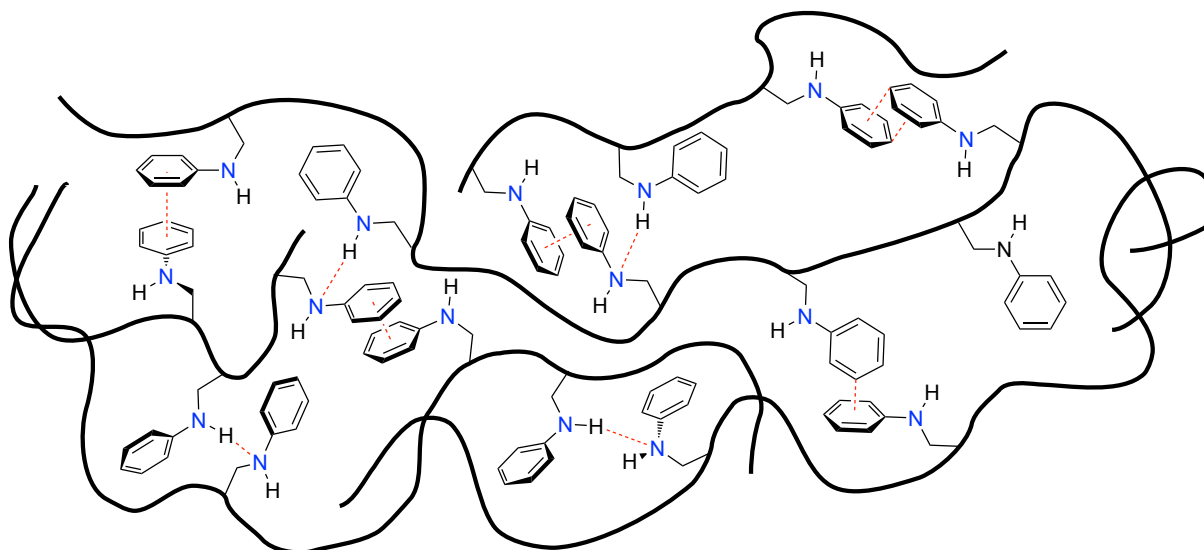


Figure 1-3. General illustration of backbone entanglement, H-bonding, π - π interactions in APO matrix.

As a result, APOs exhibit enhanced adhesion, as the amine moiety can interact with a wide range of polar and non-polar substrates. They also exhibit self-healing behaviour, driven by reversible H-bonding and π - π stacking. The structure of the polymeric backbone and the degree of amination of the alkene within the polymer matrix allows further tunability of mechanical properties, making this class of polymers responsive to subtle structural variations.²⁵ APOs extend the uses of commodity polyolefins into the realm of specialty materials with dynamic performance characteristics.

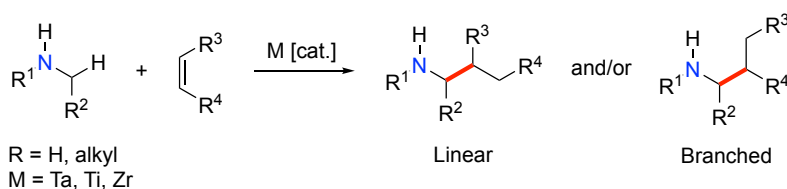
However, APOs face the recyclability challenges unique to specialty plastics as well. Existing EoL treatments fail to preserve these advanced features, leading to compromised performance and shortened product lifetimes. APOs exemplify the broader sustainability issues associated with specialty polymers and provide a valuable model system for developing tailored recycling strategies and advancing molecular design principles.

1.4.2 Synthetic Strategies

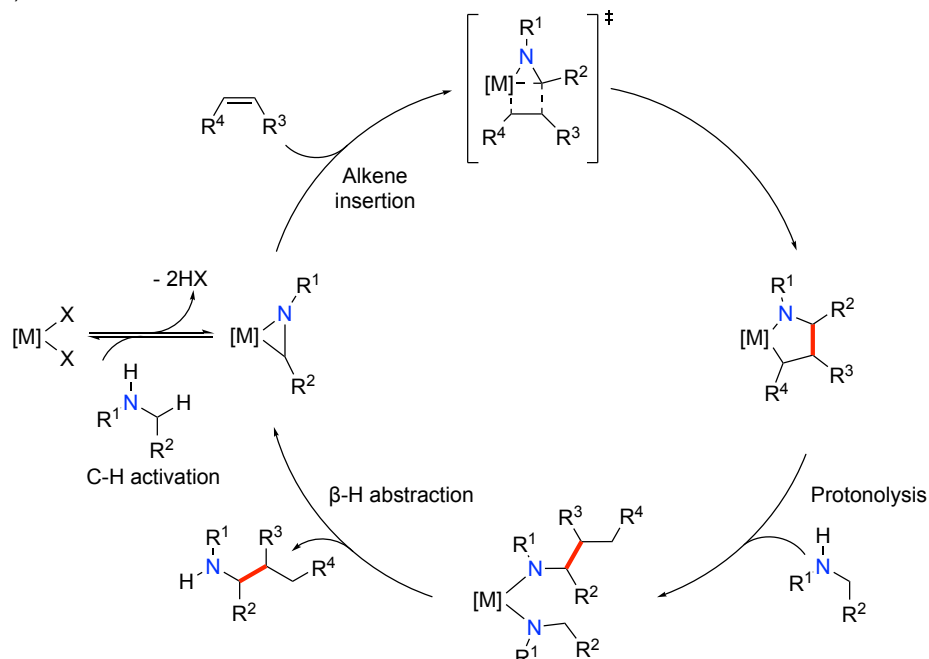
The Schafer group has pioneered a catalytic synthetic route to introduce amine moieties into alkenes, with an emphasis on aniline derivatives. The development of early transition metal-catalyzed hydroaminoalkylation (HAA) has been studied extensively (Scheme 1-1, a).²⁴ This

catalytic synthetic protocol allows a transformation that directly couples secondary amines with alkenes via the C–H activation, formation of metallaaziridine, alkene insertion, protonolysis, and β -hydrogen abstraction (Scheme 1-1, b). This transformation is highly atom-economic, as it forms a new carbon–carbon single bond at the alkyl α -carbon of the amine without generating stoichiometric by-products and simplifies a multi-step synthesis. Earth abundant early transition metal catalysts like tantalum, titanium, and zirconium were used for HAA, making this process more environmentally friendly. Some late transition metals like nickel, ruthenium, and iridium have also been studied as catalysts for HAA.^{26–28}

a) General reaction scheme of hydroaminoalkylation (HAA)

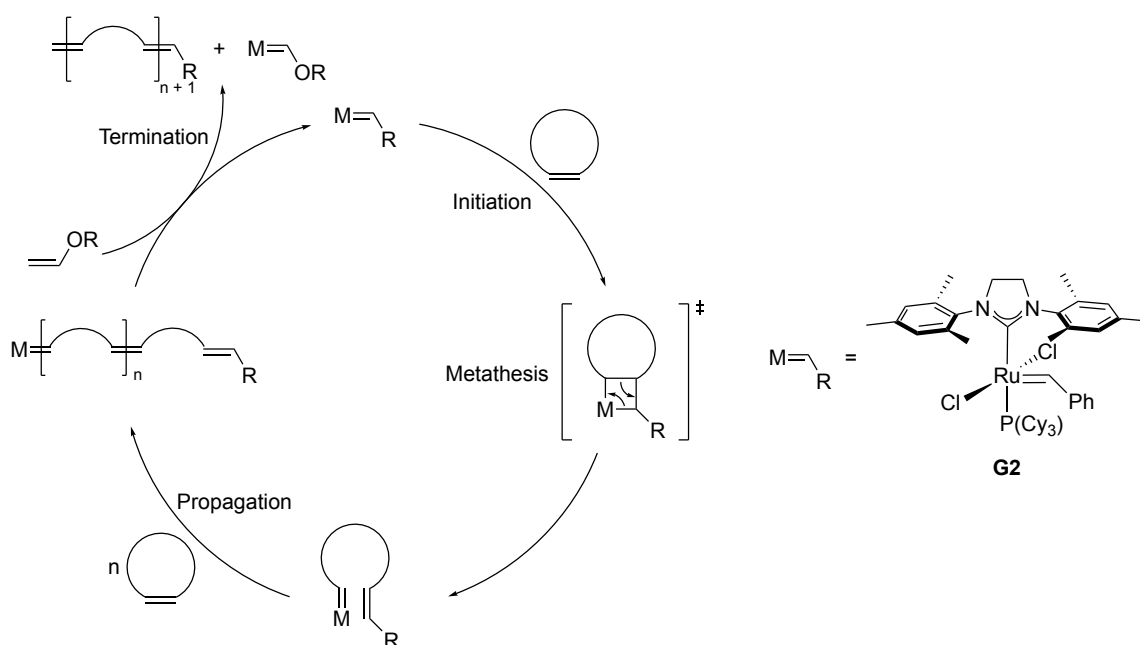


b) General reaction mechanism of HAA



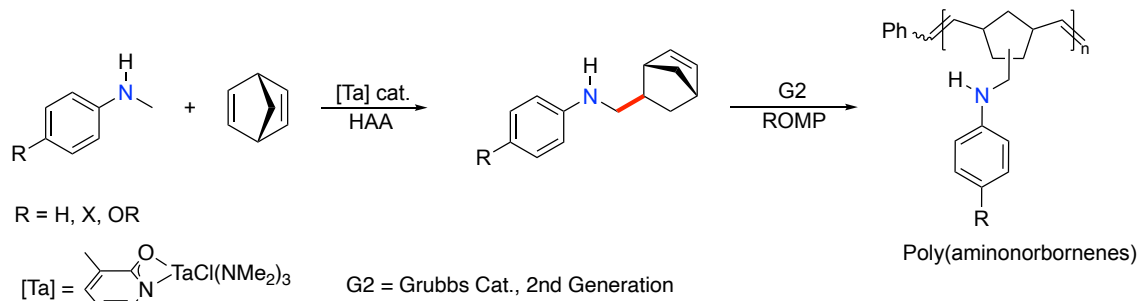
Scheme 1-1. Hydroaminoalkylation to form C-C bond at α -carbon to the amine and the alkene.²⁴

There are two synthetic routes to prepare APOs. The first involves the HAA of one of the alkenes in cyclic-diene monomers to yield amine functionalized monomers consisting of one alkene using Scheme 1-1. The monomers are subsequently subjected to ring-opening-metathesis polymerization (ROMP) using Grubbs catalyst 2nd generation (G2) to yield linear polyolefins containing 2° amine (Scheme 1-2).^{29,30}

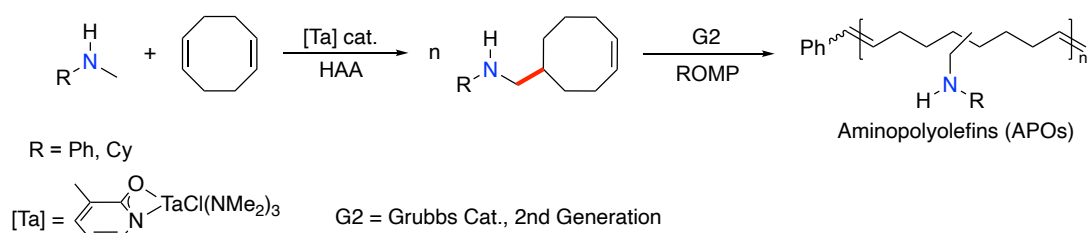


Scheme 1-2. General reaction scheme of ring-opening-metathesis polymerization (ROMP) using G2.

This synthetic strategy affords linear polymers with well-defined architectures and the controlled distribution of amine groups along the backbone. Our group first synthesized poly(aminonorbornenes) using this protocol in 2016, as shown in Scheme 1-3.²⁹ The resultant polymers showed tunable thermal and rheological properties due to H-bonding and π - π stacking interactions based on the electronic properties of the amine. In 2021, our group further applied this synthetic strategy to cyclooctadiene with 2° amine derivatives to yield aminopolyolefins (APOs) with increased flexibility in the polymer backbone compared to poly(aminonorbornenes), enabling rapid self-healing behaviour (Scheme 1-4).³⁰

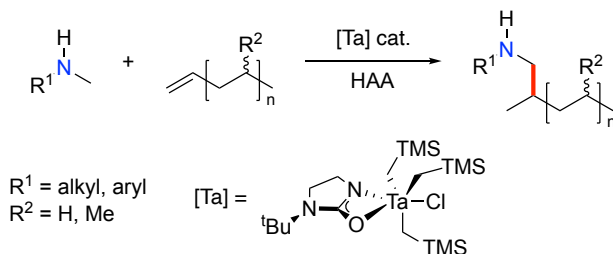


Scheme 1-3. General synthetic protocol of poly(aminonorbornenes).

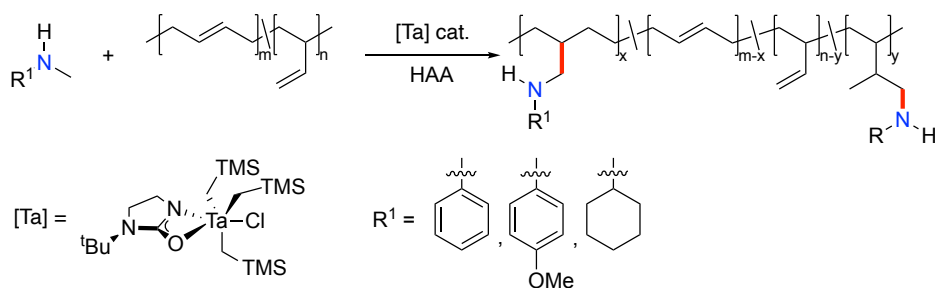


Scheme 1-4. General synthetic protocol of aminopolyolefins (APOs).

The second synthetic strategy uses post-polymerization modification (PPM), where commodity polyolefins are functionalized with amine groups through HAA, as illustrated in Scheme 1-1. Using this synthetic protocol, our group successfully aminated vinyl-terminated polypropylene in 2021 and vinyl-terminated polyethylene in 2024.^{31,32} Furthermore, the internal alkene of 1,2-polybutadiene were aminated via HAA with the presence of directing group, as shown in Scheme 1-6.²⁵ PPM takes advantage on the wide availability of commodity polymers, and transform their physicochemical properties through the incorporation of reversible non-covalent interactions.



Scheme 1-5. General synthetic scheme of HAA of vinyl-terminated polyolefins.



Scheme 1-6. General synthetic scheme of HAA of 1,2-polybutadiene.

Together, these strategies highlight the versatility of HAA and establish a robust framework for transforming simple alkenes or existing polyolefins into materials with dynamic behaviour. By combining catalyst design and monomeric structural control, the Schafer group's methodology broadens the scope of functionalized polyolefins.

1.4.3 Material Properties and Potential Applications

The incorporation of amine functionalities into the polyolefin backbone introduces distinct material properties that differentiate aminopolyolefins (APOs) from conventional polyolefins. APOs display self-healing behaviour arising from reversible non-covalent interactions, particularly dynamic H-bonding and π - π stacking, which allows polymer chains to re-associate after mechanical damage.³⁰ These reversible physical crosslinkages within the bulk material contribute to enhanced cohesive strength. Furthermore, the presence of pendant amines increases the surface energy of APOs compared to unfunctionalized polyolefins, thereby improving adhesion to a range of substrates, including steel, aluminum, and PTFE.³⁰

The combination of self-healing behaviour and substrate adhesion renders APOs as promising candidates for applications where commodity polyolefins are inadequate. Potential applicable areas include coatings, interlayer primers, and functional polymer blends that require improved interfacial adhesion. Preliminary studies within the Schafer group on polymer blending with ethylene-vinyl acetate (EVA) have shown that APOs promote compatibility and mechanical reinforcement in mixed polymer systems, underscoring their potential in the design of advanced blends. Collectively, these attributes establish APOs as valuable materials for specialty markets where the enhanced performance justify the higher cost of specialty polymers.

1.4.4 Economic Value and Recyclability Challenges

As specialty polymers, APOs occupy a distinct niche in the polymer economy. Their high-performance features, including tunable adhesion, self-healing behaviour, and recyclability potential, justify their higher economic value compared to commodity polyolefins and make them attractive for applications demanding advanced functionality. However, these attractive polymeric features complicate their EoL treatment, as outlined in Section 1.3. In particular, the H-bonding interactions that enable their adhesive strength and dynamic reversibility drive the system toward its thermodynamic minimum, forming H-bonding-induced clustering at its equilibrium state.³³ This process results in a gradual loss of adhesion and the formation of insoluble aggregates, compromising reprocessability and recyclability. Unlike conventional thermoplastics that tolerate high-temperature remelting or mechanical grinding during common EoL treatments, APOs require

mild processing conditions to retain their advanced properties. Without mild approaches, their functionalities are lost and their higher production costs become difficult to justify.

This highlights the need for sustainable recycling strategies designed for APOs. Effective solutions must retain functional properties while minimizing material degradation during processing and preserve the economic value of APOs by preventing downcycling into lower economic value products. Conventional EoL treatments, including WtE incineration and mechanical recycling, fall short of these requirements despite being lower cost. Physical recycling through solvent extraction offers a viable recycling method, since it avoids harsh thermal and mechanical conditions and can maintain the integrity of dynamic non-covalent interactions, thereby preserving APOs' functional properties. Therefore, an ideal physical recycling process for APOs was investigated. The efficiency and effectiveness of the physical recycling process were evaluated by comparing the pristine and recycled APO in terms of their purities, polymeric properties, thermal stabilities, and rheological properties.

Unfortunately, solvent extraction alone cannot fully ensure the long-term sustainability of APOs. To extend product lifetimes and align with circular economy principles, reprocessability must be embedded directly at the molecular level. This prompted the development of new materials that would allow for reversible depolymerization and repolymerization. By enabling APOs to be broken down and reformed without significant loss of function, such molecular design strategies could address both the environmental and economic challenges of specialty polymers. APOs not only exemplify the difficulties of recycling advanced polymers but also provide a model framework for innovating next-generation sustainable materials.

1.5 Molecular Design Principles for Recyclable Polymers

Molecular design offers a promising pathway to address the recycling challenges of specialty polymers by embedding chemical functionalities that enable controlled degradation or reprocessability. Two major strategies have been widely explored. The first is the development of biodegradable polymers and the second is the incorporation of dynamic covalent networks (DCNs) into polymer backbones or crosslinking sites.^{34,35}

The first approach focuses on designing polymers that undergo degradation under environmental or biological conditions. Notable examples include polylactic acid (PLA) and polycaprolactone (PCL), which are used in packaging and 3D printing materials. While effective in shortening natural degradation process in the environment, biodegradable polymers face significant limitations in specialty applications. Their mechanical strength and thermal stability are often insufficient for demanding uses such as adhesives, coatings, or structural components.³⁶ Moreover, degradation pathways are uncontrolled, and result in unpredictable product lifetimes and compromised reliability. The requirement for specific environmental conditions at their EoL stage restricts large-scale implementation and undermines economic feasibility.^{37,38} Biodegradable polymers tackle disposal challenges at the EoL stage, however, they do not inherently enable closed-loop recycling.

An alternative design principle lies in the use of dynamic covalent networks (DCNs), which combine the mechanical robustness of thermosets with the reprocessability of thermoplastics shown in Figure 1-4.³⁹ There are two general mechanisms that govern dynamic covalent bond exchange: dissociative and associative DCNs. In dissociative DCNs, covalent bond cleavage occurs first, followed by bond re-formation at another site. This is a two-step mechanism analogous to an S_N1 mechanism (Figure 1-5).⁴⁰ This reduces crosslink density once activated, leading to temporary decreases in viscosity. In contrast, associative DCNs undergo simultaneous bond cleavage and formation. This is a one-step process analogous to an S_N2 mechanism, thereby maintaining constant crosslink density and viscosity throughout the exchange.⁴⁰ This associative mechanism enables vitrimer behaviour, where materials remain malleable at elevated temperatures while preserving their network connectivity. The concept of “vitrimerization” was first introduced by Leibler and co-workers in 2011, and numerous DCN systems have been developed and incorporated into polymers to-date.⁴¹

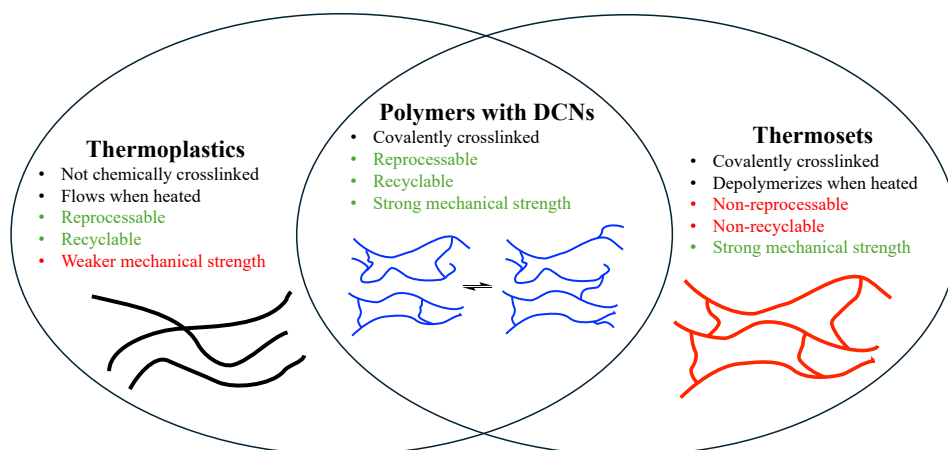


Figure 1-4. Venn diagram of thermoplastic, thermoset, and polymer with DCNs.

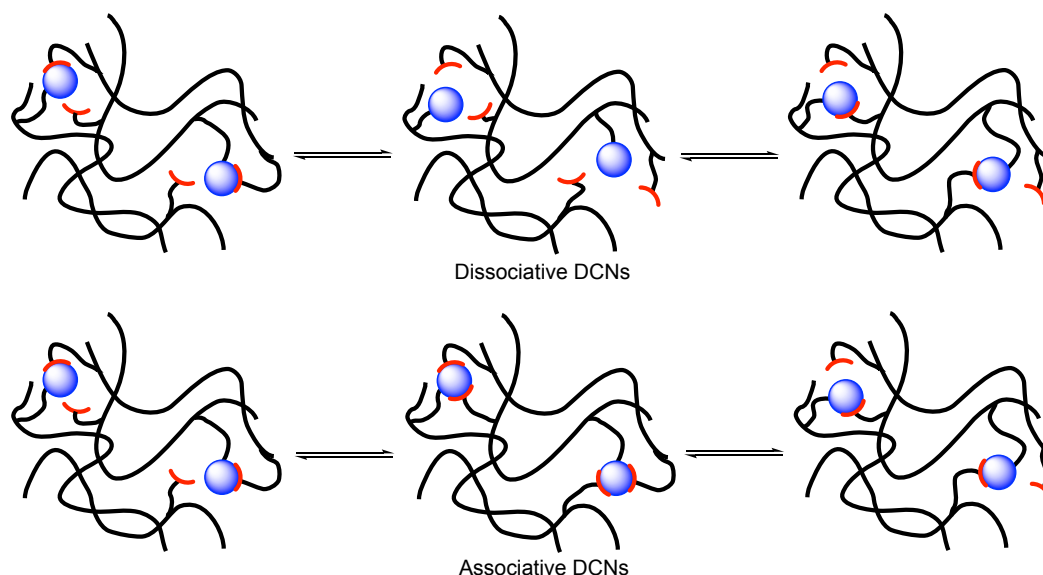


Figure 1-5. Illustration of dynamic covalent networks.

DCNs introduce additional advantages beyond reprocessability. Selective bond cleavability enables depolymerization or network breakdown under controlled conditions, allowing recovery of polymer fractions without irreversible degradation or side reactions. Furthermore, the reversibility of bond exchange grants self-healing capability, as damaged regions can repair through reversible bond shuffling. The synergistic combination of dynamic covalent networks with reversible non-covalent interactions enhances cohesive strength and adhesion.⁴² Collectively, polymers incorporating DCNs represent a powerful design framework for advancing circularity in specialty polymers by combining durability, recyclability, and functional versatility within a single material.

These design principles are directly relevant to APOs. As outlined in Section 1.4, APOs suffer from gradual loss of adhesion and recyclability challenges due to the formation of H-bonding-induced clustering. Embedding DCNs into APOs as crosslinking sites offers a pathway to mitigate these issues. Crosslinking can anchor polymer segments, restrict backbone mobility, and suppress aggregation, while cleavable covalent bonds preserve reprocessability. Motivated by this framework, this thesis investigates strategies to enhance the sustainability of APOs.

1.6 Scope of Thesis

The sustainability of specialty polymers such as APOs requires both process-level and molecular-level solutions. APOs are valuable for their adhesion, self-healing, and dynamic reversibility, but their recyclability is compromised by aggregation that reduces adhesion and shortens material lifetimes. Addressing these challenges calls for approaches that recover APOs from EoL stage while embedding reprocessability at the molecular scale. This thesis pursues two complementary research directions. The first is a process-level approach, which is the development of a solvent-based recycling method for APO-containing blends. The second is a molecular-level approach, which is the design of cleavable crosslinked APOs that retain self-healing and adhesive behaviours while enabling controlled recyclability.

The first research direction, presented in Chapter 2, focuses on developing a physical recycling strategy for APOs through selective solvent extraction. By exploiting differences in solubility, APOs can be separated from polymer blends and recovered with minimal loss of functional properties. This process-level solution addresses a critical gap in specialty polymer recycling by enabling the preservation of adhesion, mechanical integrity, and self-healing properties during recovery. At the same time, it provides a practical pathway for extending the usable lifetime of APOs, though challenges such as solvent recovery and process scalability remain to be addressed.

The second research direction, presented in Chapter 3, explores a molecular-level solution through the incorporation of cleavable DCN into APO. These crosslinking networks anchor polymeric segments to suppress aggregation while enabling controlled cleavage under specific stimuli, allowing for polymer fractions to be recovered. This approach aims to optimize crosslink

density to balance adhesion strength, mechanical performance, and recyclability. By characterizing the thermal, mechanical, and rheological properties of these systems, this work demonstrates how molecular design principles can be applied to create APOs that are high-performance during use and remain recyclable at end-of-life.

Together, these two strategies illustrate how sustainability can be advanced for specialty polymers using complementary approaches. By addressing the dual challenges of performance retention and recyclability, this thesis positions APOs as a model platform for next-generation specialty plastics that combine durability, functionality, and sustainability.

Chapter 2: Physical Recycling of APO-Containing Blend

2.1 Introduction

2.1.1 APO/EVA-40 Blend

The Schafer group has been developing polymer blends that incorporate APOs as functional additives to commodity polymers introducing adhesive and self-healing properties. This blending strategy represents a promising route to reduce the overall cost of APO-based materials while broadening their property profile for diverse applications such as coatings and adhesives. However, previous studies have demonstrated that APOs face significant challenges associated with the aggregation of amine functional groups as shown in Figure 2-1.³³ Thermodynamically, this phenomenon arises because supramolecular interactions shift toward the lowest-energy equilibrium state, favouring insoluble clustered arrangements over time. Non-covalent interactions like H-bonding drive the association of amines into clusters, resulting in diminished adhesion, reduced solubility, and compromised processability, which all lead to reduced lifetime of APOs.⁴³ These effects become particularly problematic at the EoL stage, where aggregation hinders the reprocessability and recyclability of APOs.

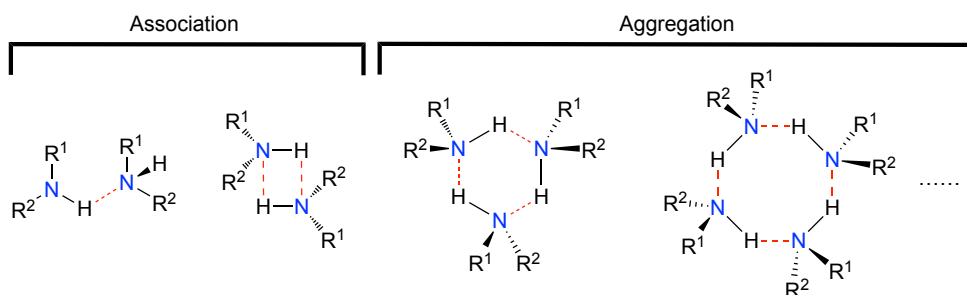
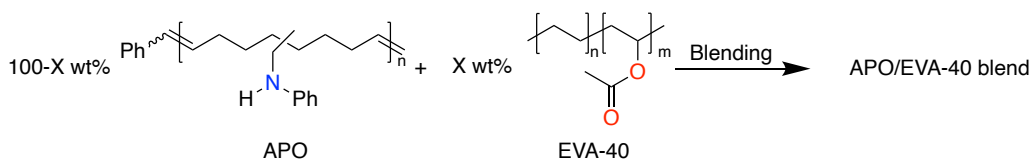


Figure 2-1. Illustration of dynamic H-bonding between 2° amines.

Several strategies have been explored to suppress or control this aggregation. For instance, Moradinik and co-workers studied the effect of quenching agents used during ROMP of APOs and observed that potassium tri(1-pyrazolyl) borohydride (KTP), as a bulkier quenching agent, hindered the formation of aggregates by sterically shielding the amine groups as a H-bond acceptor.⁴⁴ Similarly, studies of supramolecular polymers more broadly have shown that H-bonding networks can be disrupted by either steric hindrance or dilution effects, thereby maintaining functional properties over longer timescales.^{45,46} These insights suggest that matrix modification or additive introduction can play a critical role in extending APO lifetime.

Blending APOs with other polymers represents a particularly attractive strategy. Beyond lowering material costs, blending a secondary polymer into the APO matrix provides an effective way to modify the physical properties. The addition of a secondary polymer to “dilute” the first polymer disperses the amine functionalities in APO. This dilution effect, therefore, reduces the tendency to aggregate.⁴⁷ The secondary polymer can essentially acts as a plasticizer, which increases the free volume of the primary polymer chain, making the local segmental movement of the polymer chain more flexible. The enhanced flexibility of polymer chain reflects to the bulk properties, such as decrease in glass transition temperature (T_g) or increase in adhesion.^{47,48} This suppression of aggregation helps preserve the adhesive and self-healing properties of APOs at their EoL stage. Similar approaches have been widely reported for other functionalized polyolefins, where miscible blends prevent phase segregation and stabilize dynamic interactions.^{45,47}

One particularly promising blending candidate with APO is an ethylene–vinyl acetate copolymer containing 40 wt% vinyl acetate (EVA-40), illustrated in scheme 2-1. EVA-40 is composed of a non-polar polyethylene backbone with polar acetate side groups, which provides a similar polarity that suggests compatibility with APOs, as shown in Figure 2-2. Previous studies of EVA blends with other polymers have consistently demonstrated excellent miscibility and suppression of phase separation due to favourable enthalpic interactions between acetate groups and complementary functional moieties.^{49,50} In the case of APOs, blending APO with EVA-40 in weight ratios ranging from 70/30 to 80/20 resulted in full miscibility without detectable phase separation, a critical factor for maintaining uniform mechanical and adhesive performance. Thermal analyses of the 70/30 and 80/20 wt% APO/EVA-40 blend further support this conclusion, as the T_g of APO and EVA-40 converge into a single transition temperature, a classic indication of fully miscible polymer blend.



Scheme 2-1. Physical blending of APO and EVA-40.

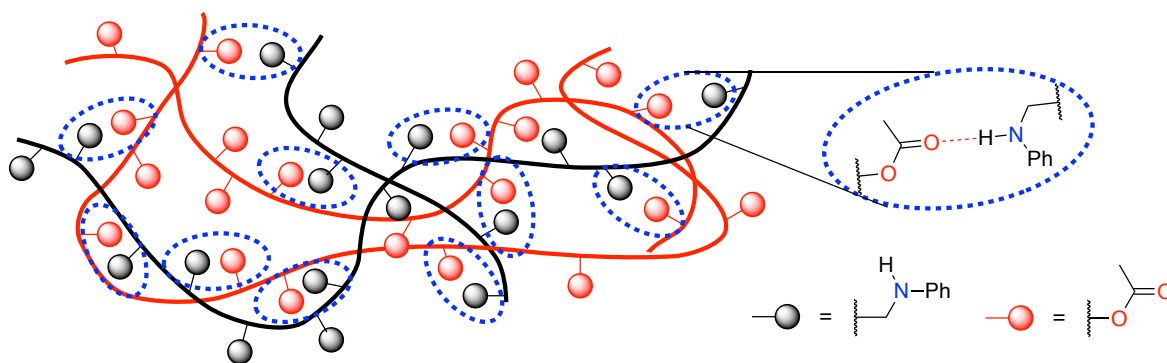


Figure 2-2. General illustration of backbone entanglement and H-bonding interactions between APO and EVA-40.

Beyond compatibility, EVA-40 also broadens the performance profile of APOs. While pristine APOs exhibit adhesion to substrates with different surface energies, such as aluminum, steel, and PTFE, the presence of EVA-40 eases the difficulty to handle plain APO. When coating substrates with plain APO, defects are often observed due to variable concentration, rate of casting, or non-consistent solvent evaporation. These cause the pure APO coating to have non-uniform thickness, poor wettability, and potential delamination. Precise control over the variables when dropcasting APO was important to achieve uniform films. The blending of EVA-40 with APO helps to create a more uniform coating with less defects, thereby expanding the applicability in diverse industrial contexts. EVA copolymers are already widely employed in adhesives, compatibilizers, and hot-melt formulations due to their favourable processing characteristics, mechanical flexibility, and interfacial adhesion properties.⁴⁹ This industrial precedent highlights EVA-40 as not only a compatible blending component for APO but also one with clear scalability and commercialization potential.

Despite these advantages, the development of new polymer-based materials with advanced functionalities must be accompanied by careful consideration of sustainability. While commodity polymers face limited recyclability, specialty polymers encounter even greater challenges, with virtually none being recovered at the EoL stage. Because specialty polymers are often integrated within complex products, their non-recyclable nature can compromise the recyclability of the entire material system. This exacerbates plastic waste accumulation and imposes a disproportionate environmental burden relative to their production volume.^{10,21-23} For APOs in particular, whose economic and functional values derive from their amine groups and associated

dynamic non-covalent properties, preserving these features during recycling is essential for extending their service lifetime and enabling circular material use. Establishing strategies to effectively separate APOs from the material system while retaining their functionality during recycling is, therefore, a prerequisite for both the sustainable deployment and the eventual large-scale adoption of APO-based blends.

Among the common recycling methods, including mechanical and WtE incineration approaches, solvent extraction has emerged as an industrially viable technique, offering several advantages over conventional mechanical recycling, particularly for multi-polymer blends or multilayer plastics.^{19,20} By leveraging solubility differences of the polymers, solvent extraction enables selective separation of targeted polymers under mild conditions. Therefore, the recovered recyclates are typically purer than those obtained from the mechanical recycling method. In contrast, recyclates from mechanical recycling suffer from detrimental thermomechanical degradation during the remelting step. However, solvent extraction avoids high temperatures, allowing recyclates to retain properties close to their pristine state. Furthermore, undesirable additives or plasticizers can be removed during the extraction process which the mechanical recycling process cannot effectively achieve. Overall, solvent extraction is considered a more favourable approach than conventional mechanical recycling when dealing with mixed polymeric systems.

2.1.2 Scope of Chapter

In this chapter, we investigate a physical recycling strategy for APO/EVA-40 blends as a representative model system, focusing on selective dissolution and precipitation. This system was chosen because EVA-40 is a common additive and highly miscible with APOs, making it an ideal platform to examine the challenges and opportunities of recovering specialty polymers from commodity blends. Selective dissolution has previously been applied to multilayer packaging films and other mixed polymer systems, where one component can be preferentially solubilized while the other remains in the solid phase.¹⁸ Unlike conventional WtE or mechanical recycling, which often lead to chain scission or loss of functionality, selective dissolution offers a pathway to preserve the dynamic non-covalent properties of APOs while maintaining the integrity of EVA-40.

To evaluate this approach, various solvents were tested to either dissolve APOs or EVA-40 while leaving the other insoluble, enabling the isolation of APO fractions with minimal structural or functional degradation. The extent of material preservation in both recovered APO and EVA-40 was assessed through changes in molecular weight, dispersity, thermal properties, morphology, and rheology compared with their pristine ones. Additionally, the efficiency of the solvent extraction process was quantified and optimized. Beyond recovery rate, the retention of adhesion and self-healing behaviours of APO was also examined qualitatively as an indicator of how effective the recycling process is.

By establishing this methodology, this chapter provides a process-level solution for recycling high-value APO-containing blends while preserving their unique material characteristics. Importantly, the process optimized here is not limited to APO/EVA-40 but is broadly transferable to other specialty/commodity polymer systems. In doing so, this work bridges the gap between advanced polymer innovation and sustainable end-of-life strategies, contributing to the design of future circular material systems.

2.2 Results and Discussion

2.2.1 Solvent Screening

To facilitate recyclability of an APO/EVA-40 blend, the separation of the two polymers by dissolving was envisioned. Therefore, a systematic solvent screening was conducted to identify solvents capable of selectively dissolving either APO or EVA-40 while leaving the other one undissolved and intact.

Polymer solubility is governed by the interplay between polymer architecture and solvent characteristics. Both APO and EVA-40 are expected to display intermediate polarity, owing to their non-polar hydrocarbon backbones combined with polar functional side groups. Effective solvents were therefore hypothesized to have intermediate polarity, balancing interactions with polar moieties while accommodating the hydrophobic backbone. To conduct systematic solvent screening, dissolution tests were first performed on pure APO and pure EVA-40.

To rationalize solvent behaviour, three complementary parameters were considered: polarity indices, dipole moment (μ), and dielectric constant (ϵ).⁵² Two polarity indices are commonly used in the chemistry community. The first is based on the Reichardt scale (E_T^N), in which solvatochromic dyes are used to determine relative solvent polarity, normalized from 0.0 for tetramethylsilane to 1.0 for water.⁵³ The second is the Snyder scale, which extends beyond simple “like-dissolves-like” considerations by incorporating solvent selectivity. This index accounts for functional interactions such as proton donation, proton acceptance, and dipole-dipole interactions, all of which strongly influence solute-solvent compatibility.⁵⁴ Snyder’s approach quantifies these parameters using standard solutes (ethanol for proton donor, dioxane for proton acceptor, and nitromethane for dipole interactor) in the tested solvent to determine the gas-liquid partition coefficients, which are expressed as X_e , X_d , and X_n , respectively. The combined effect defines the empirical polarity index (P'), providing a more nuanced ranking of solvent polarity.⁵⁴

In addition to polarity indices, the dipole moment (μ) reflects the magnitude of charge separation within a solvent molecule, directly influencing dipole-dipole interactions with polar functional groups in polymers. The dielectric constant (ϵ) of solvents, on the other hand, is particularly important for polymers containing polar groups, as it describes the ability of a solvent to stabilize polar or H-bonding interactions by screening electrostatic forces at macroscopic scale.⁵⁴ Together, these three parameters provide a comprehensive framework for evaluating solvent-polymer compatibility, particularly in APO/EVA-40 systems, where selective dissolution depends on a delicate balance between solvating one while excluding the other one.

Table 2-1. Solubility of APO and EVA-40 in common organic solvents listed based on ascending Reichardt polarity index at 25 °C for 24 hours.

	Reichardt Polarity index (E_T^N)	Snyder Polarity index (P')	$X_e; X_d; X_n$	Dipole moment (μ)	Dielectric constant (ϵ)	APO solubility	EVA-40 solubility
Hexane	0.009	0.1	-	0.08 D	1.9	N.S.	N.S.
Toluene	0.099	2.4	0.25; 0.28; 0.47	0.31 D	2.4	S.	S.
THF	0.207	4.0	0.38; 0.20; 0.42	1.75 D	7.6	S.	S.S.
EtOAc	0.228	4.4	0.34; 0.23; 0.43	1.88 D	6.0	S.	S.
CHCl₃	0.259	4.1	0.25; 0.41; 0.33	1.15 D	4.8	S.	S.
DCM	0.309	3.1	0.29; 0.18; 0.53	1.14 D	8.9	S.	S.
NMP	0.355	6.7	0.40; 0.21; 0.39	4.09 D	32.2	S.	N.S.
Acetone	0.355	5.1	0.35; 0.23; 0.42	2.69 D	20.7	S.	S.
DMF	0.386	6.4	0.39; 0.21; 0.40	3.86 D	36.7	S.	S.
DMSO	0.444	7.2	0.39; 0.23; 0.39	4.10 D	46.7	S.	S.
MeCN	0.460	6.2	0.31; 0.27; 0.42	3.44 D	37.5	S.S.	N.S.
<i>n</i>-BuOH	0.586	3.9	0.59; 0.19; 0.25	1.75 D	7.8	N.S.	N.S.
MeOH	0.762	5.1	0.48; 0.22; 0.31	2.87 D	32.7	N.S.	N.S.

(S. = solubilized, S.S. = Slightly solubilized, and N.S. = Not solubilized)

The solvent screening results, summarized in Table 2-1, clearly highlight the importance of balancing polarity, dipole moment, and dielectric constant in achieving selective dissolution. Non-polar solvents such as hexane ($E_T^N = 0.009$, $\epsilon = 1.9$) were unable to dissolve either polymer due to low polarity. At the opposite extreme, highly polar protic solvents such as methanol ($E_T^N = 0.762$, $\epsilon = 32.7$) and *n*-butanol ($E_T^N = 0.60$, $\epsilon = 17.8$) also failed, as their strong H-bonding networks hindered interactions with the hydrophobic backbones.

Solvents with intermediate polarity, such as tetrahydrofuran (THF), ethyl acetate (EtOAc), chloroform (CHCl₃), and dichloromethane (DCM), with E_T^N ranging from 0.207 to 0.309 dissolved both APO and EVA-40, which offered no selectivity. Similarly, polar aprotic solvents such as acetone, dimethyl formamide (DMF), and dimethyl sulfoxide (DMSO), with E_T^N ranging from 0.355 to 0.444 and ϵ ranging from 20 to 47, also dissolved both polymers, reflecting their ability to stabilize APO's amine and EVA-40's acetate moieties. These solvents were therefore unsuitable for selective recovery.

Notably, *N*-methyl-2-pyrrolidone (NMP) uniquely dissolved APO while leaving EVA-40 as an insoluble, gel-like solid. NMP occupies an intermediate-to-high polarity regime, as its E_T^N is 0.35, slightly more polar than DCM ($E_T^N = 0.31$), but its ϵ is 32.2, which is close to DMF ($\epsilon = 36.7$).

Importantly, its Snyder selectivity parameters ($X_e = 0.40$, $X_d = 0.21$, $X_n = 0.39$) reveal a balanced strength of H-bond accepting and dipole-dipole interaction without excessive proton donation. This makes NMP well-suited to solvate APO's amine functionalities, which rely on H-bonding, remaining less compatible with EVA-40's acetate groups. In contrast, solvents such as DMF and DMSO, with ϵ greater than 36, stabilize both amines and acetates, thereby eliminating selectivity.

However, acetonitrile (MeCN), despite being a highly polar aprotic solvent comparable to DMSO, only showed trace solubility to dissolve APO and failed to dissolve EVA-40. The reasons for this remain unclear and require further investigation.

2.2.2 Recycling Procedure Development and Optimization

With the selective solvent identified for APO, methanol (MeOH) was chosen as the anti-solvent based on prior studies.^{30,55} MeOH serves as an excellent anti-solvent for APO purification, as it readily induces APO precipitation while simultaneously solubilizing unreacted reagents and quenched catalyst. To evaluate this solvent/anti-solvent system, pure APO was first dissolved in NMP at room temperature and stirred for 24 hours to ensure complete dissolution. The resulting NMP solution was then added dropwisely into vigorously stirring cold MeOH (-8 °C), promoting rapid mixing and efficient precipitation.

Precipitation efficiency was found to be highly dependent on APO concentration in NMP. At concentrations below 150 mg/mL, APO failed to precipitate and instead formed a dark grey, opaque solution, likely due to emulsion formation. This suggests that the polymer remained too dilute to effectively phase separate during anti-solvent addition. By contrast, when the APO concentration exceeded 200 mg/mL, precipitation occurred readily, producing off-white solid clumps of APO initially, which turn dark purple over time. This concentration-dependent behaviour underscores the importance of solution saturation and polymer chain entanglement while in anti-solvent to enable efficient precipitation from the NMP/MeOH system.

Building on this optimization, the recycling procedure was applied to an 80/20 wt% APO/EVA-40 blend. The blend solution in THF was first drop-cast onto a PTFE mold and dried under ambient conditions overnight, followed by vacuum drying at 30 °C for 24 hours. The resulting film was cut into small pieces with approximate dimensions of 25 mm × 25 mm. Next, 250 mg of the APO/EVA-40 blend pieces were introduced into a round bottom flask (RBF) containing a stir bar. A minimal volume of NMP (1 mL) was added to generate a saturated APO solution as shown in Figure 2-3 (Step A). The EVA-40, which is insoluble in NMP, remained as a gel-like solid and was separated via filtration in Step B (Figure 2-3).

To recover APO, the solution containing APO was dispensed with 10 mL of cold MeOH to precipitate APO as an off-white clump of solid (Figure 2-3, Step C). The precipitate was collected by removing the solvent mixtures and vacuum drying (Step D). The remaining EVA-40 gel was further washed with NMP to remove residual APO (Step E). To reduce solvent

consumption, the resulting NMP/MeOH mixtures were separated using rotary evaporator and recycled back into Steps C and E. After repeating this process for two rounds, 72.6% of the initial APO was successfully recovered in high purity. The third precipitation cycle failed to isolate additional APO, likely due to the reduced concentration of APO remaining in the blend falling below the precipitation threshold. This indicates that while high recovery can be achieved, the process efficiency is constrained by the solubility limits of APO in NMP and the minimum concentration required for effective anti-solvent precipitation.

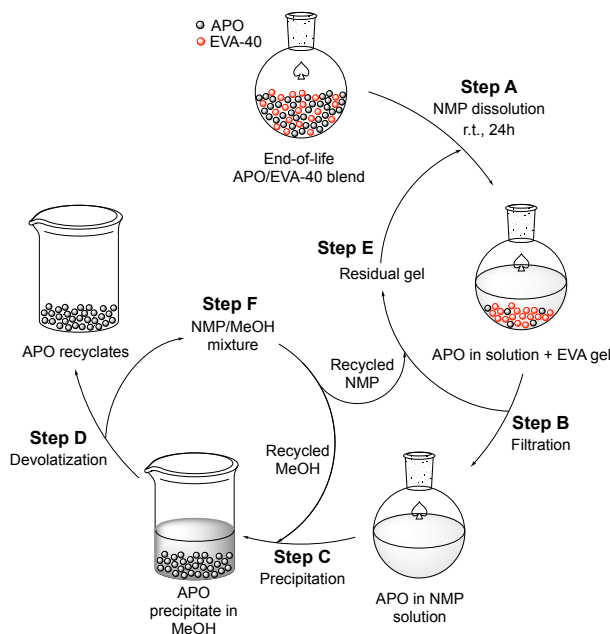


Figure 2-3. Schematic APO/EVA-40 solvent-based recycling steps using NMP and MeOH.

2.2.3 Characterization of Recovered Polymers

The recycled APO and EVA-40 were comprehensively characterized to validate the effectiveness of the developed recycling process. Structural integrity and purity were examined by ^1H NMR and FT-IR spectroscopy, and polymeric molecular weight and dispersity of recyclates were evaluated by GPC. Thermal properties were tested using DSC and TGA, and surface morphologies were investigated with AFM and SEM. In addition, rheological measurements were carried out to assess differences in viscoelastic behaviour between pristine and recycled samples. The self-healing and adhesive behaviour of recycled APO was evaluated qualitatively. Together, these complementary techniques provide a robust assessment of whether the recovered polymers retain properties comparable to their pristine counterparts.

2.2.3.1 Comparison of Pristine APO and Recycled APO

The ^1H NMR spectra of APO provided clear evidence for the effective purification of the recycled APO. The characteristic signals corresponding to EVA-40, such as the singlet at 2.04 ppm for the methyl group and the intense but broad peak at 1.25 ppm for the ethylene backbone, were reduced to nearly negligible levels in the recycled sample, as depicted in Figure 2-4. In contrast, all indicative resonances corresponding to APO were preserved with no observable changes in chemical shift or splitting pattern. These results confirm that the selective dissolution-precipitation process successfully separated APO from EVA-40.

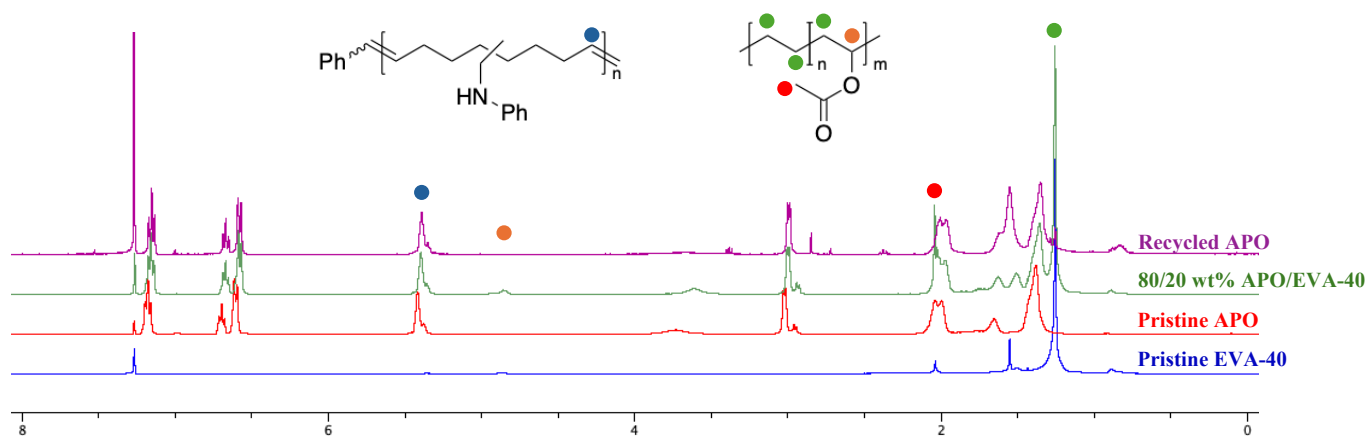


Figure 2-4. ^1H NMR spectra (400 MHz, 298 K, CDCl_3) of pristine APO, recycled APO, APO/EVA-40 blend, and pristine EVA-40 (Spectra intensity adjusted based on signal at 5.41 ppm).

Despite the good separation efficiency of APO and EVA-40, a trace amount of NMP was consistently detected in the recycled APO, even after multiple purification cycles with MeOH and 72 hours of drying under vacuum. This observation suggests that a fraction of solvent becomes trapped within the polymer matrix.

The molecular weight (M_w) and dispersity ($\mathcal{D}$) of the recycled APO, measured by GPC, were found to be comparable to those of the pristine APO as listed Table 2-2 and shown in Figure S2-1. Only a slight increase in M_w of around 5 Kg/mol was observed, while $\mathcal{D}$ remained nearly identical between the two samples. These results confirm that the recycling protocol does not induce significant chain scission or degradation, and the chemical integrity of the polymer backbone is well preserved throughout the process.

Interestingly, the glass transition temperature (T_g) of the recycled APO shifted upward from -23 °C to -11 °C, as listed in Table 2-2 and shown in Figure S2-2. This increase, along with the slight rise in M_w , can be attributed to enhanced intermolecular H-bonding interactions and clustering of amine groups in the absence of EVA-40 or aging due to the recycling process. It is noteworthy that the recycled APO sample prepared for DSC measurement was vacuum-dried for 72 hours prior to the measurement to minimize the amount of residual NMP, which might also contribute to the rise in T_g as non-covalent interactions reach thermodynamically stable equilibrium. In the pristine blend, EVA-40 acted as a steric filler, mitigating such associations by interfering with direct H-bonding between amines. Once EVA-40 was removed, APO chains could engage more strongly in amine-amine interactions, leading to a decrease in free volume and chain mobility and a higher T_g .

Table 2-2. Comparison of molecular weight and thermal properties of pristine and recycled APO based on GPC, DSC, and TGA measurements.

	M_w (Kg/mol)	dispersity ($\bar{D}$)	T_g (°C)	$T_{d,5\%}$ (°C)	$T_{d,50\%}$ (°C)
Pristine APO	75.2	1.6	-23.2	257.0	434.0
Recycled APO	80.8	1.3	-11.3	146.0	439.0

The thermal stability of APO before and after recycling, shown in Figure 2-5, further highlighted differences between pristine and recovered samples. The recycled APO exhibited a significantly lower $T_{d,5\%}$, with a decrease of nearly 110 °C relative to the pristine one. This substantial shift is most likely attributable to residual NMP entrapped within the polymer matrix, which reduces the onset temperature of weight loss as its boiling point is 202 °C. In contrast, the $T_{d,50\%}$ values of both pristine and recycled APO remained nearly identical, indicating that the thermal decomposition pathway of the polymer backbone was unaffected and that no significant polymer degradation occurred during the recycling process.

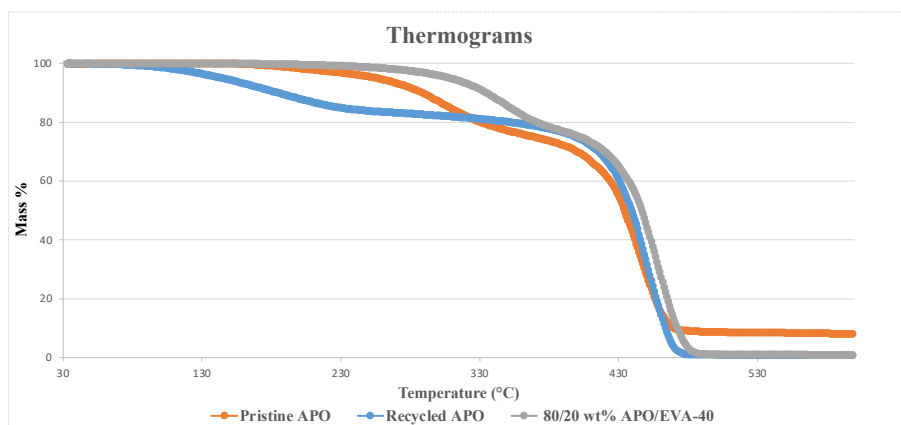


Figure 2-5. Thermograms of pristine and recycled APO, and 80/20 wt% APO/EVA-40 blend.

The surface topographies of pristine and recycled APOs were evaluated and compared with that of the APO/EVA-40 blend using SEM. As shown in Figure 2-6, both pristine and recycled APO exhibited smooth and uniform surface topography, with no evidence of microphase separation or aggregation, indicating good homogeneity after the recycling process. In contrast, the SEM image of the APO/EVA-40 blend revealed distinct fibre-like features approximately 2.5 μm in width, homogeneously distributed throughout the background matrix. These features are likely attributable to the EVA-40 component, consistent with the blend weight ratio, suggesting partial microphase separation between the APO and EVA-40 phases.

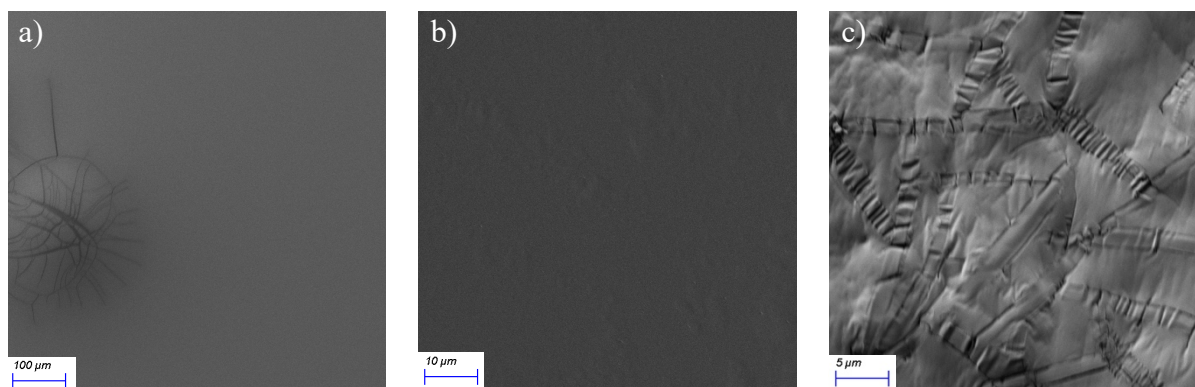


Figure 2-6. SEM images of a) pristine APO, b) recycled APO, and c) 80/20 wt% APO/EVA-40.

AFM imaging further supports these findings as shown in Figures 2-7, revealing fibre-like domains of similar width dispersed across the APO matrix. In the AFM topography images, these domains appear as elevated white regions, corresponding to areas of greater stiffness or Young's modulus, while the surrounding APO phase appears in orange, indicating a softer surface. The contrast arises from variations in viscoelastic properties between the EVA-40 and APO phases, as the ethylene segments of EVA-40 impart higher crystallinity and mechanical rigidity compared to the amorphous APO matrix.⁵⁶ The uniform distribution of EVA-40 domains suggests good interfacial compatibility and effective dispersion, which likely contributes to improved mechanical performance and adhesion of the blend compared to pristine APO. AFM topography measurements of pristine and recycled APO, however, failed to resolve distinct features due to the softness of these materials. This observation further confirms that the incorporation of EVA-40 increases the overall stiffness of the polymer blend, consistent with the topographical evidence obtained from SEM and AFM analyses.

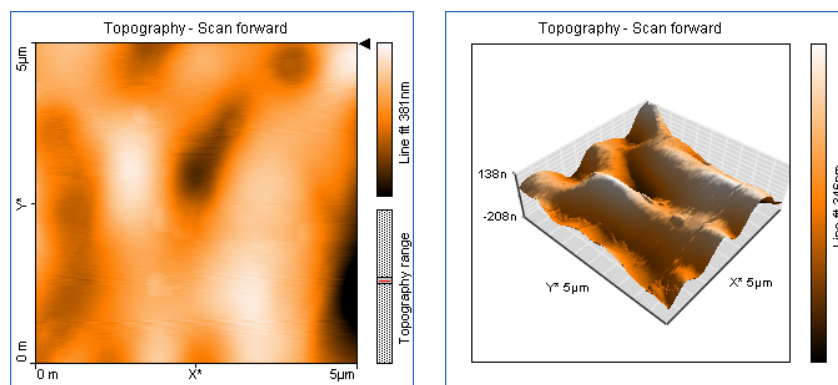


Figure 2-7. AFM image of 80/20 wt% APO/EVA-40.

Rheological measurements were conducted to further evaluate the viscoelastic behaviour of pristine and recycled APOs. Figure 2-8 presents the master curves constructed from frequency sweep experiments ranging from 10^{-1} to 10^2 rad/s at a reference temperature of 30 °C. The recycled APO exhibited a similar response in the high-frequency region compared to the pristine one, indicating the similar segmental dynamics before and after the recycling process. However, differences showed in the rubbery (10^0 to 10^4 rad/s) and terminal region (below 10^0 rad/s), where significant increases in both storage (G') and loss (G'') moduli were observed. The pristine sample exhibited a typical liquid-like behaviour in the terminal region, evidenced by the presence of crossover of G' and G'' at 10^0 rad/s and the power law scaling of $G' \sim \omega^2$, $G'' \sim \omega^1$ below 10^{-2} rad/s. However, the G' surpassed G'' across the entire frequency range, and a secondary plateau emerged at low frequencies, suggesting the formation of a long-range network structure which restricted polymer chain relaxations. This behaviour implied enhanced intermolecular association, likely due to H-bonding between amine groups, which effectively transformed the recycled APO into a more elastic and rubbery network. Similar rheological behaviour has been reported in a previous study where APO samples were intentionally aged at elevated temperatures, further supporting the hypothesis that stronger intermolecular interactions contribute to the observed increase in elasticity and the disappearance of terminal relaxation in the recycled material.³³

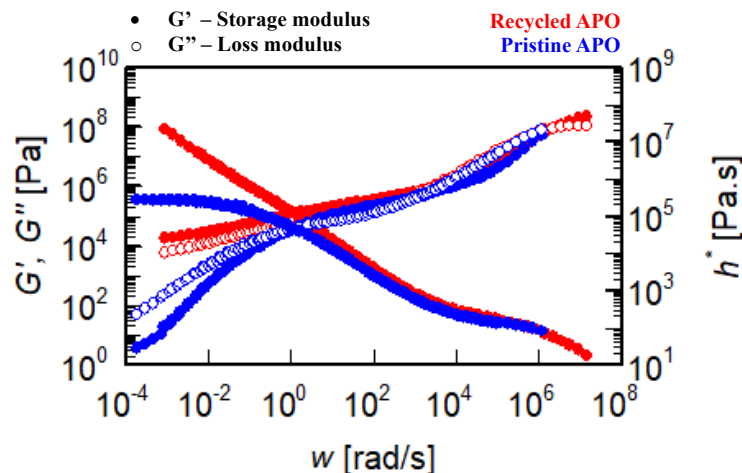


Figure 2-8. Master curves of pristine and recycled APO from frequency sweep experiments at $T_{ref} = 30\text{ }^{\circ}\text{C}$.

Taken together, these results indicate that the recycling process effectively retrieves APO back from the blend, but with moderate aging. The comparable molecular weight and preserved thermal stability at $T_{d,50\%}$ confirm that the polymer backbone remains intact throughout the process. However, the noticeable decrease in $T_{d,5\%}$ highlights a limitation in solvent removal, suggesting that residual NMP persists in the recycled material. This finding points to the need for improved drying or purification strategies, such as extended vacuum treatment or better solvent exchange to fully eliminate residual solvent and restore the thermal performance to that of pristine APO. Moreover, the enhanced aging and increased elastic behaviour observed after recycling is likely due to the absence of EVA-40, which originally acted as a H-bond disrupting component in the blend. A mild reduction of adhesion of the recycled APO was observed, but the self-healing behaviour was preserved. This indicates that while the recovery of APO is successful, its physical properties are altered and cannot be fully restored.

2.2.3.2 Comparison of Pristine EVA-40 and recycled EVA-40

In addition to evaluating the recycled APO, the insoluble gel of EVA-40 was also analyzed and compared with its pristine counterpart, despite EVA-40 being a lower-value commodity polymer. This comparison provides insight into the integrity and purity of the EVA-40 fraction recovered from the APO/EVA-40 blend after the recycling process. To have a comprehensive evaluation of the EVA-40, several techniques were applied, including FT-IR, NMR, GPC, DSC, TGA, and SEM.

FT-IR spectroscopy was used to first evaluate the recycled EVA-40. Figure 2-9 presents a comparison of the spectra for pristine APO, the 80/20 wt% APO/EVA-40 blend, and the recycled EVA-40. A broad stretching vibration signal corresponding to the N–H bond of secondary amines appears near 3400 cm^{-1} across all spectra. However, its transmittance intensity is substantially weaker in the recycled EVA-40, consistent with only trace amounts of APO remaining. Similar trends are observed for the C–H stretching vibrations of alkene protons at 3024 and 3000 cm^{-1} , which are also less intense in the recycled EVA-40 spectrum. Additionally, characteristic APO bands at 1600 and 1505 cm^{-1} , corresponding to C=C stretching and N–H bending, respectively, also display reduced transmittance intensity relative to both pristine APO and the blend. These results collectively confirm that only minor APO residues are present in the recycled EVA-40 fraction, supporting the effectiveness of the physical separation and recovery process in isolating EVA-40 from the blend.

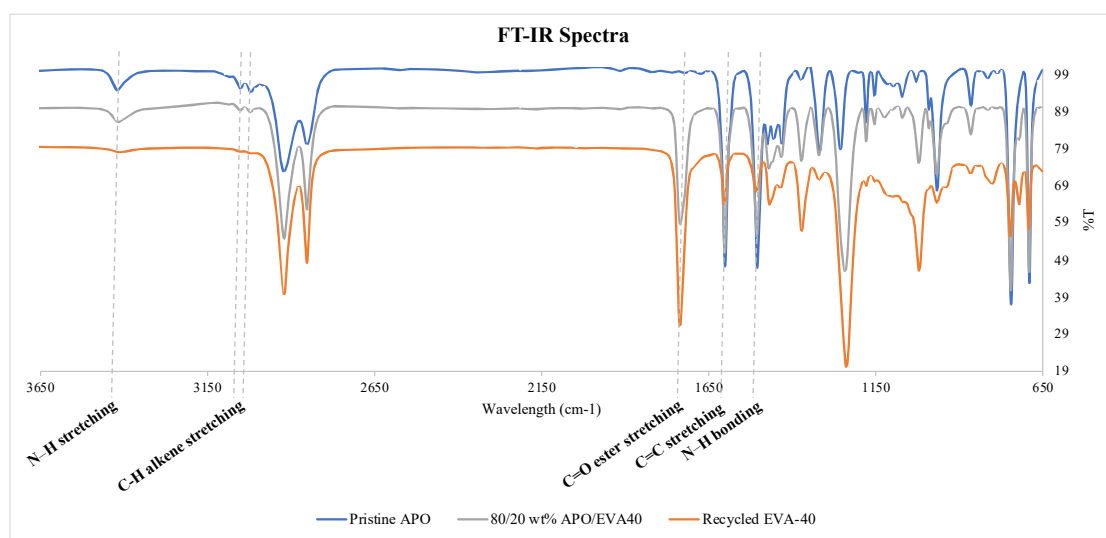


Figure 2-9. FT-IR spectra of pristine APO, 80/20 wt% APO/EVA-40 blend, and recycled EVA-40.

^1H NMR analysis was used to further validate the conclusion made from the FT-IR analysis. The ^1H NMR spectrum of recycled EVA-40, shown in Figure 2-10, was compared with those of pristine EVA-40 and the 80/20 wt% APO/EVA-40 blend. The spectrum of the recycled EVA-40 exhibits trace signals attributable to residual APO. Specifically, a triplet at 5.39 ppm corresponds to the internal alkene protons from the APO backbone, while aromatic resonances between 6.5 and 7.2 ppm indicate the presence of phenyl groups from APO. These minor APO signals likely arise from the saturation of APO in NMP during the dissolution step.

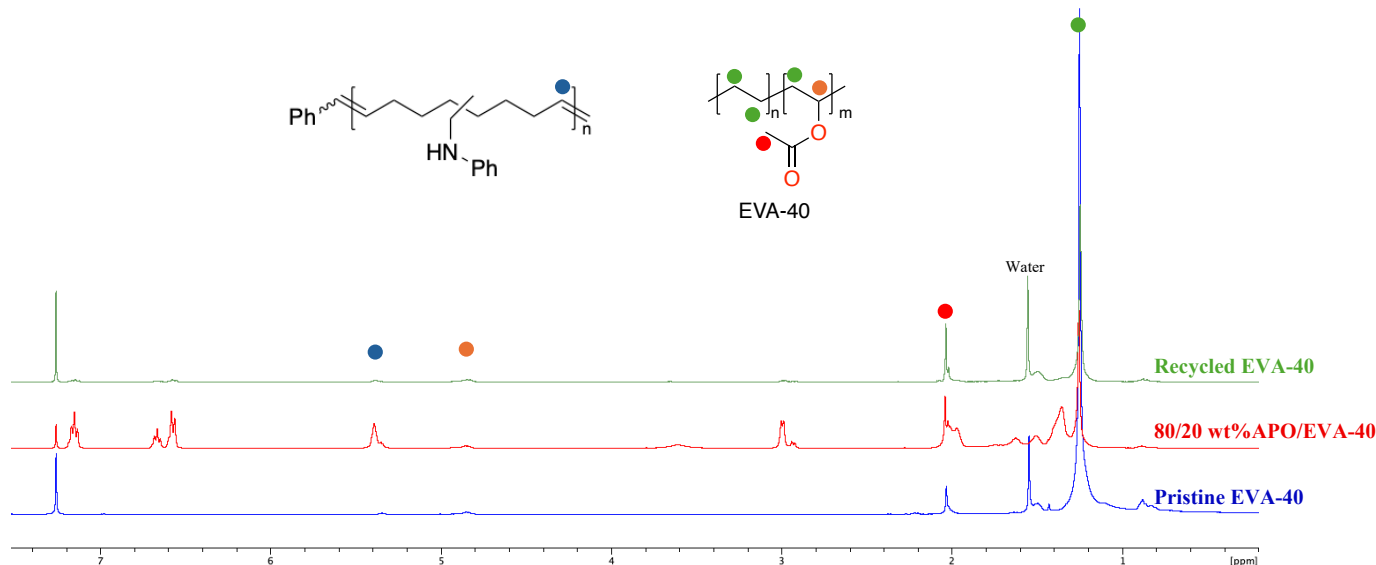


Figure 2-10. ^1H NMR spectra (400 MHz, 298 K, CDCl_3) of pristine EVA-40, recycled EVA-40, and 80/20 wt% APO/EVA-40 blend (Spectra intensity adjusted based on signal at 4.86 ppm).

Nevertheless, the dominant features of the recycled EVA-40 spectrum match well to pristine EVA-40, including the broad singlet at 1.26 ppm assigned to the methylene protons of the polyethylene backbone, the singlet at 2.04 ppm corresponding to the acetoxy methyl group, and a small broad multiplet, ranging from 4.80 to 4.95 ppm, attributed to the $-\text{CH}-$ group adjacent to the acetate functionality. The intense EVA-40 peaks indicate that EVA-40 constitutes the majority of the recovered insoluble gel-like material. This result demonstrates the overall effectiveness of the physical separation and recovery process in isolating EVA-40 from the APO/EVA-40 blend while maintaining its structural integrity.

To evaluate the structural integrity, GPC was used to examine any change in the polymer molecular weight. THF serves as the eluent for GPC measurement, which is completed at room temperature. However, pristine EVA-40 was not subjected to GPC analysis because it is only readily soluble in THF at 60 $^{\circ}\text{C}$, with slight solubility at room temperature. In contrast, the molecular weight (M_w) and dispersity ($\bar{D}$) of the recycled EVA-40 were successfully determined by GPC, yielding an M_w of 344.1 Kg/mol and a $\bar{D}$ of 1.15, as summarized in Table 2-3 and shown in Figure S2-3. The recycled EVA-40 exhibited better solubility in THF at room temperature, which can be attributed to the presence of trace APO residues. These residual APO components likely interfere with the crystalline packing of EVA-40 chains, reducing the overall degree of crystallinity and making the recycle more amorphous and soluble.

The 80/20 wt% APO/EVA-40 blend was also analyzed by GPC to provide a reference for correlating molecular weight data with the EVA-40 fraction. The GPC retention peak of the blend displayed a single broad elution peak rather than distinct bimodal peaks, indicating that the molecular weights of APO and EVA-40 overlap within the detectable range (Figure S2-4). Although this prevents deconvolution of the two polymers' molecular weights, the data implies a general idea regarding the molecular weight of EVA-40 when the APO molecular weight is known to be 75.2 Kg/mol in Table 2-2. Based on GPC measurements, the blend exhibits an M_w of 137.5 Kg/mol with a $\bar{D}$ of 1.12, which is considerably lower than that of the recycled EVA-40. The higher M_w of the recycled EVA-40 may result from mild physical aggregation occurring during recovery.

These intermolecular interactions, such as aggregation, restrict chain mobility and decrease the polymeric free volume. As the free volume decreases, the T_g of the polymer increases due to their inverse relationship. To further confirm the effect of physical aggregation of the recycled EVA-40, DSC was used to measure its thermal properties and compared with its pristine counterpart. As shown in Table 2-3 and Figure S2-5, the T_g of EVA-40 increased from -31.0 °C to -25.0 °C, confirming the aggregation.

Noticeably, the T_m stayed identical at 28 °C for both pristine and recycled EVA-40. However, the enthalpy required to melt the pristine EVA-40 is measured to be 10.3 J/g, but the recycled EVA-40 only required 3.0 J/g to melt. The decrease in enthalpy indicates that the degree of crystallinity of EVA-40 decreased, potentially due to the presence of residual APO in trace amount. Thus, while the residual APO reduces the intrinsic crystallinity and improves solubility, post-recycling aggregation of EVA-40 chains contributes to the observed increases in M_w and T_g .

Table 2-3. Comparison of molecular weight and thermal properties of pristine and recycled EVA-40, and APO/EVA-40 blend based on GPC, DSC, and TGA measurements.

	M_w (Kg/mol)	dispersity ($\bar{D}$)	T_g (°C)	T_m (°C)	$T_{d,5\%}$ (°C)	$T_{d,50\%}$ (°C)
Recycled EVA-40	344.1	1.15	-25.0	28.0	300	446
Pristine EVA-40	N/A	N/A	-31.0	28.1	355	462
APO/EVA-40 blend	137.5	1.12	-24.7	29.3	310	446

The thermal stability of EVA-40 also exhibited notable differences between the pristine and recovered samples as listed in Table 2-3 and shown in Figure 2-11. The recycled EVA-40

displayed a $T_{d,5\%}$ of 300 °C, approximately 55 °C lower than that of the pristine one, and a $T_{d,50\%}$ that was 24 °C lower. These downward shifts are most likely due to residual APO in the recycled EVA-40 fraction, consistent with the incomplete APO recovery yield mentioned in the previous section. The presence of residual APO, which has lower $T_{d,5\%}$ and $T_{d,50\%}$, triggers early-stage thermal degradation as it disrupts the crystalline domains of EVA-40, thereby lowering its decomposition temperatures. Supporting this interpretation, the thermogram of the recycled EVA-40 closely resembles that of the original APO/EVA-40 blend, where the incorporation of APO reduced both $T_{d,5\%}$ and $T_{d,50\%}$ in the same manner. This observation suggests that despite successful APO recycling, minimal APO residue remains sufficient to alter the thermal stability profile of the recycled EVA-40.

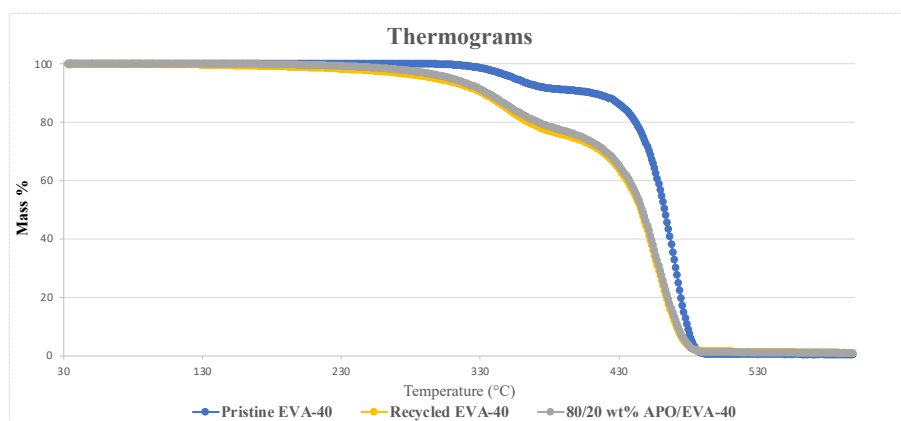


Figure 2-11. Thermograms of pristine and recycled EVA-40, and the blend.

In addition to the thermal stability differences, the residual APO and the recycling process may also influence the surface topography of EVA-40. The amine groups within APO can interact with acetate moieties in EVA-40 through non-covalent interactions, potentially affecting the polymer chain packing and crystallinity. These effects can manifest in observable changes in surface topography, which are further examined through SEM imaging.

The surface topographies of pristine and recycled EVA-40 were compared to assess potential microstructural changes resulting from the recycling process. As shown in Figure 2-12 a) and b), the pristine EVA-40 exhibits a relatively rough and irregular surface topography, whereas the recycled EVA-40 displays a more regular and patterned surface aggregation, suggesting polymer chain rearrangement and partial recrystallization during solvent exposure and subsequent drying.

This transition from a rough to a more patterned topography may be attributed to the influence of residual APO and the processing history. The presence of residual APO likely increases the amorphous character of EVA-40, promoting localized aggregation into numerous small crystalline domains. These topographical changes indicate that, while the bulk chemical structure of EVA-40 remains intact, alterations in chain packing and crystallinity at microscopic level occurred during recycling, affecting its thermo-physical properties like T_g and T_m .

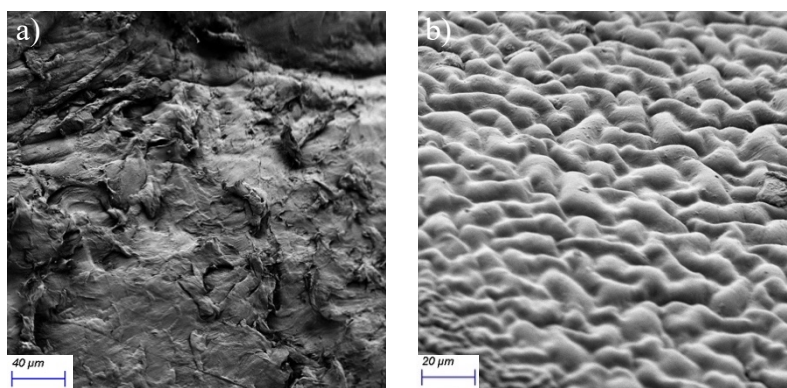


Figure 2-12. SEM images of a) pristine EVA-40 and b) recycled EVA-40.

2.2.4 Discussion

The sustainability of the recycling process was further enhanced through efficient solvent recovery and reuse. Both NMP and methanol were successfully recovered at the end of each recycling cycle using a rotary evaporator. The recovered solvents were subsequently reused across multiple cycles without noticeable loss in performance, confirming their reusability and demonstrating the robustness and circularity of the process.

This approach shows strong potential for industrial scalability due to its operational simplicity, low energy demand, and adaptability to various polymer systems. The use of green methanol, produced via biomass gasification, could further minimize the environmental footprint of the process. Although NMP is not classified as a green solvent owing to its reproductive toxicity, it remains widely utilized in industries such as textiles, lithium-ion battery manufacturing, and polymer membrane production because of its low volatility, high thermal stability, and excellent solvating power.⁵⁷ In these industrial contexts and in the recycling method developed here, NMP is thoroughly removed or recovered before the final product stage.

Nevertheless, from a long-term sustainability perspective, identifying green alternatives to NMP is desirable. Promising substitutions such as dihydrolevoglucosenone (Cyrene) and γ -valerolactone (GVL) exhibit comparable polarity profiles, lower toxicity, and biodegradability.⁵⁸ However, large-scale production of these solvents remains at an early stage, limiting their immediate applicability. Hence, NMP remains the most practical solvent choice for achieving effective APO recycling.

2.3 Conclusion

Overall, the results of this study highlight physical recycling as an effective and practical approach for recovering specialty polymer blends such as APO/EVA-40, which are typically difficult to recycle with preserved functional properties using conventional EoL treatments. The NMP/MeOH solvent/anti-solvent system enabled selective dissolution and recovery of the polymer components under mild conditions, allowing efficient phase separation without significantly compromising material quality. Future work should focus on optimizing NMP removal, improving recycled EVA-40 purification, and identifying sustainable alternatives to NMP to further advance the feasibility of this process for real-world implementation.

2.4 Experimental

2.4.1 Materials

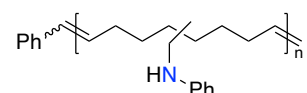
General Unless otherwise indicated, all air- and/or water-sensitive reactions were performed under inert atmosphere using a Schlenk double manifold equipped with N₂ and high vacuum (10⁻³ mbar) or a glovebox filled with N₂. All glassware used was heated above 160 °C in an oven overnight prior to use. Reactions were performed in threaded 20 mL scintillation vials equipped with a PTFE-coated magnetic stir bar and a PTFE-lined PP cap.

Reagents All reagents were purchased from commercial sources unless otherwise specified. Cyclooctadiene (Aldrich) and N-methyl aniline (NMA) (Aldrich) were dried over CaH₂ overnight, purified by vacuum distillation into PTFE-sealed Straus flasks, and subsequently handled using standard Schlenk techniques. The [Ta] catalyst for ACC-NMA synthesis was prepared according to literature procedures.³⁰ Grubbs Catalyst™ 2nd Generation (G2) (Sigma-Aldrich) was obtained commercially, stored under N₂ atmosphere at -8 °C in a glovebox, and used without further

purification. Amine-containing cyclooctene *N*-methyl aniline (ACC-NMA) was either synthesized according to literature procedures³⁰, or was provided by GreenCentre Canada. ACC-NMA was vacuum-dried overnight on a Schlenk line prior to use. Ethylene vinyl acetate-40 (EVA-40) (Sigma-Aldrich), ethyl vinyl ether (EVE) (Aldrich), and *cis*-4-octene (Sigma-Aldrich) were used as received. Solvents for air- and moisture-sensitive reactions were dried over CaH₂ overnight, purified by vacuum distillation into PTFE-sealed Straus flasks over activated 4 Å molecular sieves for 48 hours, and degassed by sparging with N₂ for 30 minutes. Solvents for non-sensitive reactions were used as received.

Synthesis and Characterization

Poly-ACC-NMA (APO). Based on literature reported procedure, vacuum-dried ACC-NMA monomer was dissolved in DCM and charged with a magnetic stir bar.³⁰ To the solution, a DCM solution of G2 catalyst



was added, followed by a DCM solution of *cis*-4-octene, in sequential order under stirring. The reaction vessel was capped with a rubber septum, vented with a needle, and left stirring at room temperature for 20 hours. During the reaction, the colour of the mixture gradually changed from transparent burgundy to opaque dark purple, accompanied by an increase in viscosity. The reaction was quenched by adding an excess amount of EVE and allowed to stir for an additional 24 hours. The resulting polymer was isolated by precipitation, in which the reaction mixture was slowly added to the center of a vortex of cold methanol under stirring. The mother liquid was decanted, and the precipitated polymer was collected and vacuum-dried to remove residual solvent prior to further processing and characterization. ¹H NMR (400 MHz, CDCl₃): δ 7.17 (t, 2H, 2 × ArH), δ 6.69 (t, 1H, ArH), δ 6.60 (d, 2H, 2 × ArH), δ 5.41 (br s, 2H, RHC=CHR), δ 3.73 (br s, 1H, NH), δ 3.01 (d, 2H, CH₂), δ 2.00 (m, 4H, CH₂), δ 1.65 (br m, 1H, CH), δ 1.37 (br m, 6H, CH₂). IR (solid, cm⁻¹, intensity): 3419 brr w, 3024 w, 3019 w, 2908 m, 2844 m, 1601 s, 1505 s, 966 s.

2.4.2 Instrumentation

Nuclear magnetic resonance (NMR) spectroscopy NMR spectra were recorded on a Bruker Avance 300 MHz and 400 MHz spectrometer at 25 °C. Chemical shifts are reported in parts per million (ppm) versus residual proteo-solvents in deuterated solvents as follows: δ 1.94 CD₃CN, δ

3.58 THF- d_8 , δ 7.26 $CDCl_3$. Abbreviations for NMR assignments are as follows: s = singlet; d = doublet; t = triplet; q = quartet; m = multiplet; br = broad.

Fourier-Transform Infrared (FT-IR) Spectroscopy FT-IR measurements were recorded at room temperature on a Perkin Elmer FT-IR Single-Range Spectrometer with attenuated total reflectance (ATR) for direct measurement on polymeric materials. Measurements were collected from 4000-600 cm^{-1} . Bands are reported in wavenumbers (cm^{-1}) and assigned with the abbreviations s = strong, m = medium, w = weak, br = broad.

Gel Permeation Chromatography (GPC) Polymer M_n , M_w and dispersity ($\bar{D}$) were obtained using triple detection gel permeation chromatography coupled with laser light scattering (GPC-LLS) using a Malvern OMNISEC GPC instrument equipped with an Agilent 1200 series isocratic pump and autosampler, Viscotek T3000 and T6000 GPC columns packed with poly(styrene-*co*-divinylbenzene) particles, three Phenogel 5 μm narrow bore columns (4.6×300 mm, 500, 1000 and 10000 Å pore size), a Wyatt Optilab differential refractometer, Wyatt miniDAWN TREOS (multiangle laser light scattering detector) and a Wyatt ViscoStar viscometer. The flow rate was 0.5 mL/min, and the column temperature was 40 °C. Polymer samples were dissolved in THF (ca. 1.0 mg/mL) and filtered through 0.2 μm PTFE syringe filter prior to measurement. The measurements were carried out at a laser wavelength of 690 nm, at 25 °C. The data was analyzed using the Astra[®] processing program¹⁵⁷ provided by Wyatt Technology Corp. The dn/dc (differential refractive index increment) values of the polymers studied were calculated assuming a 100% mass recovery method from Malvern OMNISEC software version 11.4.

Differential Scanning Calorimetry (DSC) Thermograms were recorded on a TA Instruments DSC Q2000 equipped with a TA Instruments Refrigerated Cooling System 90, and calibrated using indium before analyte measurements. Analyses were performed with a constant flow of N_2 gas at a rate of 45 mL/min with sample masses of approximately 5-10 mg in an aluminum pan sealed with aluminum lid. A heating/cooling rate of 5 or 10 °C/min was used for each run in the range of -70 °C to 200 °C. All analyses were conducted in duplicate after the completion of one heating/cooling cycle to remove thermal history. The glass transition temperatures (T_g) and/or melting temperature (T_m) were determined from the second heating ramp.

Thermogravimetric Analysis (TGA) Thermal stability of the polymers was determined by using Netzsch TGA209 F1 Libra® thermogravimetric analyzer. Samples were vacuum-dried overnight at room temperature using Schlenk line. Measurements were carried out with sample masses of approximately 2-5 mg in a alumina crucible under constant purging with high purity N₂ gas (99.5%) into the ceramic micro furnace at a flow rate of 50 mL/min at atmospheric pressure and room temperature. Weight loss as a function of time was continuously monitored in the temperature range from 25 to 600 °C at a heating rate of 10 °C/min.

Scanning Electron Microscope (SEM) Topographical analyses were recorded using Zeiss Crossbeam 350 FIB-SEM operating under vacuum (4.6×10^{-5} mbar). Images were acquired at acceleration voltage of 1.0 or 2.0 kV and magnification ranging from 100× to 3000×. Samples were sputter-coated with a 1 µm layer of Au/Pd under vacuum (2×10^{-5} mbar) using Leica EM ACE600 Coater to enhance conductivity.

Atomic Force Microscope (AFM) Topographical analyses were recorded using Nanosurf EasyScan 2 AFM operating in dynamic mode at room temperature and under atmospheric pressure. Aspire™ CT170R-10 single crystal silicon cantilever (170 kHz resonant frequency, 50 N/m spring constant, conical, 8 nm tip radius) were employed, and the actual resonance frequency was determined to be approximately 190 kHz by frequency sweep prior to imaging. The vibration amplitude was adjusted to 200 mV, and the setpoint amplitude was maintained at 50% of the free amplitude to minimize tip-sample interaction. Samples were prepared on VWR VistaVision™ microscope slide and mounted onto magnetic pucks with adhesive tape. After laser and photodetector alignment, the cantilever was engaged onto the surface under feedback control, and scans were acquired at typical sizes of 1–10 µm with resolutions of 256 pixels at scan rates of 0.5–1.0 Hz. Height and phase images were collected simultaneously, and raw data were processed by line-by-line correction using the Nanosurf software.

Rheological Measurements Polymer films were prepared for rheological testing. The precipitated polymer was solubilized in THF and drop casted onto PTFE sheet. Solvent was allowed to evaporate at room temperature at ambient pressure for 12 hours, followed by drying in a vacuum

oven (30 to 60 °C). Films could then be carefully removed from the sheet and used for analysis. Linear Visco-Elastic (LVE) measurements were performed using a stress-controlled rotational rheometer (Anton Paar, MCR302e) equipped with 8 mm parallel-plate and Peltier heating system for temperature control with accuracy of ± 0.1 °C. The frequency sweep tests were conducted over a wide range of angular frequency ($\omega = 0.1$ to 100 rad/s) at temperatures from -10 to 50 °C at 20 °C intervals and strain within the LVE region (γ depending on torque value at different temperatures). The time-Temperature Superposition (tTS) principle was applied to construct the master curves of storage (G') and loss (G'') moduli, and complex viscosity, $|\eta^*(\omega)|$.

Chapter 3. Design of Cleavable Crosslinked APO

3.1 Introduction

The physical recycling of APO has been shown to be an effective approach at the process-level to improve APO's sustainability. However, in order to achieve long-term circularity of polymer materials, sustainability principles need to be embedded into the molecular design stage, so that intrinsic reprocessability and recyclability can help address the current plastic waste crisis.

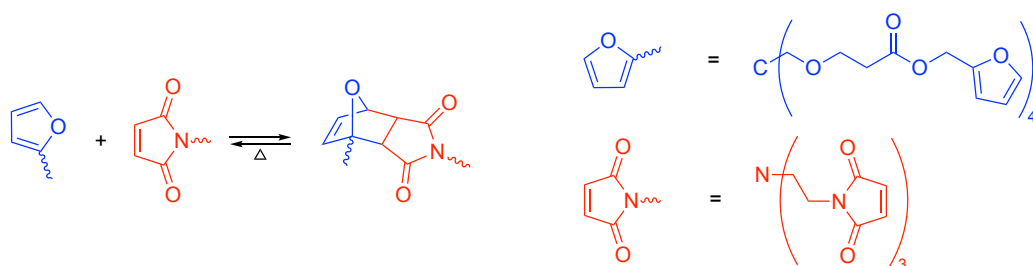
The introduction of dynamic covalent networks (DCNs) into polymeric systems has emerged as a powerful approach to combine the characteristics of thermosets and thermoplastics, offering the mechanical robustness of the former while retaining the reprocessability and recyclability of the latter, as illustrated in Figure 1-4. In this chapter, DCNs were used to transform the linear thermoplastic APO into a crosslinked but recyclable material. This molecular design improves the sustainability of APO beyond process optimization by integrating recyclability directly into the polymer architecture without losing the mechanical strength.

3.1.1 Dynamic Covalent Networks (DCNs) in Polymers

The incorporation of DCNs into various polymer architectures installs covalent bonds that reversibly cleave and reform under certain external stimuli such as heat, light, or changes in pH.⁵⁹ In contrast to traditional thermosets, in which permanently fixed covalent crosslinks prevent melting and flow, polymers with DCNs can be reshaped via exchange mechanisms when activated. There are two types of exchange mechanisms, dissociative and associative, that are commonly observed in polymers with DCNs.

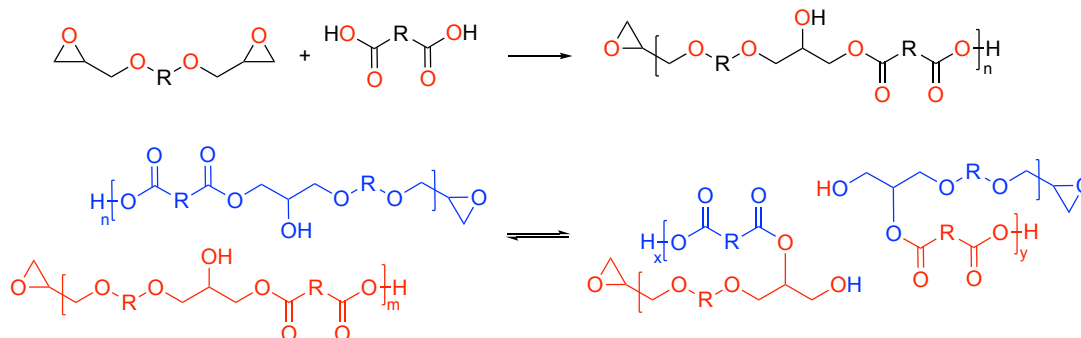
The dissociative exchange mechanism proceeds through two steps, involving temporary bond cleavage and subsequent reformation, allowing network rearrangement and recyclability under mild conditions. One of the earliest polymer examples reported with this mechanism utilized a thermally reversible Diels-Alder reaction as shown in Scheme 3-1.⁶⁰ The polymer was synthesized using multi-furan and multi-maleimide monomers to form a highly crosslinked network. The crosslinkage cleaves upon heating, resulting in loss of dimensional stability and decrease in viscosity. This allows the once crosslinked polymer material to be reprocessed like

thermoplastics. The crosslinking sites reform upon cooling, restoring the viscosity with an increase in mechanical strength similar to cured epoxy networks.⁶⁰ This reversible two-step mechanism allows the polymer to be healable and, therefore, reprocessable. Various DCNs with dissociative mechanisms have been employed into crosslinking polymer materials, including systems based on click chemistry between triazolinedione with indole, dynamic hindered urea bonding, dynamic oxime-carbamate bonding, and dynamic alkoxyamine bonding enabled by nitroxide-mediated polymerization.^{61–64}



Scheme 3-1. Polymer with DCN using reversible Diels-Alder reaction.⁶⁰

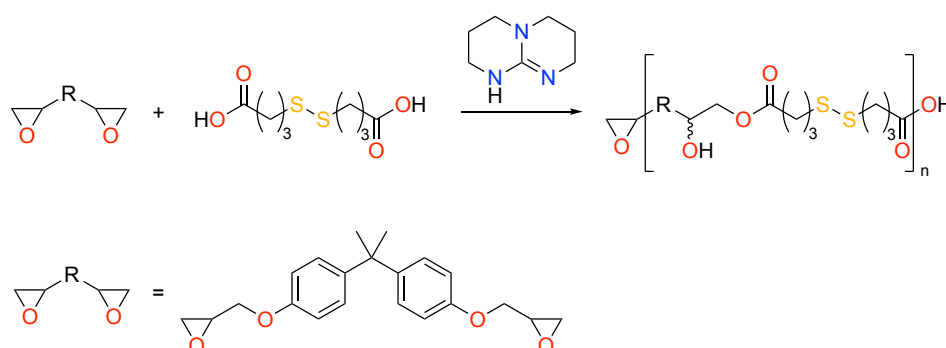
On the other hand, the associative exchange mechanism only proceeds with a single step where new bonds form as existing ones cleave, similar to a S_N2 mechanism. This preserves the overall crosslink density, which prevents the loss of dimensional stability and rapid decrease in viscosity at high temperature. Polymers that undergo associative exchange have strong mechanical performance and are insoluble like traditional thermosets but remain reprocessable like thermoplastics. When the crosslinking sites are activated by external stimuli, it grants the material malleability, similar to vitreous silica.⁴¹ This unique property was coined as “vitrimization”, and the polymers exhibiting this property are called “vitrimers.”⁴¹ The vitrimerization mechanism was observed in a polymer that was crosslinked with diepoxide, dicarboxylic acid, and tricarboxylic acid. Within the crosslinking site, transesterification occurs at high temperature as shown in Scheme 3-2. Great amounts of attention have been focused on the development of vitrimers utilizing different DCNs to potentially substitute commodity non-recyclable thermosets. Other than transesterification, other systems like transamination, transimination, transacetalation, transalkylation, and disulfide metathesis, have also exhibited associative dynamic characteristics in polymeric materials, making this synthetic strategy very versatile and adaptive to different functional groups.^{65–69}



Scheme 3-2. Exchange reaction via transesterification in epoxide carboxylic acid crosslinked networks.⁴¹

3.1.2 Dual System of DCNs and Non-covalent Interactions in Polymers

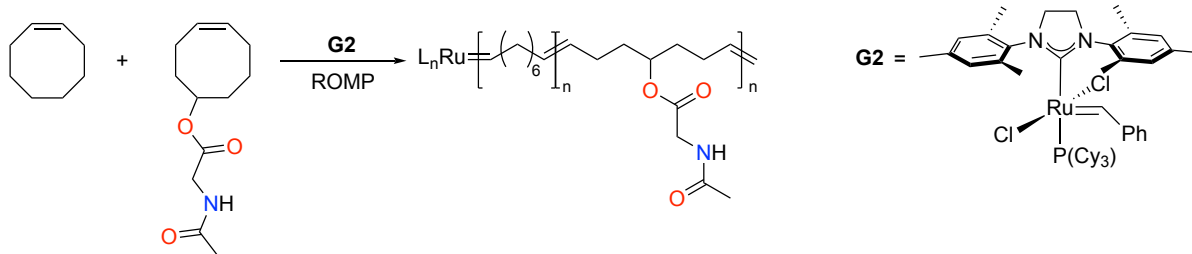
The incorporation of a single DCN system within a polymer material often limits the operational service window of the product, as certain DCNs are prone to hydrolysis while others exhibit low activation temperatures for bond exchange.⁴² As a result, polymer systems containing two or more distinct networks have been increasingly explored to enhance or tailor material properties for broader and more desirable working conditions.⁷⁰ For example, an epoxy vitrimer, shown in Scheme 3-3, consisting of disulfide metathesis and carboxylate transesterification was developed in 2019 to demonstrate the synergistic effect of dissociative and associative mechanisms.⁷¹ The dual-DCN system showed rapid stress relaxation and a low activation temperature for the exchange reaction, making the first example of utilizing two DCNs within a single polymer matrix.



Scheme 3-3. Dual-DCN polymer matrix containing disulfide metathesis and carboxylate transesterification.⁷¹

Beyond covalent exchanges, non-covalent interactions, like H-bonding and metal-ligand coordination, have also been used along with DCNs to further enhance the dimensional stability and dynamic response. The synergistic effects between DCNs and non-covalent interactions provide greater tunability in viscoelasticity, toughness, and adhesion, thereby expanding the service window of dynamic polymers. While DCNs primarily maintain structural integrity of the

network, non-covalent interactions serve as sacrificial motifs that dissipate energy and promote molecular self-assembly. For instance, the Guan group incorporated 2° amide H-bonding into G2-mediated self-healing olefin-containing network, shown in Scheme 3-4, to achieve a significant improvement in polymer toughness due to the intermolecular H-bonding interactions.⁷² The presence of G2 as the terminal group on polymer chain effectively recovered mechanical damage via olefin cross-metathesis, allowing the polymer to be more durable. Similarly, the Konkolewicz group developed an interpenetrating network in 2022 that combined a thiol-Michael DCN with dynamic H-bonding interaction, resulting in self-healing behaviour over a wide temperature range.⁷³ The Zhu group also reported enhanced creep resistance by introducing metal complex crosslinkers into a Schiff-base-containing polyimine vitrimer.⁷⁴



Scheme 3-4. Polymer containing 2° amide H-bonding site and G2-mediated self-healing olefin-containing network.⁷²

The DCNs preserve the reprocessability of the polymer, while the non-covalent interactions enhance its mechanical strength by reinforcing intermolecular associations. The synergistic integration of multiple networks within a single polymer matrix has therefore emerged as an effective strategy to achieve a balance between material performance and sustainability. Inspired by this approach, the incorporation of dual-network architecture into APO is proposed to improve its mechanical strength while maintaining its reprocessability even when aggregated.

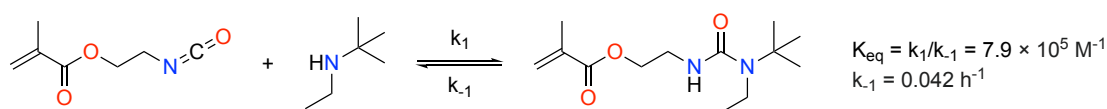
3.1.3 Incorporating DCNs into APO

APO, which contains 2° aromatic amine side groups capable of forming dynamic H-bonding and π - π stacking interactions, tends to undergo aggregation over time, leading to reduced adhesion strength and limited reprocessability. To address this issue from a polymer architecture approach, the incorporation of controlled crosslinking through DCNs into the APO matrix aims to suppress this phenomenon by confining polymer segmental mobility, while simultaneously enhancing mechanical strength through the formation of strong covalent linkages. When

introducing a moderate crosslinking density, non-covalent interactions can remain effective for rapid self-healing behaviour. Moreover, when the N–H bonds are transformed into the crosslinking site, the overall 2° amine density in APO matrix would decrease and the aggregation would be suppressed. This hybrid network design, therefore, provides a balance between dimensional stability and dynamic behaviour under external stimuli.

From a structural perspective, the repeating unit of APO consists of a chemically stable aliphatic backbone with internal alkenes and pendant 2° aromatic amine side groups. Among these, the 2° amine is the most reactive site and can be readily transformed into various functional groups such as urea, urethane, and aminal groups, which serve as potential crosslinkers.

Hindered urea bonds, in particular, are known for their dynamic behaviour when the nitrogen substituents are sterically bulky. This steric hindrance makes the bond less stable than a traditional urea bond as the planar geometry of conjugated urea bond distorts and weakens.^{75–77} Consequently, such bonds can undergo reversible bond exchange under mild conditions, making them attractive as dynamic crosslinkers for amine-containing polymers. However, bulky substituents used to form hindered urea bonds are usually alkyl groups so that the urea bonds are more reactive. For example, the dynamic hindered-urea-bond-containing poly(urea), developed by Cheng group in 2014, utilized only alkyl-substituted amines as candidates, and identified 1-(*tert*-butyl)-1-ethylurea as the most dynamic linkage, as it has high K_{eq} and k_{-1} values at 37 °C (Scheme 3-5).⁷⁵ The high K_{eq} value for this system suggests that the formation of urea bond happens readily. In contrast, the high k_{-1} value suggests that the urea bond can be cleaved under mild condition, making this reaction a dynamic equilibrium.



Scheme 3-5. Equilibrium between isocyanate, amine, and corresponding urea.⁷⁵

APO contains aniline as the 2° amine moieties, in which the lone pair on nitrogen is further delocalized through conjugation with the phenyl substituent. If APO is crosslinked via a urea linkage, the extended conjugation suppresses the reversible bond exchange reaction, making it less dynamic. Therefore, hindered urea bonds might not be suitable for APO system.

Aminal (*N,N*-acetal) bonds, on the other hand, consisting of two amine groups connected through a methylene linkage, have demonstrated dynamic exchange reactivity via dissociative transamination upon treatment with acid catalyst, as shown in Figure 3-1. Such dynamic behaviour makes the aminal functionality a potential dynamic crosslinker for APO.

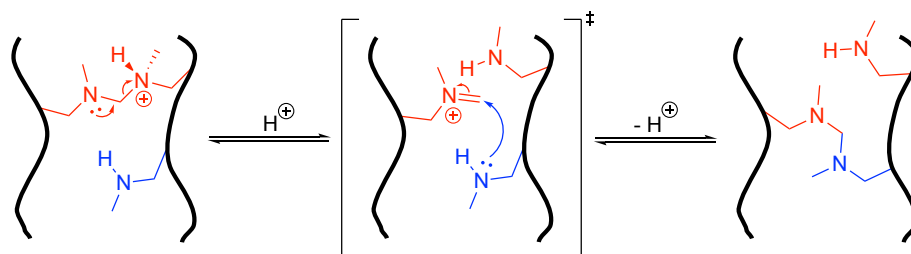
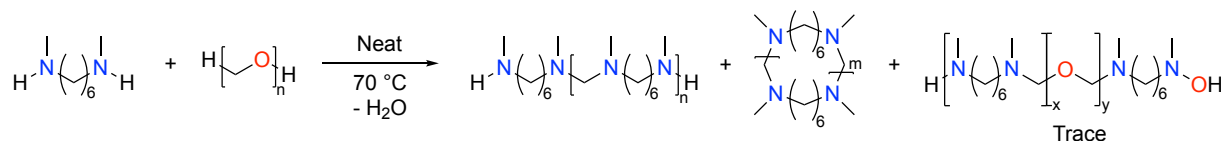


Figure 3-1. Transamination between 2° amines via formation of iminium ion.⁷⁸

A recent study by the Zhang group reported the preparation of poly(aminals) through condensation of 2° diamines and paraformaldehyde, as shown in Scheme 3-6, forming covalent bonds capable of undergoing reversible transamination in the presence of mild acid catalyst.⁷⁸ The reaction predominantly yielded linear and cyclic poly(aminals) as the major products, while polymers containing hemiaminal ether linkages were only shown in trace amounts based on NMR. Due to the dynamic nature of aminal bond, the linear poly(aminal) could readily convert into the cyclic poly(aminal) via intramolecular chain transfer, and vice versa, establishing an equilibrium between the two species. Furthermore, triamines could also be used as crosslinkers to form networks along with the diamines and paraformaldehyde to tune the material properties. The crosslinked poly(aminal) networks exhibited thermal reprocessability and recyclability. Furthermore, the recycled crosslinked poly(aminal) networks maintained mechanical strength comparable to their pristine states. More recent studies have employed dynamic aminal bonds into polymer systems to improve the reprocessability, highlighting their growing potential as an attractive motif for DCNs.^{79–82}



Scheme 3-6. Synthesis of poly(aminals) using diamine and paraformaldehyde.⁷⁸

In the case of APO, the incorporation of aminal bonds as crosslinking sites introduces a DCN into the polymer matrix. Such structural integration may not only preserve the inherent reprocessability and recyclability of APO but also enhance its mechanical robustness and mitigate

amine aggregation. The formation of crosslinks partially restricts the segmental motion of polymer chains, thereby reducing chain entanglement and suppressing aggregation, while maintaining reversible bond exchange dynamics under mild conditions. Moreover, the presence of dynamic non-covalent interactions, including H-bonding and π - π stacking among the residual free amine groups, can further reinforce the network. These synergistic effects may improve creep resistance and enable rapid stress relaxation at ambient temperature, thereby extending the functional temperature range of the modified APO.

3.1.4 Scope of Chapter

In this chapter, APO was crosslinked with a C₁ source, forming dynamic aminal bonds to enhance its mechanical strength and reprocessability via reversible transamination (Figure 3-2). The aminal functionality was selected due to its proven dynamic exchange reactivity and chemical compatibility with the pendant 2° amine moieties in other polymer systems. Small-molecule model reactions were first conducted using *N*-methyl-aniline (NMA) as a benchmark compound to evaluate the feasibility of aminal bond formation and to identify optimal synthetic routes. The targeted compound, *N,N'*-dimethyl-*N,N'*-diphenylmethylenediamine (**a1**) in Scheme 3-7, was synthesized and characterized by NMR, GC-MS, and XRD analyses to confirm the aminal linkage formation, and to assess bond geometry. The dynamic nature of the aminal bond was subsequently tested under both thermal and acid-catalyzed conditions using GC-MS to verify its reversible transamination behaviour.

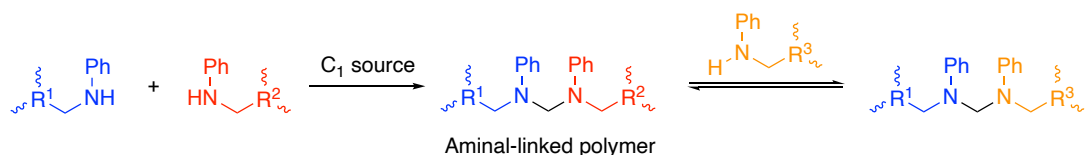


Figure 3-2. General illustration of aminal-linked polymer followed by reversible transamination.

Building upon these model reactions, two approaches were investigated to incorporate the aminal linkage into APO systems. The first synthetic strategy was to convert ACC-NMA into its aminal-linked dimer and subsequently polymerized to yield an APO matrix containing aminal crosslinkers *in-situ*. The second synthetic strategy was to introduce network formation directly into linear APO under mild conditions as a post-polymerization modification. The resulting dimers and polymers from both approaches were comprehensively characterized using spectroscopic, thermal, and rheological analyses to evaluate the incorporation of dynamic crosslinks, network integrity,

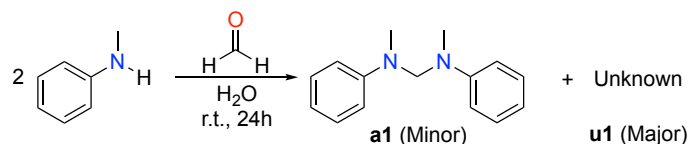
thermal stability, and viscoelastic properties. Different crosslinking densities were made to assess optimal viscoelasticity and processability.

Overall, this chapter establishes the conceptual and experimental framework for incorporating aminal-based DCNs into APO. The findings presented here lay the groundwork for understanding how molecular-level dynamic crosslinking can be used to simultaneously enhance mechanical strength and tune thermal property in aminated polyolefin systems.

3.2 Results and Discussion

3.2.1 Small-Molecule Reaction with Formaldehyde

To investigate the feasibility of incorporating dynamic aminal linkages into the APO matrix, a series of small molecule model reactions was performed using NMA and its derivatives as representative 2° amine substrates. The screening aimed to identify an optimal synthetic route to form the targeted aminal-linked diamines, which could subsequently serve as a crosslinking motif in APO. Three synthetic approaches were evaluated using formaldehyde, diiodomethane, and carbon dioxide as methylene sources.



Scheme 3-7. Reaction scheme of NMA and formaldehyde.

Formaldehyde was tested as the C₁ electrophilic source to react with NMA based on the synthetic procedure reported in literature to yield aminal species with alkyl 2° amines.⁷⁸ This reaction proceeded readily at room temperature through the condensation of NMA with 37% aqueous formaldehyde solution under mild conditions to generate methylene-bridged diamines, as shown in Scheme 3-7. However, analyses of the crude product mixture indicated the formation of a mixture of products as listed in Table 3-1 entry 1. The chromatogram of the crude mixture obtained by GC-MS (Figure S3-1) revealed the formation of *N,N'*-dimethyl-*N,N'*-diphenylmethylenediamine (**a1**) as the minor product with a [M + 1] fragmentation ion signal of 227.2 m/z. The major product (**u1**) with the [M + 1] fragmentation ion signal of 241.2 m/z, remains unknown. Simple vacuum distillation of the reaction mixture afforded aminal-linked NMA in high purity based on ¹H and HSQC NMR spectra (Figure S3-2 and S3-3).

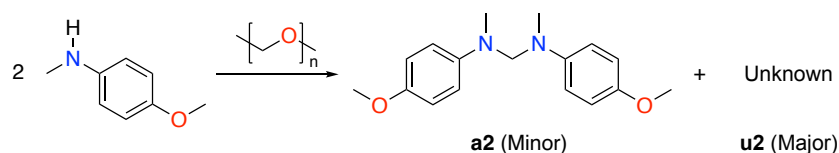
Table 3-1. Summary of reactions of aniline derivatives with formaldehyde under different conditions.

Entry	Aniline	Aniline:CH ₂ O Molar ratio	Time (hr)	Solvent	Temp. (°C)	Conversion to aminal species	Conversion to unknown species
1	NMA	2:1	24	H ₂ O	r.t.	12 % (a1)	46 % (u1)
2	4-MeO-NMA	2:1 ^a	24	H ₂ O	r.t.	30 % (a2)	56 % (u2)
3	4-MeO-NMA	2:1 ^a	24	Neat	r.t.	15 % (a2)	71 % (u2)
4	ACC-NMA	2:1 ^a	24	H ₂ O	r.t.	No reactivity	
5	ACC-NMA	2:1 ^a	24	H ₂ O/Tol ^b	130	No reactivity	
6	ACC-NMA	2:1 ^a	24	Neat	90	No reactivity	
7	ACC-NMA	2:1 ^a	72	THF	60	No reactivity	
8	ACC-NMA	2:1 ^a	72	THF ^c	60	No reactivity	

a = paraformaldehyde was used. b = Dean-Stark apparatus was used. c = MgSO₄ was used as drying agent.

The ¹H NMR spectrum (Figure S3-2) of **a1** exhibited a diagnostic singlet at $\delta = 4.79$ ppm corresponding to the methylene protons of the aminal linkage, that matched the literature-reported values.⁸³ The integration ratio of methylene:methyl:phenyl protons is roughly 1:3:5, consistent with the formation of **a1** structure. Upon cooling of the distillate overnight, amber coloured crystals formed, which were measured using single-crystal X-ray diffraction (XRD). The resulting crystal structure (Figure S3-4) confirmed the methylene connectivity between two NMA units, verifying the successful synthesis of the targeted small-molecule model compound.

To further investigate the influence of electronic effects towards aminal formation, 4-methoxy-*N*-methylaniline (4-MeO-NMA) was used as an alternative aniline derivative to react with paraformaldehyde, as illustrated in Scheme 3-10. Paraformaldehyde (PFA) was selected as the C₁ source instead of the conventional 37% aqueous formaldehyde solution to minimize the loss of volatile formaldehyde to the headspace of the reaction vessel and to provide a more controlled release of the substrates. The introduction of a methoxy substituent at the *para*-position enhances the electron density of the aromatic ring through resonance, thereby increasing the nucleophilicity of the amine nitrogen. As a result, the nitrogen lone pair is more activated for nucleophilic attack toward the carbonyl on formaldehyde.



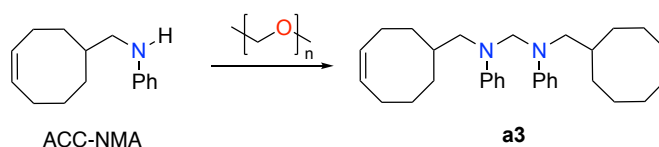
Scheme 3-8. Reaction scheme of 4-MeO-NMA and formaldehyde.

The reactivity and product selectivity of 4-MeO-NMA and PFA under two different reaction conditions are summarized in table 3-1 entries 2 and 3. The formation of **a2** was confirmed

by the ^1H NMR spectrum of the crude product mixture (Figure S3-5), which showed the diagnostic singlet at $\delta = 4.57$ ppm corresponding to the methylene protons of the aminal linkage, consistent with literature reported data.⁸³ In addition, the broad singlet at $\delta = 1.56$ ppm, attributed to the water formed through the condensation reaction. Reactions using 4-MeO-NMA proceeded faster than with NMA, as the consumption of 4-MeO-NMA is higher based on their GC chromatograms (Figure S3-6 and S3-7). This observation supports the expected increase in reactivity.

However, despite the higher conversion, product selectivity toward **a2** remained poor. GC-MS chromatograms and mass spectra of the crude mixtures revealed that the formation of **u2** ($m/z = 300.2$) was favoured over the desired **a2**, both in the presence and absence of solvent. Although the incorporation of an electron-donating methoxy substituent on the aromatic ring enhanced the overall conversion, it did not improve the product selectivity toward aminal linkage formation under the tested conditions.

To tailor toward APO modification, ACC-NMA was subjected to condensation with PFA under various reaction conditions to synthesize the targeted aminal-linked product **a3**, as illustrated in Scheme 3-9. Multiple reaction parameters, including solvent, temperature, and reaction duration, were varied in an attempt to favour aminal bond formation, as summarized in Table 3-1 entries 4 to 8. Despite these efforts, no detectable conversion of ACC-NMA was observed under any of the tested conditions, as confirmed by GC-MS and ^1H NMR analyses of the crude reaction mixtures (Figure S3-8 and S3-9).



Scheme 3-9. General reaction between ACC-NMA and PFA with target product a3.

ACC-NMA was first attempted to react with PFA in water at room temperature, as shown in Table 3-1 entry 4. However, no reactivity was observed based on both GC-MS and NMR analyses. To promote the formation of **a3**, higher reaction temperature and constant water removal were applied to drive the condensation reaction forward. Therefore, a Dean-Stark apparatus was used to facilitate higher reaction temperature (130 °C) and constant removal of by-product water (Table 3-1 entry 5), but no **a3** formed. A reaction in neat conditions, where ACC-NMA was used

as the solvent, also showed no reactivity and no dissolution of the PFA at 90 °C (Entry 6). Further attempts by replacing water with THF as the reaction solvent were conducted to improve the solubility of ACC-NMA (Entry 7). However, only trace amounts of PFA were dissolved in THF at 60 °C, and no formation of **a3** was observed. Anhydrous magnesium sulfate (MgSO₄) was introduced as a drying agent in THF to push the reaction toward product formation by continuously removing the water by-product (Entry 8). However, the inclusion of MgSO₄ did not result in any observable product formation, and the GC-MS spectra remained identical to that of the starting material, indicating that the reaction was unfavourable under all tested conditions.

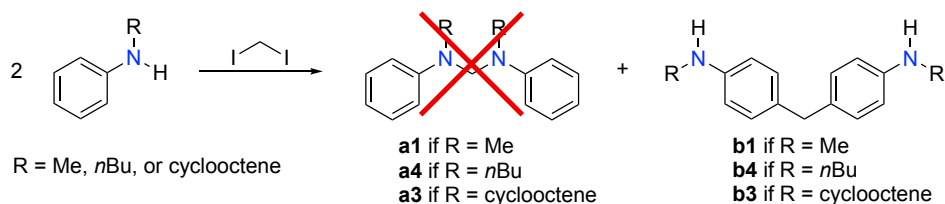
Although ACC-NMA contains a weakly electron-donating cyclooctene alkyl substituent that could activate the nitrogen toward nucleophilic attack, the lack of reactivity is likely due to steric effects. The bulky cyclooctene ring adjacent to the 2° amine imposes significant steric hindrance around the nitrogen atom, thereby potentially restricting the approach of formaldehyde to the reactive site. This steric hindrance inhibits nucleophilic attack on the carbonyl carbon, preventing the formation of the desired aminal linkage under the examined conditions. Therefore, other potential C₁ sources like diiodomethane and CO₂ were investigated as alternatives.

3.2.2 Small-Molecule Reaction with Diiodomethane

The second synthetic route investigated for the formation of aminal-linked species involved the use of diiodomethane (CH₂I₂) as the C₁ source. The presence of the two iodide leaving groups in CH₂I₂ was expected to facilitate the nucleophilic substitution with amine-containing compounds, and, thus, promote the formation of the aminal species. This approach was inspired by a previously established procedure in which the 2° amines in ACC-NMA were methylated into 3° amines using methyl iodide (CH₃I).³⁰ In that reaction, the N–H in ACC-NMA was first deprotonated by a base to generate an anilide anion, which subsequently underwent a substitution reaction with CH₃I, yielding the tertiary amine.

NMA and *N*-*n*-butylaniline (NBA) were used as substrates to evaluate this synthetic route, as shown in Scheme 3-10. The reaction with NMA served as a benchmark reaction, while the reaction with NBA was meant to be a comparative model reaction to examine the effect of increased steric

bulk on reactivity. The bulkier *n*-butyl substituent on the nitrogen provided insight into how steric hindrance influences nucleophilic substitution during the formation of the aminal-linked product.



Scheme 3-10. General reaction between aniline derivatives and CH₂I₂ with targeted aminal products and by-products.

The results are summarized in Table 3-2, with Entries 1 and 2 corresponding to NMA and NBA, respectively. When CH₂I₂ was used as the C₁ source, the reaction with NMA proceeded readily under the tested reaction conditions. However, instead of yielding the desired aminal-linked species (**a1**), the main product was identified as 4,4'-methylene-bis(*N*-methylamine) (**b1**) with high conversion of 80.2% based on GC-MS and ¹H NMR analyses.

The purified product was first analyzed by GC-MS, which showed a retention peak corresponding to a [M + 1] fragmentation ion signal of 227.2 m/z (Figure S3-10), consistent with the theoretical molecular weight of **a1** (226.3 g/mol). Since **a1** and **b1** are constitutional isomers, the isolated product was initially presumed to be **a1** based solely on the mass spectrum. However, further characterization by ¹H NMR spectroscopy (Figure S3-11) revealed that the isolated compound did not match the diagnostic chemical shifts of **a1** (Figure S3-2).

The proton integration ratio between methylene, methyl, and phenyl in the ¹H NMR spectrum was roughly 1:3:4, which differs from the expected ratio for the aminal structure. Moreover, the aromatic region exhibited two doublets at 6.55 ppm and 7.00 ppm, suggesting a *para*-substituted phenyl ring, which hints at the potential *para-para* methylene linkage. The singlet peak at $\delta = 3.78$ ppm was assigned to the methylene (–CH₂–) bridge based on the HSQC NMR spectrum (Figure S3-12). This peak did not correspond to the characteristic chemical shift of the aminal methylene group in **a1**, but instead matched literature-reported spectra for **b1**.⁸⁴ Collectively, the reaction of NMA with CH₂I₂ was confirmed to favour the formation of **b1** rather than the aminal product **a1**.

Table 3-2. Summary of reactions of aniline derivatives with CH₂I₂ under different conditions.

Entry	Aniline	Aniline:CH ₂ I ₂ Molar ratio	Base	Time (hr)	Solvent	Temp. (°C)	Conversion to amination species (%)	Conversion to diphenyl methylene species (%)
1	NMA	2:1	N/A	120	MeCN	80	0 (a1)	80.2 (b1)
2	NBA	2:1	N/A	168	MeCN	80	0 (a4)	51.7 (b4)
3	ACC-NMA	2:1	N/A	72	DCM	r.t.	No reactivity	
4	ACC-NMA	2:1	N/A	120	THF	60	No reactivity	
5	ACC-NMA	2:1	NEt ₃ ^a	24	THF	60	No reactivity	
6	ACC-NMA	2:1	K ₂ CO ₃ ^a	24	DMF	150	No reactivity	
7	ACC-NMA	2:1	NaHMDS ^a	120	Toluene	100	No reactivity	

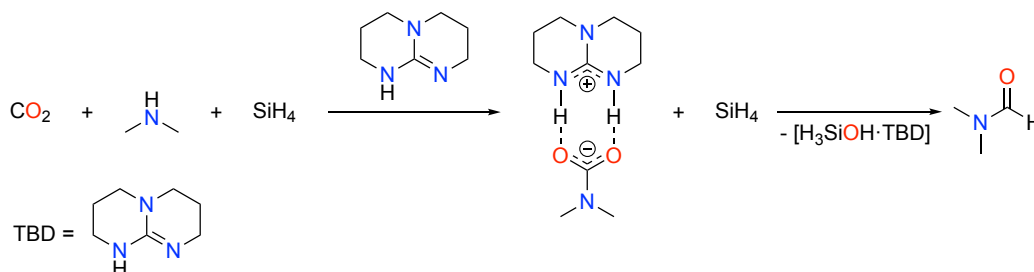
a = excess equivalents of base were added to the ACC-NMA solution at 0 °C, followed by the addition of CH₂I₂.

When the substrate was changed from NMA to NBA, the conversion of the corresponding diphenylmethylene species (**b4**) decreased relative to **b1**. However, **b4** was obtained with 51.7% conversion as confirmed by the GC-MS and ¹H NMR analyses. The GC chromatogram and mass spectrum showed a retention peak corresponding to **b4** with a [M + 1] fragmentation ion signal at m/z = 311.2, matching with its theoretical molecular weight (310.5 g/mol), as shown in Figure S3-12. The diagnostic singlet for methylene linkage at δ = 3.91 ppm in the ¹H NMR spectrum further confirmed the formation of **b4** (Figure S3-13).

ACC-NMA was subsequently employed as the substrate to react with CH₂I₂ under various reaction conditions to examine whether the corresponding amination species (**a3**) could be formed, as summarized in Table 3-2 (Entries 3–7). Due to the poor solubility of ACC-NMA in acetonitrile, the reaction was carried out in other solvents instead. Reaction temperatures were set according to the reflux temperature of each solvent used. Several bases were tested to promote deprotonation of the amine and facilitate nucleophilic substitution. However, no observable product formation was detected under any of the conditions tested. The reactions were monitored by ¹H NMR spectroscopy, and no conversion of ACC-NMA to either **a3** or **b3** was shown. The results suggested that the amine remained unreactive toward CH₂I₂ under all examined conditions, which may be due to the steric hinderance ACC-NMA imposed.

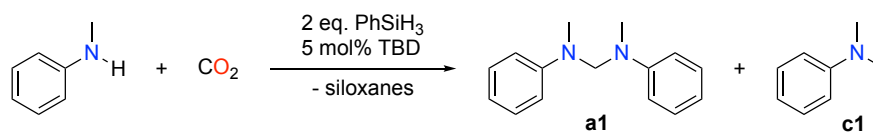
3.2.3 Small-Molecule Reaction with CO₂

The third synthetic route investigated for the formation of aminal-linked species involved the reductive functionalization of CO₂ as the C₁ source by using an organocatalyst to bridge two amines. This methodology was first reported by Canat et al. in 2012 to react CO₂ in a single step with a catalyst, a functionalization agent, and a reducing agent as shown in Scheme 3-11.⁸⁵ 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD) was used to facilitate CO₂ insertion into the N–H bonds of aliphatic amines, forming a CO₂-base adduct, which subsequently underwent reduction with silanes to yield formamide.⁸³



Scheme 3-11. Single-step reductive functionalization of CO₂ using TBD organocatalyst to yield formamide.⁸⁵

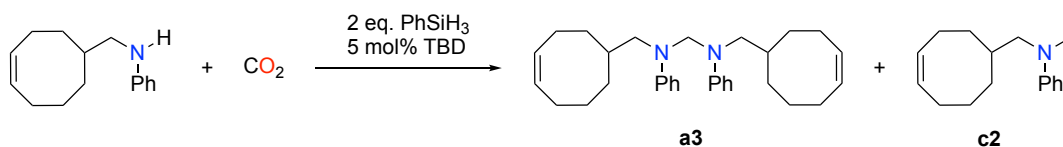
This reaction protocol was modified in 2015 to favour the formation of aminal species by optimizing the amine and silane substrates and the stoichiometric ratio between CO₂, amine, and silane as shown in Scheme 3-12.⁸² It was reported that aliphatic amines favour the formation of formamides while aromatic amines favour the formation of aminals due to differences in nucleophilicity. Under 1 bar of CO₂ pressure, NMA was converted into the aminal species with 93% yield in 2.5 hours at 80 °C, making this synthetic strategy suitable for the scope of this chapter.



Scheme 3-12. Single-step reductive functionalization of CO₂ using TBD organocatalyst to yield aminal **a1.⁸³**

To validate the reproducibility of this synthetic strategy, the reaction in Scheme 3-12 was repeated under identical conditions. NMA was reacted with CO₂ in the presence of TBD as the organocatalyst and phenylsilane as the reductant. Under these reaction conditions, the desired aminal product (**a1**) was obtained. The formation of the aminal linkage was confirmed by ¹H NMR analysis, where the diagnostic resonance of the aminal moiety appeared as a singlet at $\delta = 3.79$ ppm (Figure S3-15). The corresponding peak in the HSQC spectrum (Figure S3-16) further confirmed the presence of the –CH₂– group, consistent with the proposed aminal structure of **a1**.

However, the formation of dimethylaniline (**c1**) as a side product was also observed. This was confirmed by the appearance of a small yet distinct singlet at $\delta = 2.90$ ppm in the proton NMR spectrum, corresponding to the methyl substituents on the 3° amine. The generation of **c1** likely arises from methylation of the amines by an excess amount of CO₂. Similar observations were also reported in the reductive functionalization of CO₂ using betaine as the organocatalyst.⁸⁶ This could be attributed to the larger head space of the J-Young NMR tube, which allowed more CO₂ to be loaded. Interestingly, the major product became **c1** when the reaction time was prolonged from 2.5 hours to 20 hours, and **a1** was only observed in trace amounts. Despite the methylation reaction, the major product was determined to be **a1** when the reaction time was kept short (1 to 2.5 hours). Therefore, this synthetic route is an effective strategy for aminal formation under mild conditions with short reaction times.



Scheme 3-13. Single-step reductive functionalization of CO₂ using TBD organocatalyst to yield aminal **a3.**

With the successful synthesis of **a1** using this strategy, the same synthetic strategy was subsequently applied to ACC-NMA, as illustrated in Scheme 3-13. However, the desired **a3** was not synthesized under the conditions listed in Table 3-3 Entry 1. Instead, the reaction yielded the methylated ACC-NMA derivative (**c2**) as the solo product, as confirmed by both NMR and GC-MS analyses. In the HSQC NMR spectrum (Figure S3-17), a distinct and intense cross-peak corresponding to the methyl substituent (–CH₃) on the 3° amine was observed. The GC-MS spectrum (Figure S3-18) showed a fragmentation ion signal at $m/z = 230.2$, consistent with the theoretical molecular weight of **c2** (229.4 g/mol).

Reducing the reaction time to 1 hour resulted in an identical outcome, as listed in Table 3-3 Entry 2, with **c2** remaining the only product. This finding indicates that, under the established conditions reported in the literature, the reaction pathway favours methylation of ACC-NMA, consistently yielding **c2** rather than **a3**. To further investigate this synthetic strategy, the reaction temperature was varied to 30, 50, and 70 °C (Table 3-3 Entry 3 to 5). A noticeable singlet corresponding to the methyl group on the 3° amine appeared at $\delta = 2.91$ ppm as the most intense

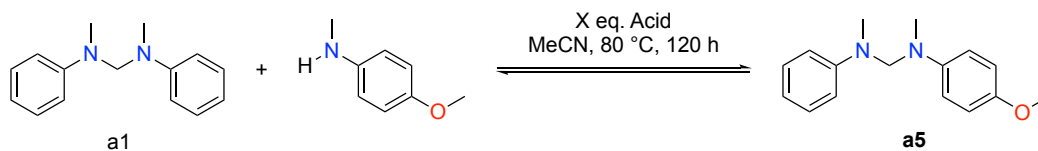
peak in each case. However, a weak signal at $\delta = 4.84$ ppm in the ^1H NMR spectrum, characteristic of the aminal methylene group, also emerged under all three conditions, suggesting the formation of **a3** in trace amounts. The signal was further confirmed to be the methylene ($-\text{CH}_2-$) group based on the HSQC NMR spectrum (Figure S3-19). Among the tested temperatures, 70 °C showed the highest relative intensity of this signal and was therefore selected for subsequent optimization experiments.

Table 3-3. Summary of reactions of ACC-NMA with CO₂ under different conditions.

Entry	ACC-NMA (eq.)	PhSiH ₃ (eq.)	TBD (mol%)	Time (hr)	Temp. (°C)	a3:c2 relative ratio
1	1	2	5	2	80	Only c2 was obtained
2	1	2	5	1	80	Only c2 was obtained
3	1	2	5	1	70	1:3.5
4	1	2	5	2.5	50	1:3.8
5	1	2	5	20	30	1:13.6
6	2	2	5	2	70	1:2.9
7	4	2	5	2	70	1:20.9
8	8	2	5	2	70	1:10.4
9	8	8	5	4	70	2.1:1

With the reaction temperature optimized, the stoichiometric ratios of ACC-NMA and the silane reductant were varied while keeping the CO₂ loading consistent (Table 3-3 Entries 6 to 8). When 2, 4, and 8 equivalents of ACC-NMA were used with 2 equivalents of the silane, trace formation of **a3** was observed in reactions with 4 and 8 equivalents. In contrast, when the molar ratio of ACC-NMA to silane was adjusted to 1:1, as listed in Entry 6, the formation of the aminal species increased notably. This observation suggests that excess amounts of CO₂ and silane in the reaction vessel promote methylation, leading to preferential formation of the **c2** rather than the desired aminal **a3**. This is consistent with the proposed reaction pathway of reductive functionalization of CO₂ reported in literature as shown in Figure 3-3, where the formation of stable siloxane drives the reaction forward and yields methylated product.^{83,86} Based on this hypothesis, both the amine and silane were used in 8 equivalents while keeping the CO₂ loading constant. Under these optimized conditions, **a3** became the major product over **c2**, with a relative ratio of approximately 2.3:1. The yield of **a3** under this condition could be as high as 54% based on internal standard as shown in Figure S3-20. The methylene signal at $\delta = 4.84$ ppm was confirmed to be the aminal moiety based on HSQC and COSY 2D NMR analyses, where the corresponding cross-peaks were determined to be ($-\text{CH}_2-$) and do not correlate with other protons, respectively (Figure S3-21 and S3-22).

aminal linkage, a transamination model reaction was conducted, as illustrated in Scheme 3-14. Aminal-linked NMA (**a1**) was reacted with 4-MeO-NMA to determine whether the aminal bond could undergo exchange and form asymmetric aminal species, thereby validating its dynamic nature. However, no formation of asymmetric aminal (**a5**) was observed without an acid catalyst.



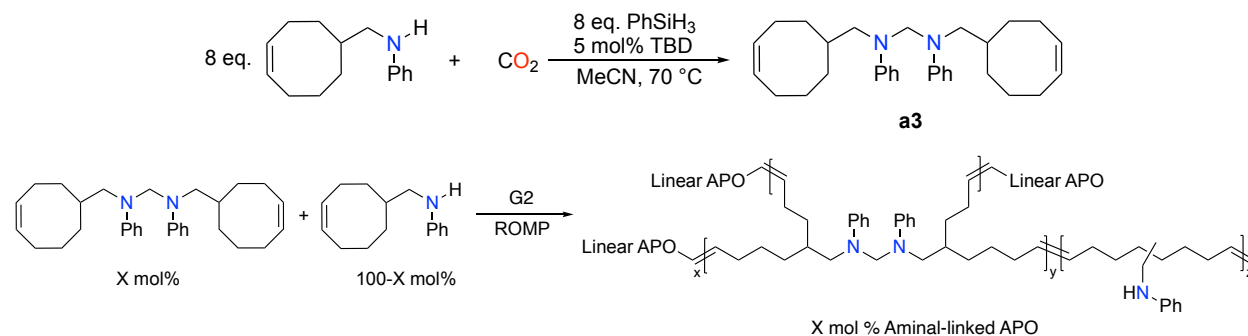
Scheme 3-14. Formation of asymmetric aminal species.

To promote the dynamic exchange of aromatic aminal linkage, a catalytic amount of trifluoroacetic acid (TFA, 0.05 eq.) was introduced to activate C–N bond cleavage and the iminium formation. The formation of **a5** in trace amounts was observed by GC-MS analysis, where the molecular ion signal at $m/z = 256.1$ was detected (Figure S3-23). This result indicated that transamination occurred but was limited under the tested conditions. Increasing the TFA loading to 1 or 2 equivalents did not show significant improvement. Moreover, the use of a stronger acid such as HCl resulted in the complete cleavage of the aminal bond, converting back to the parent aniline compound (NMA). These findings demonstrate that aromatic aminal linkages are less dynamic than the aliphatic ones. Further optimization of reaction parameters, including temperature and acid catalyst loading, are needed to optimize this feature. Nevertheless, the aromatic aminal bond can be readily cleaved by the addition of a strong acid, making such a cleavable linkage useful as a reversible crosslinker in APO to transform the linear thermoplastic APO into a recyclable thermoset.

3.2.5 Crosslinking APO with Aromatic Aminal Linkage

To evaluate how incorporating aromatic aminal bonds into APO changes its material properties, the ACC-NMA monomer was first converted into its aminal-linked dimer (**a3**) using the CO₂ reductive functionalization approach. The crosslinking density was subsequently tuned by changing the molar percentage of ACC-NMA monomer to the aminal crosslinker before ROMP, as illustrated in Scheme 3-15. This synthetic strategy allowed precise control over the aminal content within the crosslinked APO network. However, **a3** was used directly without further purification, due to the inherent instability of the aminal species. Attempts to isolate **a3** proved challenging and often led to degradation. As a result, the subsequent ROMP reaction was carried

out *in-situ*, as the G2 catalyst is known to have good tolerance toward functional groups and impurities.



Scheme 3-15. *In-situ* reaction of aminal crosslinked APO.

Two different crosslinking densities, 40 mol% and 5 mol%, were targeted to prepare aminal-linked APO, representing the upper and lower bounds of crosslinking. When 40 mol% of aminal (**a3**) was used, the subsequent ROMP reaction gelled within approximately 30 minutes, forming a spherical solid as shown in Figure 3-4. The colour of the solid gradually changed from transparent light yellow to opaque dark purple as the reaction progressed. The resulting 40 mol% aminal-linked APO was rigid, brittle, and found to be insoluble in both THF and DCM. The product did not show self-healing or adhesive behaviour and crumbled into pieces once vacuum-dried at 30 °C for 12 hours. The resulting T_g was measured to be -11.5 °C, consistent with the crosslinked network (Figure S3-24), as parent APO has a T_g of -21.0 °C. This brittle material was not subjected to rheological, NMR, or GPC analyses.

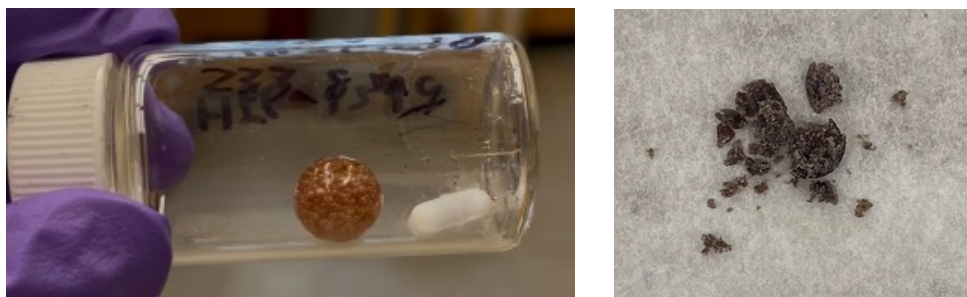


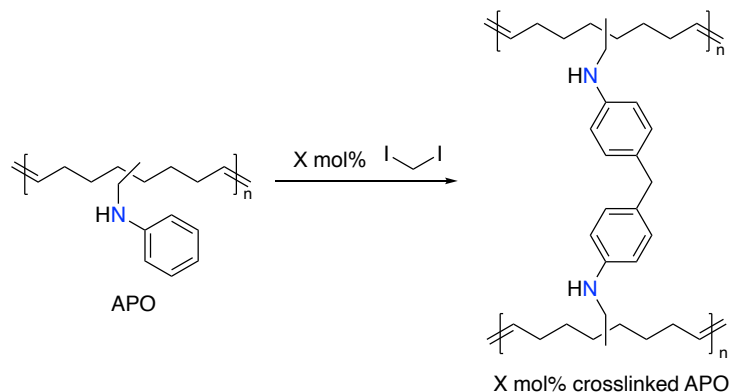
Figure 3-4. Gelled 40 mol% aminal-linked APO.

In contrast, the 5 mol% aminal-linked APO remained processable, exhibiting self-healing and adhesive properties comparable to the linear APO. The product was soluble in common organic solvents such as DCM and chloroform. The ^1H NMR analysis confirmed the presence of aminal crosslinkages, with a diagnostic resonance at $\delta = 4.75$ ppm, corresponding to the methylene protons of the aminal moiety (Figure S3-25). However, rheological analysis of the 5 mol% crosslinked APO revealed no significant change in viscoelastic behaviour compared to linear APO

(Figure S3-26), suggesting that the lower crosslinking density did not substantially alter the material properties.

3.2.6 Crosslinking APO with Diiodomethane

In addition to incorporating the amination moieties into APO through CO₂ fixation as discussed above, CH₂I₂ was also investigated as a direct crosslinking agent for linear APO. This post-polymerization modification (PPM) approach aimed at converting the pendant amine groups in APO into amination species *in-situ*, eliminating the need to pre-synthesize the amination crosslinker prior to polymerization. Since the product of the small molecule model reaction was originally anticipated to be *N,N'*-dimethyl-*N,N'*-diphenylmethylenediamine (**a1**), this approach was explored before the unexpected finding that 4,4'-methylene-bis(*N*-methylamine) (**b1**) formed readily when reacting NMA with CH₂I₂. Therefore, the same synthetic strategy was applied to APO, as illustrated in Scheme 3-16 to evaluate the effect of PPM with different loading of CH₂I₂.



Scheme 3-16. General reaction of crosslinking APO with CH₂I₂.

Low-molecular-weight APO, 20.3 kg/mol, was synthesized for subsequent crosslinking experiments to ensure that the structural and property changes before and after crosslinking could be more distinguished. The linear APO was precipitated twice using cold MeOH to ensure complete removal of residual quenched Grubbs catalyst. The reaction parameters, including solvent, temperature, and CH₂I₂ loading, were systematically screened to test the resulting material performance. The influence of crosslinking on the material properties was evaluated by measuring the T_g, molecular weight, and viscoelastic behaviour using DSC, GPC, and rheological analysis, respectively. The results are summarized in Table 3-4.

Table 3-4. Summary of crosslinked APO using CH₂I₂ under different conditions.

Entry	Solvent	Temp. (°C)	CH ₂ I ₂ (mol%) ^a	M _w (Kg/mol)	<i>D</i>	T _g (°C)
1	-	-	0	20.3	1.7	-21.8 ^{b, c}
2	DCM	r.t.	10	22.9	1.7	-19.3
3	DCM	r.t.	25	31.5	1.5	-17.7
4	DCM	r.t.	50	22.2	1.8	-17.0
5	DCM	40	10	32.0	2.0	-14.0 ^b
6	DCM	40	25	42.3	2.0	-13.6 ^b
7	DCM	40	50	46.1	3.3	-14.3 ^b
8	THF	r.t.	10	29.9	1.4	-18.1
9	THF	r.t.	25	35.2	1.5	-19.3
10	THF	r.t.	50	30.4	1.5	-19.6
11	THF	60	10	Gelled up		-12.8 ^c
12	THF	60	25	Gelled up		-10.0 ^c
13	THF	60	50	Gelled up		-8.2 ^c

a = mol% of CH₂I₂ based on ACC-NMA monomer. b = data shown in Figure S3-27. c = data shown in Figure S3-28.

The reactions were first screened in DCM, as the previous synthetic step (ROMP) was also performed in this solvent. Reaction temperatures were tested at either room temperature or 40 °C, with different loading of CH₂I₂ (10, 25, and 50 mol%) based on the stoichiometry of the ACC-NMA monomer. When the reactions were performed in DCM at room temperature (Table 3-4 Entries 2 to 4), the molecular weight of the resulting polymers did not change significantly compared to that of the linear APO, regardless of the CH₂I₂ loading. A slight increase in T_g from -21.0 to -17.0 °C was observed when 50 mol% CH₂I₂ was used. The viscoelastic behaviour of the resulting polymers also remained unchanged, as shown by the storage (G') and loss (G'') moduli compared to that of the linear APO in Figure 3-5. In all cases, G'' suppressed G' at low frequencies, suggesting flexible polymer segmental motion and long-range relaxation, which is similar to the rheological behaviour of linear APO.

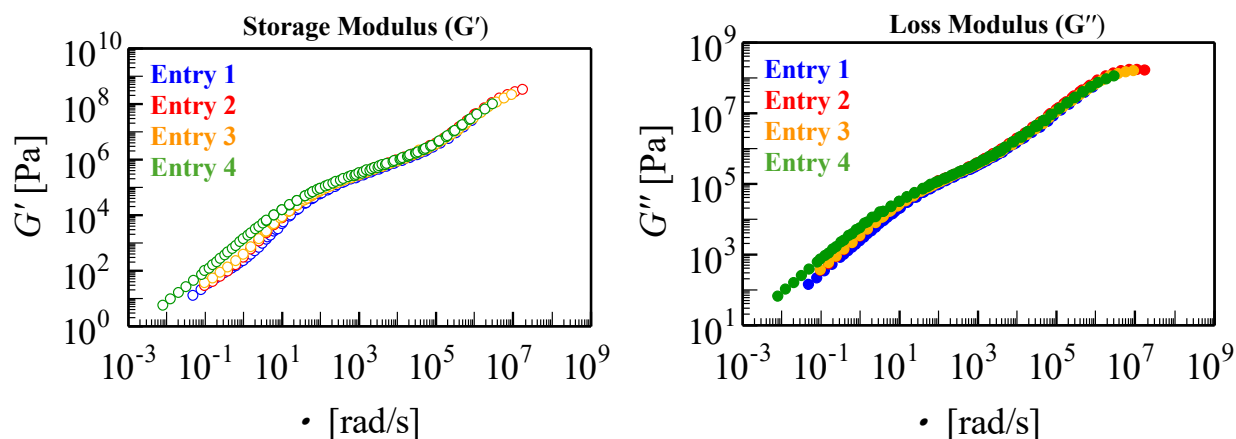


Figure 3-5. Master curves of linear and crosslinked APOs synthesized in DCM at room temperature from frequency sweep experiments at T_{ref} = 30 °C.

In contrast, reactions performed at 40 °C yielded polymers with slightly higher molecular weights, increasing from 20.3 Kg/mol to a maximum of 46.1 Kg/mol, confirming the effectiveness of crosslinking as listed in Entries 5 to 7 in Table 3-4. The dispersity ($\bar{D}$) increased from 1.7 to 3.3, suggesting that only partial crosslinking had taken place, leaving some linear APO unreacted. Furthermore, the T_g increased from -21.0 to approximately -14.0 °C across all CH_2I_2 loadings, confirming partial crosslinking and incomplete reaction conversion.

The viscoelastic behaviour of the resulting polymers showed moderate changes compared to the linear APO, as shown in Figure 3-6. Similar results were observed where G'' remained higher than G' at low frequencies for all samples, indicating that the polymers continued to exhibit liquid-like behaviour. However, G' increased progressively with higher loading of CH_2I_2 from 10 to 50 mol%. This behaviour suggests an increasing number of crosslinking before reaching the gel state at which the system contains one average crosslinking site per polymer chain. These results demonstrate that reaction temperature plays a crucial factor in facilitating crosslinking, and the material properties of APO can be tuned by adjusting the CH_2I_2 loading. Therefore, THF was later chosen as the solvent because of its higher boiling point and great solubility of APO.

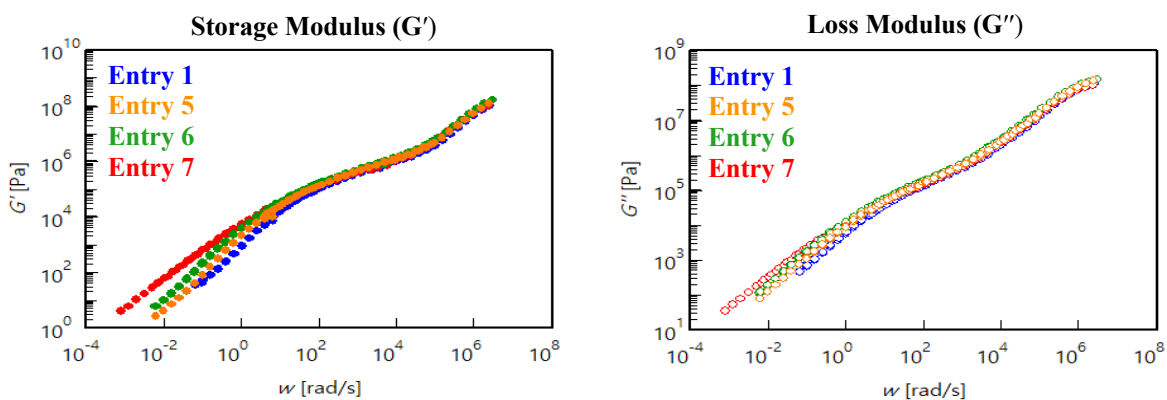


Figure 3-6. Master curves of linear and crosslinked APOs synthesized in DCM at 40 °C from frequency sweep experiments at $T_{\text{ref}} = 30$ °C.

The solvent was changed from DCM to THF based on the previous conclusion. Reaction temperatures were tested at either room temperature or 60 °C, with different loading of CH_2I_2 (10, 25, and 50 mol%) for consistency. When the reactions were performed in THF at room temperature (Table 3-4, Entries 8 to 10), only moderate increases in molecular weight (M_w) were observed in the resulting polymers. However, the changes were independent to the CH_2I_2 loadings, as all samples exhibited a M_w of approximately 30.0 Kg/mol. Similarly, the increases in T_g were nearly

constant across all samples. All of the samples have a T_g at around $-19.0\text{ }^{\circ}\text{C}$, showing no clear correlation with different CH_2I_2 loadings. The viscoelastic behaviour of the resulting polymers (Figure 3-7) also exhibited only minor variations, comparable to that of the crosslinked APO synthesized in DCM at $40\text{ }^{\circ}\text{C}$. Their storage modulus (G') increased slightly relative to the linear APO, but did not surpass the loss modulus (G'') at low frequencies, indicating that the materials retained predominantly liquid-like characteristics despite minor network formation.

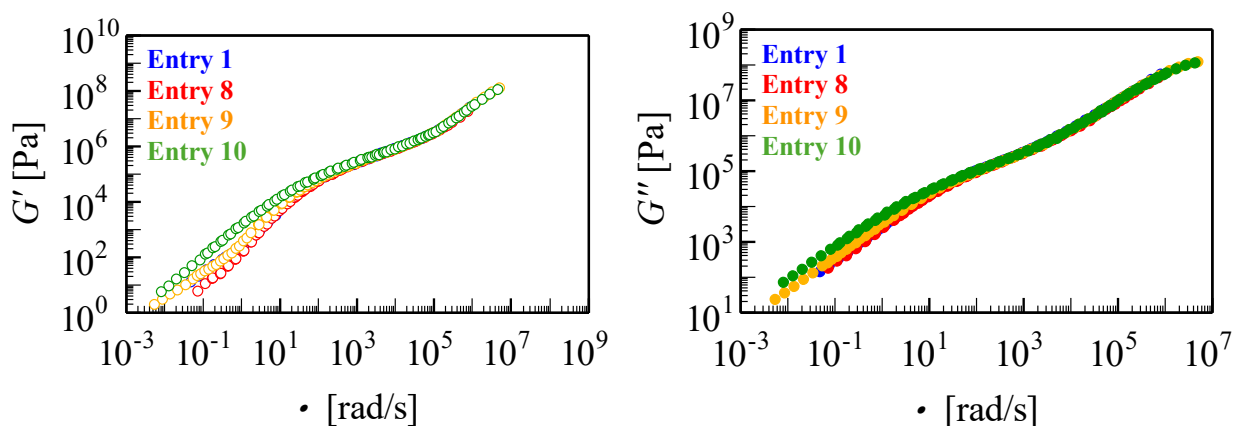


Figure 3-7. Master curves of linear and crosslinked APOs synthesized in THF at room temperature from frequency sweep experiments at $T_{\text{ref}} = 30\text{ }^{\circ}\text{C}$.

The crosslinking reaction was further evaluated in THF at $60\text{ }^{\circ}\text{C}$. The resulting polymers gelled into rubbery and elastic solids in the reaction vessels at the end of the reaction time, qualitatively confirming the effectiveness of the crosslinking process. However, none of the samples exhibited self-healing or adhesive behaviours anymore, likely due to the restricted chain mobility in the crosslinked network. The T_g increased from -12.8 to $-8.0\text{ }^{\circ}\text{C}$ with increasing CH_2I_2 loading, indicating that the thermal properties of APO can be tuned through the extent of crosslinking. The viscoelastic behaviour of the resulting polymers showed pronounced differences compared to that of the linear APO, as shown in Figure 3-8. When 10 mol% CH_2I_2 was used, the G' curve overlapped with the G'' curve at low frequencies, signifying a transition from liquid-like to more gel-like behaviour in the crosslinked APO (Entry 11). At higher CH_2I_2 loadings (25 and 50 mol%), G' became nearly independent of angular frequency in the low-frequency region, suggesting that the equilibrium modulus (G_e) was achieved. The primary plateau in the transition regime was connected by a secondary plateau immediately without a second crossover frequency, confirming the formation of an elastically dominant network structure.

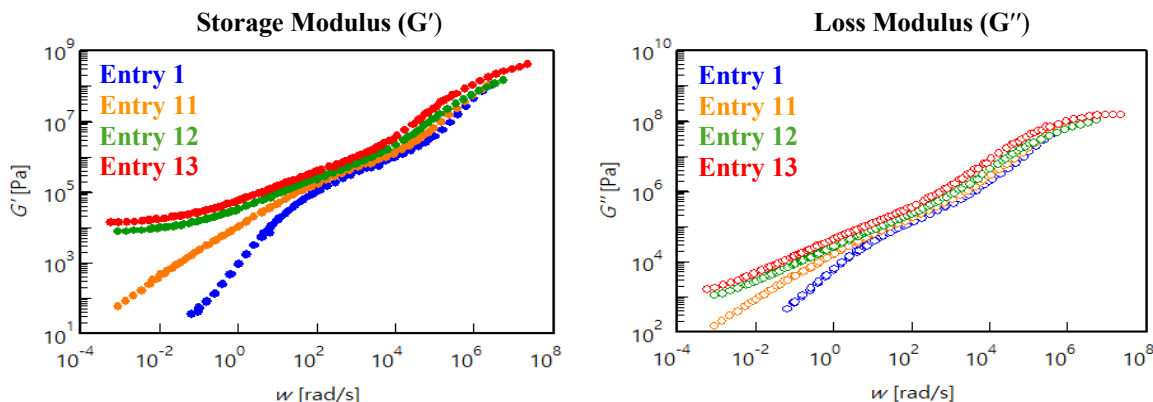


Figure 3-8. Master curves of linear and crosslinked APOs synthesized in THF at 60 °C from frequency sweep experiments at $T_{\text{ref}} = 30$ °C.

The post-polymerization modification (PPM) of APO using CH_2I_2 as a direct crosslinking agent successfully demonstrated tunable network formation and controllable thermal and viscoelastic properties. The systematic screening of reaction conditions revealed that the reaction temperature played a critical role in promoting the crosslinking reaction. Reactions carried out in DCM at room temperature showed no significant changes compared to linear APO. When elevated temperature (40 °C) was applied in DCM, slight changes were observed in their resulting M_w and T_g , suggesting partial crosslinking. However, no tunability was shown when varying the CH_2I_2 loading. The effects of crosslinking became more pronounced at 60 °C when the reaction medium was switched to THF, as the resulting materials gelled into elastic solids. The rise in T_g and the appearance of G_e in rheological measurements confirmed the formation of crosslinked networks with solid-like behaviour.

Collectively, these findings indicated that PPM using CH_2I_2 provides an effective and straightforward strategy to tailor the material properties of APO. However, the crosslinkage incorporated was not the expected DCN system. The permanent covalent crosslinking via diphenyl methylene is not cleavable nor dynamic. The loss of self-healing and adhesive functionality in these crosslinked APOs also suggests that excessive network rigidity restricts non-covalent interaction and chain mobility, which compromises the initial goal of this chapter.

3.3 Conclusion

Overall, this chapter established the groundwork for introducing crosslinking networks into APOs. Small aromatic aminal molecules were synthesized using two different C_1 sources

(formaldehyde and CO₂). *N,N'*-dimethyl-*N,N'*-diphenylmethylenediamine (**a1**) was used to evaluate the dynamic nature of the aromatic aminal bond for potential use as a DCN motif. However, the aminal linkage exhibited limited exchange behaviour, likely due to the reduced nucleophilicity of the aromatic amine nitrogen atom. Instead of demonstrating the desired dynamic exchange nature, the aminal bond was found to be cleavable by acid, suggesting its potential applicability in the design of cleavable thermosets.

An aromatic aminal species derived from ACC-NMA was successfully synthesized via CO₂ fixation, further enhancing the sustainability of the APO system. The resulting aminal crosslinker was subsequently polymerized *in-situ* with additional ACC-NMA monomers to afford an aminal-linked APO. The incorporation of aminal crosslinks led to increased rigidity, and the material became brittle at a 40 mol% crosslinking ratio, confirming the formation of a networked structure.

When attempting to generate the aminal species using CH₂I₂, an unexpected formation of 4,4'-methylene-bis-*N*-methylamine (**b1**) was observed. Moreover, CH₂I₂ proved to be an effective crosslinking agent for APO through a PPM strategy, enabling tunable thermal and viscoelastic properties. However, the resulting crosslinked APOs formed insoluble gel-like solids that exhibited loss of self-healing and adhesive properties.

Future work should focus on improving the synthesis of aminal-linked crosslinkers via CO₂ reductive functionalization by fine-tuning the CO₂ pressure, as well as optimizing the crosslinking density to enhance material performance while maintaining recyclability. This chapter establishes an important step toward the development of sustainable APO-based materials by utilizing CO₂ gas as the C₁ source. With the optimized aminal crosslinked APO network, a cleavable thermoset with dynamic non-covalent interactions could be achieved.

3.4 Experimental

3.4.1 Materials

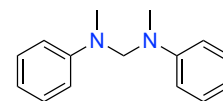
General Same as described in Section 2.4.1.

Reagents All reagents were purchased from commercial sources unless otherwise specified. *N*-methyl aniline (NMA) (Aldrich) was dried over CaH₂ overnight, purified by vacuum distillation into PTFE-sealed Straus flasks, and subsequently handled using standard Schlenk techniques. 4-methoxy-*N*-methylaniline (4-MeO NMA) (Combi-Blocks) was sublimated under vacuum at 100 °C overnight, and subsequently stored under N₂ in a glovebox. The Grubbs Catalyst™ 2nd Generation (G2) (Sigma-Aldrich) was obtained commercially, stored under N₂ atmosphere at -8 °C in a glovebox, and used without further purification. Amine-containing cyclooctene *N*-methyl aniline (ACC-NMA) was either synthesized according to literature procedures³⁰, or was provided by GreenCentre Canada. ACC-NMA was vacuum-dried overnight on a Schlenk line prior to use. Phenylsilane (Sigma-Aldrich) and 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD) were obtained commercially, stored under N₂ atmosphere in a glovebox.

Ethylene vinyl acetate-40 (EVA-40) (Sigma-Aldrich), ethyl vinyl ether (EVE) (Aldrich), *cis*-4-octene (Sigma-Aldrich), 37 wt% formaldehyde solution (Anachemia), paraformaldehyde (Alfa), diiodomethane (Sigma-Aldrich), MgSO₄ (Aldrich), *N*-*n*-butyl aniline (Sigma-Aldrich), triethyl amine (NEt₃) (Sigma-Aldrich), potassium carbonate (K₂CO₃) (Sigma-Aldrich), sodium bis(trimethylsilyl)amide (NaHMDS) (Aldrich), trifluoroacetic acid (TFA) (oakwood), 37 wt% HCl solution (Aldrich) were used as received. Solvents for air- and moisture-sensitive reactions were dried over CaH₂ overnight, purified by vacuum distillation into PTFE-sealed Straus flasks over activated 4 Å molecular sieves for 48 hours, and degassed by sparging with N₂ for 30 minutes. Solvents for non-sensitive reactions were used as received.

Synthesis and Characterization

***N,N'*-dimethyl-*N,N'*-diphenylmethylenediamine (a1).** To a vacuum-dried two-neck RBF, charged with PTFE-coated stir bar, NMA (10.0 g, 2 eq.) and 37 wt% formaldehyde_(aq) solution (4.0 mL, 1 eq.) were added under positive N₂

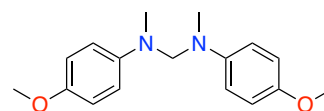


pressure. The initial amber colour solution was stirred at room temperature for 24 hours. The

reaction mixture remained amber in colour throughout the reaction time. After 24 hours, the reaction mixture was vacuum dried using a Schlenk line to remove solvent overnight. Fractional distillation under vacuum was conducted subsequently to remove residual NMA and by-products to afford an orange solid (887.0 mg, 8.4 %). **¹H NMR** (400 MHz, CDCl₃): δ 7.27 (t, ³J_{HH} = 7.40 Hz, 4H, 4 × ArH), δ 6.86 (d, ³J_{HH} = 8.40 Hz, 4H, 4 × ArH), δ 6.80 (t, ³J_{HH} = 7.56 Hz, 2H, 2 × ArH), δ 4.79 (s, 2H, CH₂), δ 2.90 (s, 6H, CH₃). **¹³C{¹H} NMR** (75 MHz, CDCl₃): δ 128.8 (CH), δ 117.8 (CH), δ 113.7 (CH), δ 70.5 (CH₂), δ 37.2 (CH₃). **GC-MS (CI)** (m/z): Calculated: 226.15; found: 226.2.

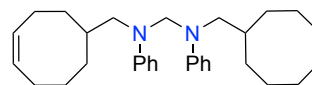
***N,N'*-bis(4-methoxyphenyl)-*N,N'*-dimethylmethylenediamine (a2).**

To a vacuum-dried two-neck RBF, charged with PTFE-coated stir bar, was added NMA (2 eq.) and paraformaldehyde (1 eq.) were added under positive N₂ pressure. The initial amber colour reaction mixture was stirred at room temperature for 24 hours. The reaction mixture remained amber colour throughout the reaction time. After 24 hours, the reaction mixture was vacuum dried using Schlenk line to remove water by-product overnight. Product mixture was used for characterization without further purification. **¹H NMR** (400 MHz, CDCl₃): δ 6.83 – 6.63 (br s, 8H, 8 × ArH), δ 4.55 (s, 2H, CH₂), δ 3.76 (s, 6H, 2 × CH₃), δ 2.82 (s, 6H, 2 × CH₃). **GC-MS (CI)** (m/z): Calculated: 286.17; found: 286.2.



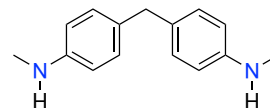
***N,N'*-diphenyl-*N,N'*-bis(4-cyclooctene-1-methyl)methylenediamine (a3).**

To a solution of TBD (5.6 mg, 5 mol%) in CD₃CN (1.0 mL) were added ACC-NMA (178.6 mg, 8 eq.), PhSiH₃ (86.6 mg, 8 eq.), and TMB (25.8 mg, 1 eq.) as internal standard. The reaction mixture was transferred into a 2.5 mL NMR tube equipped with a J. Young valve and transferred out of a glovebox. The reaction mixture underwent 3 cycles of freeze-pump-thaw to replace the N₂ atmosphere with 1 bar of CO₂. The reaction mixture was then heated to 70 °C for 2 to 4 hours to afford a light orange crude mixture. Product yield (53.2%) was determined based on the integration of internal standard. **¹H NMR** (400 MHz, CD₃CN): δ 7.18 (dd, ³J_{HH} = 7.76 Hz, ³J_{HH} = 1.28 Hz, 4H, 4 × ArH), δ 6.80 (dd, ³J_{HH} = 8.84 Hz, ³J_{HH} = 0.80 Hz, 4H, 4 × ArH), δ 6.73 (t, ³J_{HH} = 7.28 Hz, 2H, 2 × ArH), δ 5.57 (m, 4H, 2 × RHC=CHR), δ 3.05 (m, 4H, 2 × CH₂), δ 2.36 (m, 2H, 2 × CH), δ 2.14 (m, 4H, 2 × CH₂). **¹³C{¹H} NMR**



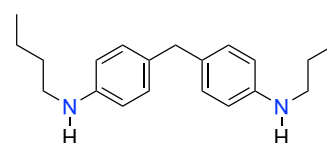
NMR (75 MHz, CD₃CN): δ 130.2 (CH), δ 129.1 (CH), δ 68.9 (CH₂), δ 56.1 (CH₂), δ 35.3 (CH), δ 33.0 (CH₂), δ 30.4 (CH₂), δ 27.8 (CH₂), δ 25.7 (CH₂), δ 24.8 (CH₂).

4,4'-methylene-bis(*N*-methylamine) (b1). To a solution of NMA (530 mg, 2 eq.) in CH₃CN (5 mL) was added CH₂I₂ (665 mg, 1 eq) under positive



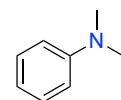
N₂ pressure. The initial transparent light-yellow solution was charged with a PTFE-coated stir bar and was stirred at 80 °C in an oil bath for 120 hours. The reaction mixture gradually turned dark purple over time. After 120 hours, the reaction mixture was cooled to room temperature. Purification was conducted via liquid-liquid extraction with saturated NaHCO_{3(aq)} (3 × 10mL), deionized water (1 × 10mL), and brine (1 × 10mL), followed by vacuum-drying at room temperature to afford dark purple oil (449.1 mg, 80.2 %). **¹H NMR** (400 MHz, CDCl₃): δ 7.00 (d, ³J_{HH} = 8.56 Hz, 4H, 4 × ArH), δ 6.55 (d, ³J_{HH} = 8.54 Hz, 4H, 4 × ArH), δ 3.78 (s, 2H, CH₂), δ 2.81 (s, 6H, CH₃). **¹³C{¹H} NMR** (75 MHz, CDCl₃): δ 129.5 (CH), δ 112.7 (CH), δ 40.9 (CH₂), δ 31.2 (CH₃). **GC-MS (CI)** (m/z): Calculated: 226.15; found: 226.2.

4,4'-methylene-bis(*N*-*n*-butylamine) (b4). To a solution of NBA (740 mg, 2 eq.) in CH₃CN (5 mL) was added CH₂I₂ (665 mg, 1 eq) under positive N₂ pressure. The initial transparent colourless solution



was charged with a PTFE-coated stir bar and was stirred at 80 °C in an oil bath for 120 hours. The reaction mixture gradually turned dark purple over time. After 120 hours, the reaction mixture was cooled to room temperature. Purification was conducted via liquid-liquid extraction with NaHCO_{3(aq)} (3 × 10mL), deionized water (1 × 10mL), and brine (1 × 10mL), followed by vacuum-drying at room temperature to afford dark purple oil (397.8 mg, 51.7 %). **¹H NMR** (400 MHz, CDCl₃): δ 7.55 (d, ³J_{HH} = 8.32 Hz, 4H, 4 × ArH), δ 7.24 (d, ³J_{HH} = 8.32 Hz, 4H, 4 × ArH), δ 3.91 (s, 2H, CH₂), δ 3.33 (t, ³J_{HH} = 8.00 Hz, 6H, 2 × CH₃), δ 1.89 (m, 4H, 2 × CH₂), δ 1.39 (m, 4H, 2 × CH₂), δ 0.91 (m, 4H, 2 × CH₂). **GC-MS (CI)** (m/z): Calculated: 310.24; found: 311.2.

***N,N*-dimethylaniline (c1).** To a solution of TBD (0.9 mg, 5 mol%) in CD₃CN (1.0 mL) were added NMA (10.3 mg, 1 eq.), and PhSiH₃ (20.5 mg, 2 eq.). The reaction

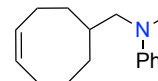


mixture was transferred into a 2.5 mL NMR tube equipped with a J. Young valve and transferred out of a glovebox. The reaction mixture underwent 3 cycles of freeze-pump-thaw to

replace the N₂ atmosphere with 1 bar of CO₂. The reaction mixture was then heated to 80 °C for 2 hours to afford a light orange crude mixture. Product yield was not determined and used for characterization directly. **¹H NMR** (400 MHz, CD₃CN): δ 7.27 (t, ³J_{HH} = 7.16 Hz, 2H, 2 × ArH), δ 6.75 (d, ³J_{HH} = 8.40 Hz, 2H, 2 × ArH), δ 6.67 (t, ³J_{HH} = 7.40 Hz, 1H, ArH), δ 2.90 (s, 6H, 2 × CH₃).

***N*-Methyl-*N*-phenyl-4-cyclooctene-1-methanamine (c2).** To a solution of TBD

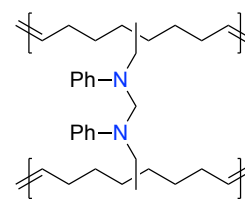
(0.9 mg, 5 mol%) in CD₃CN (1.0 mL) were added ACC-NMA (22.0 mg, 1 eq.), and PhSiH₃ (21.2 mg, 2 eq.). The reaction mixture was transferred into a 2.5 mL



NMR tube equipped with a J. Young valve and transferred out of a glovebox. The reaction mixture underwent 3 cycles of freeze-pump-thaw to replace the N₂ atmosphere with 1 bar of CO₂. The reaction mixture was then heated to 80 °C for 2 hours to afford a light orange crude mixture. Product yield was not determined and used for characterization directly. **¹H NMR** (400 MHz, CD₃CN): δ 7.19 (t, ³J_{HH} = 7.88 Hz, 2H, 2 × ArH), δ 6.71 (d, ³J_{HH} = 7.72 Hz, 2H, 2 × ArH), δ 6.63 (t, ³J_{HH} = 7.64 Hz, 1H, ArH), δ 5.66 (m, 2H, RHC=CHR), δ 2.92 (m, 3H, CH₃), δ 2.90 (m, 2H, CH₂), δ 2.36 (m, 1H, CH), δ 2.20 (m, 2H, CH₂), δ 1.91 (m, 2H, CH₂), δ 1.65 (m, 2H, CH₂), δ 1.48 (m, 2H, CH₂), δ 1.39 (m, 2H, CH₂), δ 1.18 (m, 2H, CH₂). **¹³C{¹H} NMR** (75 MHz, CD₃CN): δ 130.0 (CH), δ 128.9 (CH), δ 115.0 (CH), δ 112.0 (CH), δ 50.8 (CH₂), δ 38.7 (CH₃), δ 36.5 (CH), δ 32.8 (CH₂), δ 30.9 (CH₂), δ 30.1 (CH₂), δ 27.9 (CH₂), δ 25.6 (CH₂), δ 24.6 (CH₂). **GC-MS (CI)** (m/z): Calculated: 229.2; found: 229.4.

5 mol% amina-linked APO. To a solution of a3 (97.7 mg, 20 eq.) in DCM

(3 mL) was added ACC-NMA (950.7 mg, 400 eq.), G2 (9.4 mg, 1.0 eq.), and *cis*-4-octene (12.4 mg, 10 eq.) as chain transfer agent. The reaction



vessel was charged with a PTFE-coated stir bar, and was stirred at room temperature for 20 hours. The initial light orange solution turned dark purple solution gradually. The reaction was quenched by the addition of EVE (0.5 mL, excess) and let stirred overnight. Solvent was removed under vacuum at room temperature overnight for further characterization. **¹H NMR** (400 MHz, CDCl₃): δ 7.10 (dd, ³J_{HH} = 7.56 Hz, ³J_{HH} = 2.68 Hz, 4H, 4 × ArH), δ 6.61 (t, ³J_{HH} = 7.32 Hz, 2H, 2 × ArH), δ 6.54 (d, ³J_{HH} = 7.88 Hz, 4H, 4 × ArH), δ 5.40 (s, 2H, 2 × RHC=CHR), δ 5.36 (m, 1H, RHC=CHR), δ 4.75 (m, 0.1H, CH₂), δ 3.62 (br s, 1H, NH), δ 2.97 -

2.90 (m, 4H, 2 × CH₂), δ 2.01 – 1.35 (m, 20H, 10 × CH₂). ¹³C{¹H} NMR (75 MHz, CDCl₃): δ 130.5 (CH), δ 129.9 (CH), δ 129.1 (CH), δ 116.7 (CH), δ 112.6 (CH), δ 70.0 (CH₂), δ 51.3 (CH), δ 47.5 (CH₂), δ 35.3 (CH), δ 33.0 (CH₂), δ 32.2 (CH₂), δ 31.7 (CH₂), δ 30.0 (CH₂), δ 26.7 (CH₂), δ 24.6 (CH₂).

3.4.2 Instrumentation

NMR, GPC, DSC, and rheological measurements were the same as described in Section 2.4.2.

Gas Chromatography-Mass Spectrometry (GC-MS) Mass spectra were recorded on an Agilent 7890B with an Agilent HP-5MS column ((5%-phenyl)-methylpolysiloxane phase, 30 m length, 0.25 mm inner diameter, 0.25 μm coating thickness) coupled to a quadrupole mass filter. The instrument is equipped with an Agilent 5977 MS detector with a Chemical Ionization (CI) source of methane as the reagent gas and helium as the carrier gas. The inlet was heated to 250 °C with a total flow of 8.4 ml/min at 6.86 psi, using a 5:1 split ratio, and a split flow of 4.5 mL/min. Separation of the injected species was achieved by the following program: start at 60 °C for 1 minute, then employ a ramp of 25 °C/min until reaching 300 °C and hold for 4 minutes at the temperature. Fragment signals are given in mass per charge number (m/z). Mass scan range was between 50 to 500 m/z.

X-ray diffraction (XRD) single crystal of **a1** was mounted on MiTeGen loops. X-ray diffraction measurement was made, and data were collected using MoKα radiation (λ=0.71073 Å) or CuKα radiation (λ=1.54056 Å) on a Bruker APEX-II Duo diffractometer. The structure of **a1** was solved by using **ShelXT 2018/2** solution program using Intrinsic Phasing methods and refined by full-matrix least-squares procedures on F² (SHELXL-2018) on the OLEX2 interface.^{87,88} All non-hydrogen atoms were refined anisotropically. All hydrogen atoms were placed in calculated positions.

Chapter 4. Conclusion and Future Work

4.1 Thesis Overview

This thesis presents the development and optimization of both process-level and molecular-level approaches to improve the sustainability of aminated polyolefins (APOs). The goal was to extend APO lifetime while preserving self-healing and adhesive properties. Two different strategies were investigated in this work: physical recycling and polymer structure modification.

Chapter 2 investigated a potentially scalable physical recycling process to selectively recover APO from the APO/EVA-40 blend (80/20 wt%) using minimal organic solvent. An NMP/MeOH solvent/anti-solvent system was proven to effectively isolate APO from the blend under mild conditions with moderate to high recovery efficiency. Solvents used throughout the process were redirected back for reuse, improving the overall sustainability of the method. The recycled APO and EVA-40 were characterized comprehensively and compared to their pristine counterparts.

The recycled APO demonstrated high purity and maintained a M_w comparable to the pristine sample, confirming that the polymer structure remained intact. Thermal analyses revealed an increase in T_g attributed to stronger non-covalent interactions after the removal of EVA-40, and a lower $T_{d,5\%}$ due to trace amount of residual NMP. However, the $T_{d,50\%}$ remained comparable, supporting that the overall polymer integrity was preserved. Topographical analysis further confirmed that the surface topographies of pristine and recycled APO were largely similar, indicating minimal structural alteration. The moderate recovery percentage (72.6%) of APO from the blend was limited to the concentration of APO in NMP required for precipitation, but it could be improved by adding more APO/EVA-40 blend into the system to maintain the high concentration of APO solution.

In contrast, the recycled EVA-40 contained trace amounts of residual APO due to the saturation of APO in NMP. The presence of residual APO increased the solubility of EVA-40 in THF at room temperature and decreased its thermal stability. Interestingly, SEM imaging revealed surface aggregation and regular pattern formation on the recycled EVA-40, suggesting polymer

chain rearrangement during the physical recycling process. These topographies changes corresponded with observed increases in M_w and T_g .

Overall, EVA-40 was not restored to its pristine state using the physical recycling process conducted in this work. However, the economically valuable APO was successfully recovered with slight aging. This physical recycling process shows great potential to be scale-up due to its easy installation, reusable solvent system, and mild conditions.

Chapter 3 investigated the incorporation of dynamic covalent networks into the APO system, with the goal of chemically crosslinking the linear thermoplastic APO into a recyclable thermoset. This work was motivated by the hypothesis that partial restriction of segmental motion in APO could suppress H-bond-induced clustering, thereby extending the material's functional lifetime without sacrificing reprocessability. Aminoal-based DCNs were selected as a promising motif, as the pendant amine groups on APO can be chemically linked to form reversible aminoal crosslinks.

To test the hypothesis, small-molecule model reactions were conducted using different methylene sources, including formaldehyde, CH_2I_2 , and CO_2 , under varied reaction conditions. Aminoal-linked NMA (**a1**) was successfully synthesized through two independent synthetic routes and subsequently isolated for use in a transamination proof-of-concept experiment. Trace amounts of an asymmetric aminoal species (**a5**) were obtained via the transamination reaction in the presence of catalytic acid, supporting the intrinsic dynamic behaviour of aminoal-based DCNs. The aminoal linkage was also confirmed to be acid-labile, as introducing trace HCl readily regenerated the parent amine species, demonstrating that this linkage can serve as a chemically cleavable crosslink for APO.

Multiple synthetic routes were explored to obtain the aminoal-linked monomer ACC-NMA (**a3**), which could be polymerized via subsequent ROMP to afford an aminoal-linked APO network. **a3** was successfully synthesized using a CO_2 reductive functionalization strategy. This strategy utilized the organocatalyst TBD and an inexpensive, readily available silane reductant, thereby improving the overall sustainability of the crosslinker preparation. Using this method, 40 mol%

aminal-linked APO was obtained. However, the resulting material was rigid, brittle, and did not exhibit self-healing or adhesive properties. The T_g increased nearly 10 °C relative to the linear APO. On the other hand, 5 mol% aminal-linked APO demonstrated both self-healing and adhesive behaviour, with rheological properties comparable to that of the linear APO.

During the investigation of alternative methylene source, it was unexpectedly found that treating NMA with CH_2I_2 yielded a diphenyl methylene derivative (**b1**) in high isolated yield. This reactivity inspired a post-polymerization modification approach in which CH_2I_2 was directly reacted with linear APO. Crosslinked APOs containing 10, 25, and 50 mol% of CH_2I_2 were synthesized in THF at 60 °C, and all resulting materials became insoluble in organic solvents, indicating network formation. The thermal properties of the resulting materials showed a positive correlation with crosslinker loading, evidenced by an increase in T_g with increasing mol% of CH_2I_2 . A similar trend was observed in their viscoelastic response. The 10 mol% crosslinked polymer reached a gel point, while the 25 and 50 mol% ones displayed equilibrium modulus (G_e). G_e was characteristic when the storage modulus (G') at low angular frequencies became frequency-independent, as the network formation restricts polymer chain relaxations. The differences in their crosslinking densities suggested that the material properties could be tuned if desired.

In summary, the work in this thesis established a process-level approach to effectively recycle APO, and a molecular-level approach to change the material properties by modifying the polymer architecture. The incorporation of crosslinking networks into APO showed promising avenues for future investigation regarding the development of recyclable aminated thermoset.

4.2 Future Work

Based on the work presented in this thesis, multiple research directions can be pursued to further improve the sustainability and performance of APO materials. As demonstrated in Chapter 3, the pendant amine functionality in APO can be chemically modified to introduce dynamic covalent aminal linkages into the polymer structure. However, the reaction selectivity and yield were only moderate, so additional optimization of the reaction conditions is required. One possible approach is to vary the CO_2 pressure, making the CO_2 as the limiting reagent to favour the formation of the desired aminal species over the undesired methylated species. More efforts should

also focus on improving the purification of the aminal-linked monomer so that precise control of crosslinking density in subsequent polymerization step can be achieved.

Optimizing the balance between dynamic covalent crosslinking and the existing dynamic non-covalent interactions in APO will be critical for enhancing mechanical strength while retaining self-healing, adhesive behaviour, and reprocessability. Systematic screening of a broader range of aminal-based crosslinking densities will be essential for establishing clear structure-property relationships and identifying the optimal density for performance and recyclability. Beyond the pendant amine moieties explored in this thesis, other APO derivatives developed within the Schafer group also present opportunities for further investigation. Variations in aryl substitution on the aniline may significantly influence the electronic properties and nucleophilicity of the amine nitrogen, thereby affecting both the efficiency of CO₂ reductive functionalization and the dynamic nature of the resulting aminal linkages. Expanding the library of aminal crosslinkers derived from different aniline derivatives, followed by screening of their crosslinking densities, will provide deeper insight into how the electronic properties of amine affects macroscopic material properties in APO systems.

Moreover, stress-strain relaxation analyses and accelerated aging experiments will offer a more comprehensive understanding of the mechanical strength and long-term stability of aminal-crosslinked APO networks. Collectively, these efforts will foster the development of recyclable, mechanically tunable, and functionally versatile aminal-linked APO derivatives.

Other dynamic covalent networks, such as hindered urea bonds and urethane bonds, may also be compatible with APO system, and could also alter material performance while maintaining recyclability. In addition to DCN motifs, alternative crosslinking strategies, like metal-ligand coordination, represent promising avenues for exploration. The amine groups in APO could be used as a ligand to coordinate onto metals to form ionic crosslinked network. Moreover, blending crosslinked APO with EVA-40 to create an interpenetrated network is another compelling direction, particularly with the APO specific recycling method developed in Chapter 2.

Regarding the work presented in Chapter 2, further optimization of NMP removal from the recycled APO is necessary to mitigate the potential health risk associated with residual NMP. The long-term development of this recycling process should focus on finding alternative green solvents, such as dihydrolevoglucosenone (Cyrene) and γ -valerolactone (GVL), as replacements for NMP in the large-scale APO recycling process. Moreover, multi-cycle reprocessing should be conducted to evaluate the material durability and functionality of recycled APO after repeated physical recycling processes.

Overall, continued exploration of CO₂-derived aminated crosslinking strategies, alternative crosslinking designs, and different blending components holds strong potential to advance aminated polyolefins toward practical, recyclable, and high-performance adhesive systems.

References

- (1) Government of Canada, S. C. *Physical flow account for plastic material, by resin*. <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3810015101> (accessed 2025-10-11).
- (2) Government of Canada, S. C. *Physical Flow Account for Plastic Material, by product category*. <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3810015001> (accessed 2025-10-11).
- (3) Jiao, H.; Ali, S. S.; Alsharbaty, M. H. M.; Elsamahy, T.; Abdelkarim, E.; Schagerl, M.; Al-Tohamy, R.; Sun, J. A Critical Review on Plastic Waste Life Cycle Assessment and Management: Challenges, Research Gaps, and Future Perspectives. *Ecotoxicol. Environ. Saf.* **2024**, *271*, 115942.
- (4) *Economic Study of the Canadian Plastic Industry, Markets and Waste*; Environment and Climate Change Canada = Environnement et changement climatique Canada: Gatineau, QC, 2019.
- (5) Naderi Kalali, E.; Lotfian, S.; Entezar Shabestari, M.; Khayatzadeh, S.; Zhao, C.; Yazdani Nezhad, H. A Critical Review of the Current Progress of Plastic Waste Recycling Technology in Structural Materials. *Curr. Opin. Green Sustain. Chem.* **2023**, *40*, 100763.
- (6) Zheng, J.; Suh, S. Strategies to Reduce the Global Carbon Footprint of Plastics. *Nat. Clim. Change* **2019**, *9* (5), 374–378.
- (7) Garcia, J. M.; Robertson, M. L. The Future of Plastics Recycling. *Science* **2017**, *358* (6365), 870–872.
- (8) Ragaert, K.; Delva, L.; Van Geem, K. Mechanical and Chemical Recycling of Solid Plastic Waste. *Waste Manag.* **2017**, *69*, 24–58. <https://doi.org/10.1016/j.wasman.2017.07.044>.
- (9) Singh, N.; Walker, T. R. Plastic Recycling: A Panacea or Environmental Pollution Problem. *Npj Mater. Sustain.* **2024**, *2* (1), 17.
- (10) Al-Salem, S. M.; Lettieri, P.; Baeyens, J. Recycling and Recovery Routes of Plastic Solid Waste (PSW): A Review. *Waste Manag.* **2009**, *29* (10), 2625–2643.
- (11) Hirschberg, V.; Rodrigue, D. Recycling of Polyamides: Processes and Conditions. *J. Polym. Sci.* **2023**, *61* (17), 1937–1958.
- (12) Beyler, C.; Hirschler, M. Thermal Decomposition of Polymers. In *SFPE handbook of fire protection engineering*; 2002; Vol. 7, pp 111–131.
- (13) Bujak, J. W. Thermal Utilization (Treatment) of Plastic Waste. *Energy* **2015**, *90*, 1468–1477.

- (14) Baca, D.; Monroy, R.; Castillo, M.; Elkhazraji, A.; Farooq, A.; Ahmad, R. Dioxins and Plastic Waste: A Scientometric Analysis and Systematic Literature Review of the Detection Methods. *Environ. Adv.* **2023**, *13*, 100439.
- (15) Mangold, H.; von Vacano, B. The Frontier of Plastics Recycling: Rethinking Waste as a Resource for High-Value Applications. *Macromol. Chem. Phys.* **2022**, *223* (13), 2100488.
- (16) Yue, S.; Wang, P.; Yu, B.; Zhang, T.; Zhao, Z.; Li, Y.; Zhan, S. From Plastic Waste to Treasure: Selective Upcycling through Catalytic Technologies. *Adv. Energy Mater.* **2023**, *13* (41), 2302008.
- (17) Cheng, L.; Chen, X.; Gu, J.; Kobayashi, N.; Yuan, H.; Chen, Y. Chemical Recycling of Waste Plastics: Current Challenges and Perspectives. *Fundam. Res.* **2024**, *5* (3), 919–922.
- (18) Zhao, Y.-B.; Lv, X.-D.; Ni, H.-G. Solvent-Based Separation and Recycling of Waste Plastics: A Review. *Chemosphere* **2018**, *209*, 707–720.
- (19) Vollmer, I.; Jenks, M. J. F.; Roelands, M. C. P.; White, R. J.; van Harmelen, T.; de Wild, P.; van der Laan, G. P.; Meirer, F.; Keurentjes, J. T. F.; Weckhuysen, B. M. Beyond Mechanical Recycling: Giving New Life to Plastic Waste. *Angew. Chem. Int. Ed.* **2020**, *59* (36), 15402–15423.
- (20) Sherwood, J. Closed-Loop Recycling of Polymers Using Solvents: Remaking Plastics for a Circular Economy. *Johns. Matthey Technol. Rev.* **2020**, *64* (1), 4–15.
- (21) Yousaf, A.; Al Rashid, A.; Polat, R.; Koç, M. Potential and Challenges of Recycled Polymer Plastics and Natural Waste Materials for Additive Manufacturing. *Sustain. Mater. Technol.* **2024**, *41*, e01103.
- (22) Morici, E.; Dintcheva, N. Tz. Recycling of Thermoset Materials and Thermoset-Based Composites: Challenge and Opportunity. *Polymers* **2022**, *14* (19), 4153.
- (23) Hopewell, J.; Dvorak, R.; Kosior, E. Plastics Recycling: Challenges and Opportunities. *Philos. Trans. R. Soc. B Biol. Sci.* **2009**, *364* (1526), 2115–2126.
- (24) Ataie, S.; Schafer, L. L. Hydroaminoalkylation: A Tool of Choice for the Catalytic Addition of Amines to Alkenes in Small Molecules and Materials. *Acc. Chem. Res.* **2025**, *58* (19), 2956–2969.
- (25) Scott, S. S.; Kaur, B.; Zheng, C. H. M.; Brant, P.; Gilmour, D. J.; Schafer, L. L. Amine-Functionalized Polybutadiene Synthesis by Tunable Postpolymerization Hydroaminoalkylation. *J. Am. Chem. Soc.* **2023**, *145* (42), 22871–22877.

- (26) Pan, S.; Matsuo, Y.; Endo, K.; Shibata, T. Cationic Iridium-Catalyzed Enantioselective Activation of Secondary Sp³ C–H Bond Adjacent to Nitrogen Atom. *Tetrahedron* **2012**, *68* (44), 9009–9015.
- (27) Jun, C.-H. Chelation-Assisted Alkylation of Benzylamine Derivatives by Ru⁰ Catalyst. *Chem. Commun.* **1998**, No. 13, 1405–1406.
- (28) Yao, W.-W.; Li, R.; Chen, H.; Chen, M.-K.; Luan, Y.-X.; Wang, Y.; Yu, Z.-X.; Ye, M. Ni-Catalyzed Hydroaminoalkylation of Alkynes with Amines. *Nat. Commun.* **2021**, *12* (1), 3800.
- (29) Perry, M. R.; Ebrahimi, T.; Morgan, E.; Edwards, P. M.; Hatzikiriakos, S. G.; Schafer, L. L. Catalytic Synthesis of Secondary Amine-Containing Polymers: Variable Hydrogen Bonding for Tunable Rheological Properties. *Macromolecules* **2016**, *49* (12), 4423–4430.
- (30) Gilmour, D. J.; Tomkovic, T.; Kuanr, N.; Perry, M. R.; Gildenast, H.; Hatzikiriakos, S. G.; Schafer, L. L. Catalytic Amine Functionalization and Polymerization of Cyclic Alkenes Creates Adhesive and Self-Healing Materials. *ACS Appl. Polym. Mater.* **2021**, *3* (5), 2330–2335.
- (31) Scott, S. S.; Roşca, S.-C.; Gilmour, D. J.; Brant, P.; Schafer, L. L. Commodity Polymers to Functional Aminated Materials: Single-Step and Atom-Economic Synthesis by Hydroaminoalkylation. *ACS Macro Lett.* **2021**, *10* (10), 1266–1272.
- (32) Ataie, S.; Malmir, A.; Scott, S. S.; Goettel, J. T.; Clemens, S. N.; Morrison, D. J.; Mackie, C.; Heyne, B.; Hatzikiriakos, S. G.; Schafer, L. L. Hydroaminoalkylation for Amine Functionalization of Vinyl-Terminated Polyethylene Enables Direct Access to Responsive Functional Materials. *Angew. Chem. Int. Ed.* **2024**, *63* (49), e202410154.
- (33) Zhang, Z.; Yavitt, B. M.; Mohammadi, E.; Schafer, L. L.; Hatzikiriakos, S. G. Rheology and Aging of Amine Functionalized Polyolefins. *J. Rheol.* **2024**, *68* (6), 933–947.
- (34) Samir, A.; Ashour, F. H.; Hakim, A. A. A.; Bassyouni, M. Recent Advances in Biodegradable Polymers for Sustainable Applications. *Npj Mater. Degrad.* **2022**, *6* (1), 68.
- (35) Winne, J. M.; Leibler, L.; Du Prez, F. E. Dynamic Covalent Chemistry in Polymer Networks: A Mechanistic Perspective. *Polym. Chem.* **2019**, *10* (45), 6091–6108.
- (36) Glaskova-Kuzmina, T.; Starkova, O.; Gaidukovs, S.; Platnieks, O.; Gaidukova, G. Durability of Biodegradable Polymer Nanocomposites. *Polymers* **2021**, *13* (19), 3375.

- (37) Chamas, A.; Moon, H.; Zheng, J.; Qiu, Y.; Tabassum, T.; Jang, J. H.; Abu-Omar, M.; Scott, S. L.; Suh, S. Degradation Rates of Plastics in the Environment. *ACS Sustain. Chem. Eng.* **2020**, *8* (9), 3494–3511.
- (38) Nenadović, M.; Ponjavić, M.; Pantelic, B.; Guzik, M.; Majka, T. M.; Sourkouni, G.; Maršavelski, A.; Nikodinovic-Runic, J. Efficient Degradation of Consumer-Grade PLA by Commercial Savinase: Optimized Conditions and Molecular Dynamics Insights. *ACS Sustain. Chem. Eng.* **2025**, *13* (24), 9269–9278.
- (39) Sharma, H.; Rana, S.; Singh, P.; Hayashi, M.; Binder, W. H.; Rossegger, E.; Kumar, A.; Schlögl, S. Self-Healable Fiber-Reinforced Vitrimer Composites: Overview and Future Prospects. *RSC Adv.* **2022**, *12* (50), 32569–32582.
- (40) Tiz, D. B.; Vicente, F. A.; Kroflič, A.; Likozar, B. Lignin-Based Covalent Adaptable Network Polymers—When Bio-Based Thermosets Meet Recyclable by Design. *ACS Sustain. Chem. Eng.* **2023**, *11* (38), 13836–13867.
- (41) Montarnal, D.; Capelot, M.; Tournilhac, F.; Leibler, L. Silica-Like Malleable Materials from Permanent Organic Networks. *Science* **2011**, *334* (6058), 965–968.
- (42) Hammer, L.; Van Zee, N. J.; Nicolaÿ, R. Dually Crosslinked Polymer Networks Incorporating Dynamic Covalent Bonds. *Polymers* **2021**, *13* (3), 396.
- (43) Tajima, R.; Nakagawa, S.; Yoshie, N. Multiple Hydrogen Bonds as Tools to Enhance the Mechanical and Mechanoresponsive Properties of Polymers. *Polym. J.* **2025**, 1–13.
- (44) Moradinik, N.; Mohammadi, E.; Yavitt, B. M.; Schafer, L. L.; Hatzikiriakos, S. G. Impact of Quenching Agents on Internal Associations and Structure of Amine-Functionalized Polyolefins. *ACS Appl. Polym. Mater.* **2025**, *7* (12), 7836–7847.
- (45) Chen, N.; Zhang, J. The Role of Hydrogen-Bonding Interaction in Poly(Vinyl Alcohol)/Poly(Acrylic Acid) Blending Solutions and Their Films. *Chin. J. Polym. Sci.* **2010**, *28* (6), 903–911.
- (46) Yamada, Y.; Kakinuma, S.; Hanayama, H.; Datta, S.; Yagai, S. Competition and Cooperation between Steric Hindrance and Hydrogen Bonding of Supramolecular Discs. *Org. Biomol. Chem.* **2025**, *23* (27), 6516–6520.
- (47) Liu, J.; Qiao, J.; Zhou, K.; Wang, J.; Gui, R.; Xian, K.; Gao, M.; Yin, H.; Hao, X.; Zhou, Z.; Ye, L. An Aggregation-Suppressed Polymer Blending Strategy Enables High-Performance Organic and Quantum Dot Hybrid Solar Cells. *Small* **2022**, *18* (19), 2201387.

- (48) Huang, S.; Zhang, Y.; Wang, C.; Xia, Q.; Saif Ur Rahman, M.; Chen, H.; Han, C.; Liu, Y.; Xu, S. Mechanisms Affecting Physical Aging and Swelling by Blending an Amphiphilic Component. *Int. J. Mol. Sci.* **2022**, *23* (4), 2185.
- (49) Slonov, A.; Musov, I.; Zhansitov, A.; Khashirov, A.; Tlupov, A.; Musov, K.; Rzhetskaya, E.; Fomicheva, I.; Potapov, A.; Khashirova, S. Investigation of the Properties of Polyethylene and Ethylene-Vinyl Acetate Copolymer Blends for 3D Printing Applications. *Polymers* **2023**, *15* (20), 4129.
- (50) Khairutdinov, V.; Khabriev, I.; Khuzakhanov, R.; Garipov, R.; Yarullin, L.; Abdulagatov, I. Effects of the Blend Ratio and Operating Conditions (T and P) on the Thermal and Mechanical Properties of Polymer (EVA-113/Nitrile-Butadiene Rubber) Blends Prepared by Supercritical CO₂. *J. Chem. Eng. Data* **2024**, *69* (11), 3872–3885.
- (51) Nayanathara Thathsarani Pilapitiya, P. G. C.; Ratnayake, A. S. The World of Plastic Waste: A Review. *Clean. Mater.* **2024**, *11*, 100220. <https://doi.org/10.1016/j.clema.2024.100220>.
- (52) Katritzky, A. R.; Fara, D. C.; Yang, H.; Tamm, K.; Tamm, T.; Karelson, M. Quantitative Measures of Solvent Polarity. *Chem. Rev.* **2004**, *104* (1), 175–198.
- (53) Reichardt, C. Solvatochromic Dyes as Solvent Polarity Indicators. *Chem. Rev.* **1994**, *94* (8), 2319–2358.
- (54) Snyder, L. R. Classification of the Solvent Properties of Common Liquids. *J. Chromatogr. Sci.* **1978**, *16* (6), 223–234.
- (55) Kuanr, N.; Stoševski, I.; Wilkinson, D. P.; Schafer, L. L. Synthesis of Aminopolyolefins with Tunable Hydrophilicity and Chain Flexibility for Application as Binders in Aqueous Rechargeable Batteries. *ACS Appl. Energy Mater.* **2024**, *7* (8), 3433–3442.
- (56) Ahmine, A. N.; Bdiri, M.; Féréol, S.; Fodil, R. A Comprehensive Study of AFM Stiffness Measurements on Inclined Surfaces: Theoretical, Numerical, and Experimental Evaluation Using a Hertz Approach. *Sci. Rep.* **2024**, *14* (1), 25869.
- (57) Jouyban, A.; Fakhree, M. A. A.; Shayanfar, A. Review of Pharmaceutical Applications of N-Methyl-2-Pyrrolidone. *J. Pharm. Pharm. Sci.* **2010**, *13* (4), 524–535.
- (58) Germán, L.; Cuevas, J. M.; Cobos, R.; Pérez-Alvarez, L.; Vilas-Vilela, J. L. Green Alternative Cosolvents to N-Methyl-2-Pyrrolidone in Water Polyurethane Dispersions. *RSC Adv.* **2021**, *11* (31), 19070–19075.

- (59) Zheng, N.; Xu, Y.; Zhao, Q.; Xie, T. Dynamic Covalent Polymer Networks: A Molecular Platform for Designing Functions beyond Chemical Recycling and Self-Healing. *Chem. Rev.* **2021**, *121* (3), 1716–1745.
- (60) Chen, X.; Dam, M. A.; Ono, K.; Mal, A.; Shen, H.; Nutt, S. R.; Sheran, K.; Wudl, F. A Thermally Re-Mendable Cross-Linked Polymeric Material. *Science* **2002**, *295* (5560), 1698–1702.
- (61) Billiet, S.; De Bruycker, K.; Driessen, F.; Goossens, H.; Van Speybroeck, V.; Winne, J. M.; Du Prez, F. E. Triazolinediones Enable Ultrafast and Reversible Click Chemistry for the Design of Dynamic Polymer Systems. *Nat. Chem.* **2014**, *6* (9), 815–821.
- (62) Zhang, Y.; Ying, H.; Hart, K. R.; Wu, Y.; Hsu, A. J.; Coppola, A. M.; Kim, T. A.; Yang, K.; Sottos, N. R.; White, S. R.; Cheng, J. Malleable and Recyclable Poly(Urea-Urethane) Thermosets Bearing Hindered Urea Bonds. *Adv. Mater.* **2016**, *28* (35), 7646–7651.
- (63) Liu, W.-X.; Zhang, C.; Zhang, H.; Zhao, N.; Yu, Z.-X.; Xu, J. Oxime-Based and Catalyst-Free Dynamic Covalent Polyurethanes. *J. Am. Chem. Soc.* **2017**, *139* (25), 8678–8684.
- (64) Jin, K.; Li, L.; Torkelson, J. M. Recyclable Crosslinked Polymer Networks via One-Step Controlled Radical Polymerization. *Adv. Mater.* **2016**, *28* (31), 6746–6750.
- (65) Denissen, W.; Rivero, G.; Nicolaÿ, R.; Leibler, L.; Winne, J. M.; Du Prez, F. E. Vinylogous Urethane Vitrimers. *Adv. Funct. Mater.* **2015**, *25* (16), 2451–2457.
- (66) Taynton, P.; Yu, K.; Shoemaker, R. K.; Jin, Y.; Qi, H. J.; Zhang, W. Heat- or Water-Driven Malleability in a Highly Recyclable Covalent Network Polymer. *Adv. Mater.* **2014**, *26* (23), 3938–3942.
- (67) Cacciapaglia, R.; Di Stefano, S.; Mandolini, L. Metathesis Reaction of Formaldehyde Acetals: An Easy Entry into the Dynamic Covalent Chemistry of Cyclophane Formation. *J. Am. Chem. Soc.* **2005**, *127* (39), 13666–13671.
- (68) Hendriks, B.; Waelkens, J.; Winne, J. M.; Du Prez, F. E. Poly(Thioether) Vitrimers via Transalkylation of Trialkylsulfonium Salts. *ACS Macro Lett.* **2017**, *6* (9), 930–934.
- (69) Rekondo, A.; Martin, R.; Ruiz De Luzuriaga, A.; Cabañero, G.; Grande, H. J.; Odriozola, I. Catalyst-Free Room-Temperature Self-Healing Elastomers Based on Aromatic Disulfide Metathesis. *Mater Horiz* **2014**, *1* (2), 237–240.
- (70) Mao, J.; Wang, J.; Yan, L.; Cao, Y.; Huang, M.; Chen, W.; Chen, K.; Liang, Y.; Chen, W.; Wang, Y.; He, S.; Liu, W.; Wang, Z. Synergistic Dual Dynamic Covalent Networks Enable

- High-Performance and Self-Healing Polyurethane and Its Anticorrosion Applications. *Macromolecules* **2025**, 58 (15), 8053–8066.
- (71) Chen, M.; Zhou, L.; Wu, Y.; Zhao, X.; Zhang, Y. Rapid Stress Relaxation and Moderate Temperature of Malleability Enabled by the Synergy of Disulfide Metathesis and Carboxylate Transesterification in Epoxy Vitrimers. *ACS Macro Lett.* **2019**, 8 (3), 255–260.
- (72) Neal, J. A.; Mozhdghi, D.; Guan, Z. Enhancing Mechanical Performance of a Covalent Self-Healing Material by Sacrificial Noncovalent Bonds. *J. Am. Chem. Soc.* **2015**, 137 (14), 4846–4850.
- (73) De Alwis Watuthantrige, N.; Dunn, D.; Dolan, M.; Sparks, J. L.; Ye, Z.; Zanjani, M. B.; Konkolewicz, D. Tuning Dual-Dynamic Network Materials through Polymer Architectural Features. *ACS Appl. Polym. Mater.* **2022**, 4 (2), 1475–1486.
- (74) Wang, S.; Ma, S.; Li, Q.; Xu, X.; Wang, B.; Huang, K.; Liu, Y.; Zhu, J. Facile Preparation of Polyimine Vitrimers with Enhanced Creep Resistance and Thermal and Mechanical Properties via Metal Coordination. *Macromolecules* **2020**, 53 (8), 2919–2931.
- (75) Ying, H.; Zhang, Y.; Cheng, J. Dynamic Urea Bond for the Design of Reversible and Self-Healing Polymers. *Nat. Commun.* **2014**, 5 (1), 3218. <https://doi.org/10.1038/ncomms4218>.
- (76) Ying, H.; Cheng, J. Hydrolyzable Polyureas Bearing Hindered Urea Bonds. *J. Am. Chem. Soc.* **2014**, 136 (49), 16974–16977.
- (77) Zhang, Q.; Wang, S.; Rao, B.; Chen, X.; Ma, L.; Cui, C.; Zhong, Q.; Li, Z.; Cheng, Y.; Zhang, Y. Hindered Urea Bonds for Dynamic Polymers: An Overview. *React. Funct. Polym.* **2021**, 159, 104807.
- (78) Chao, A.; Zhang, D. Investigation of Secondary Amine-Derived Amino Bond Exchange toward the Development of Covalent Adaptable Networks. *Macromolecules* **2019**, 52 (2), 495–503.
- (79) Tang, Z.-H.; Zeng, H.; Wei, S.-Q.; Wu, S.-W.; Guo, B.-C. Structural Manipulation of Amino-Crosslinked Polybutadiene for Recyclable and Healable Elastomers. *Chin. J. Polym. Sci.* **2021**, 39 (10), 1337–1344.
- (80) Shen, Y.; Xu, N.; Adrar, Y. A.; Wang, B.; Liu, Y.; Yuan, W.; Xu, X.; Huang, Y.; Hu, Z. Imine or Secondary Amine-Derived Degradable Polyaminal: Low-Cost Matrix Resin with High Performance. *ACS Sustain. Chem. Eng.* **2020**, 8 (4), 1943–1953.

- (81) Carbajo, D.; Ruiz-Sánchez, A. J.; Nájera, F.; Pérez-Inestrosa, E.; Alfonso, I. Spontaneous Macrocyclization through Multiple Dynamic Cyclic Amino Formation. *Chem. Commun.* **2021**, 57 (10), 1190–1193.
- (82) Caliri, M.; Teotonico, J.; Irigoyen, M.; Mujika, A.; Isolabella, T.; Mantione, D.; Irusta, L.; Grignard, B.; Vidal, F.; Detrembleur, C.; Sardon, H. A Material Platform Based on Dissociative CO₂-Derived *N,O*-Acetals for Tunable Degradation of 3D Printable Materials. *J. Am. Chem. Soc.* **2025**, 147 (33), 30095–30106.
- (83) Frogneux, X.; Blondiaux, E.; Thuéry, P.; Cantat, T. Bridging Amines with CO₂: Organocatalyzed Reduction of CO₂ to Aminoamines. *ACS Catal.* **2015**, 5 (7), 3983–3987.
- (84) Mayer, R. J.; Hampel, N.; Mayer, P.; Ofial, A. R.; Mayr, H. Synthesis, Structure, and Properties of Amino-Substituted Benzhydrylium Ions – A Link between Ordinary Carbocations and Neutral Electrophiles. *Eur. J. Org. Chem.* **2019**, 2019 (2–3), 412–421.
- (85) Das Neves Gomes, C.; Jacquet, O.; Villiers, C.; Thuéry, P.; Ephritikhine, M.; Cantat, T. A Diagonal Approach to Chemical Recycling of Carbon Dioxide: Organocatalytic Transformation for the Reductive Functionalization of CO₂. *Angew. Chem. Int. Ed.* **2012**, 51 (1), 187–190.
- (86) Liu, X.-F.; Li, X.-Y.; Qiao, C.; Fu, H.-C.; He, L.-N. Betaine Catalysis for Hierarchical Reduction of CO₂ with Amines and Hydrosilane To Form Formamides, Aminoamines, and Methylamines. *Angew. Chem. Int. Ed.* **2017**, 56 (26), 7425–7429.
- (87) Dolomanov, O. V.; Bourhis, L. J.; Gildea, R. J.; Howard, J. a. K.; Puschmann, H. OLEX2: A Complete Structure Solution, Refinement and Analysis Program. *J. Appl. Crystallogr.* **2009**, 42 (2), 339–341.
- (88) Sheldrick, G. M. Crystal Structure Refinement with SHELXL. *Acta Crystallogr. Sect. C Struct. Chem.* **2015**, 71 (1), 3–8.

Appendices

Figure S2-1. GPC chromatogram for Table 2-2 Entry 2 (recycled APO).

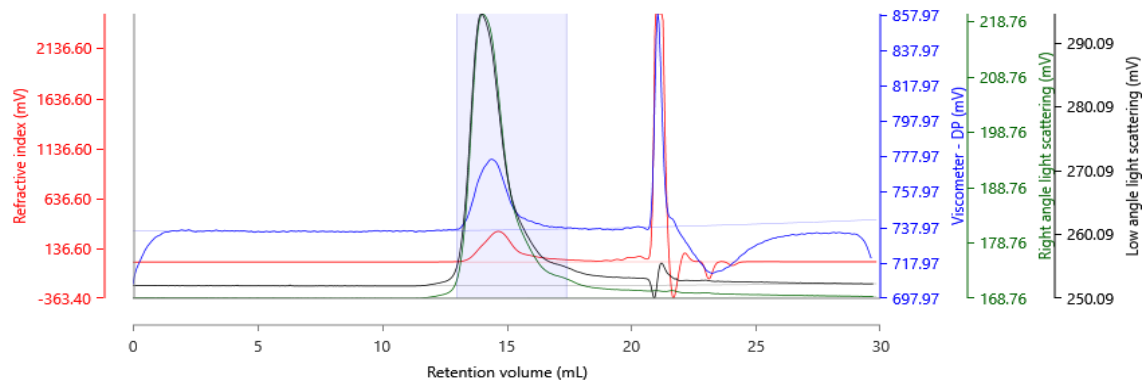


Figure S2-2. DSC thermogram for Table 2-2 Entries 1 and 2 (pristine and recycled APO).

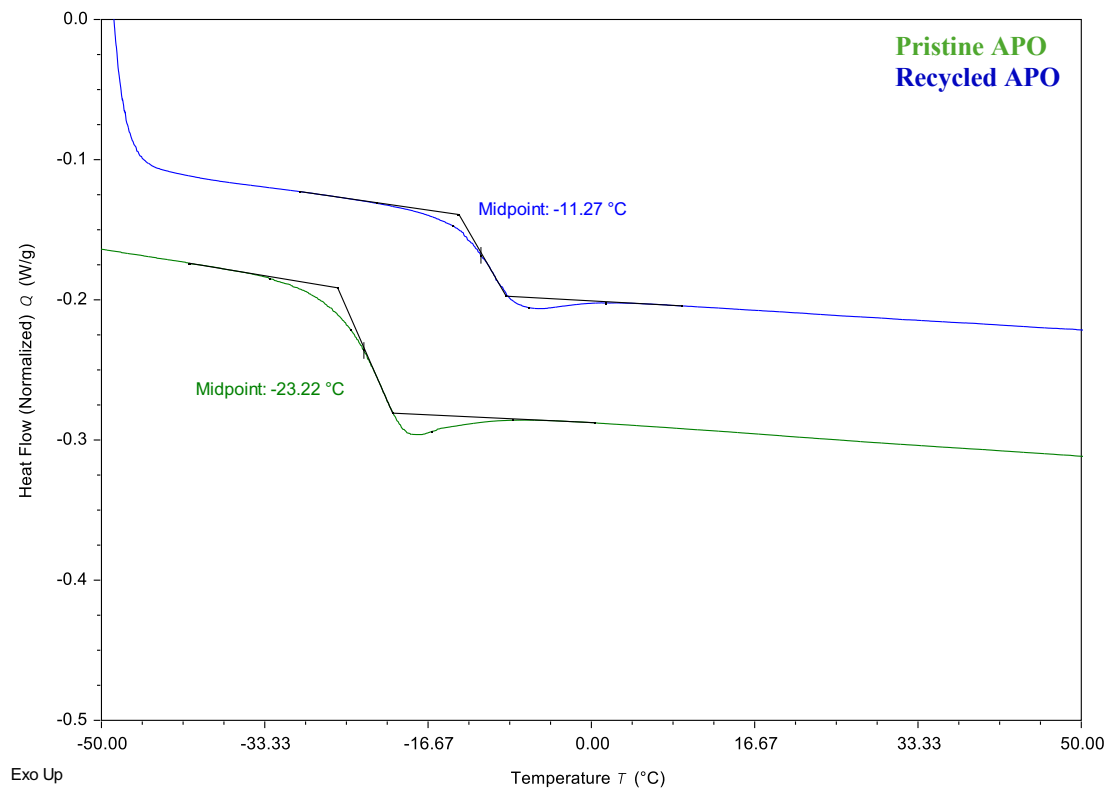


Figure S2-3. GPC chromatogram for Table 2-3 Entry 2 (recycled EVA-40).

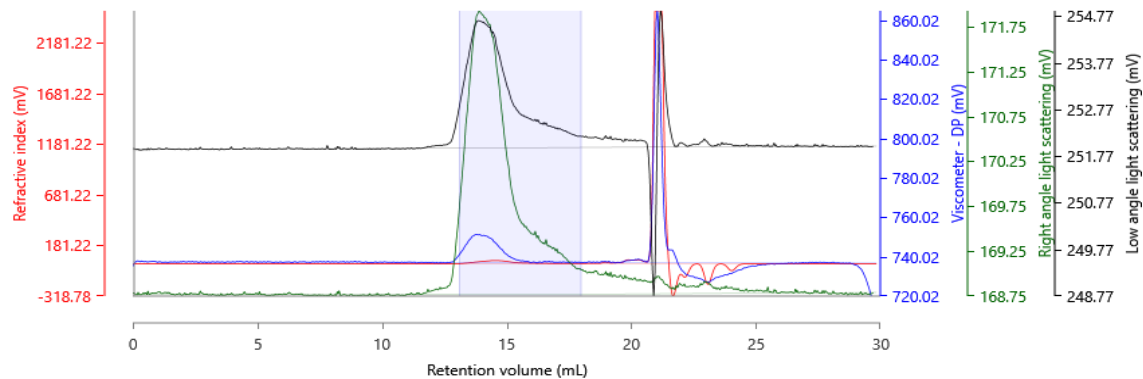


Figure S2-4 GPC chromatogram for Table 2-3 Entry 3 (80/20 wt% APO/EVA-40).

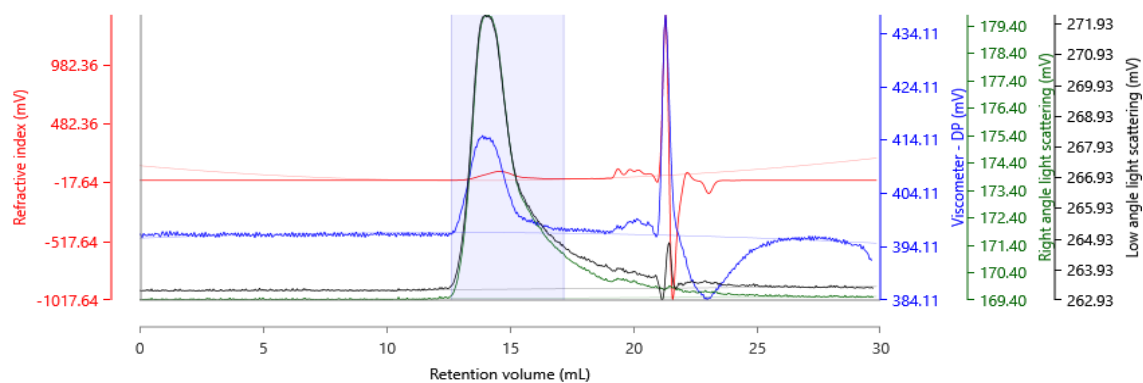


Figure S2-5. DSC thermogram for Table 2-2 Entries 1 and 2 (pristine EVA-40 and recycled EVA-40).

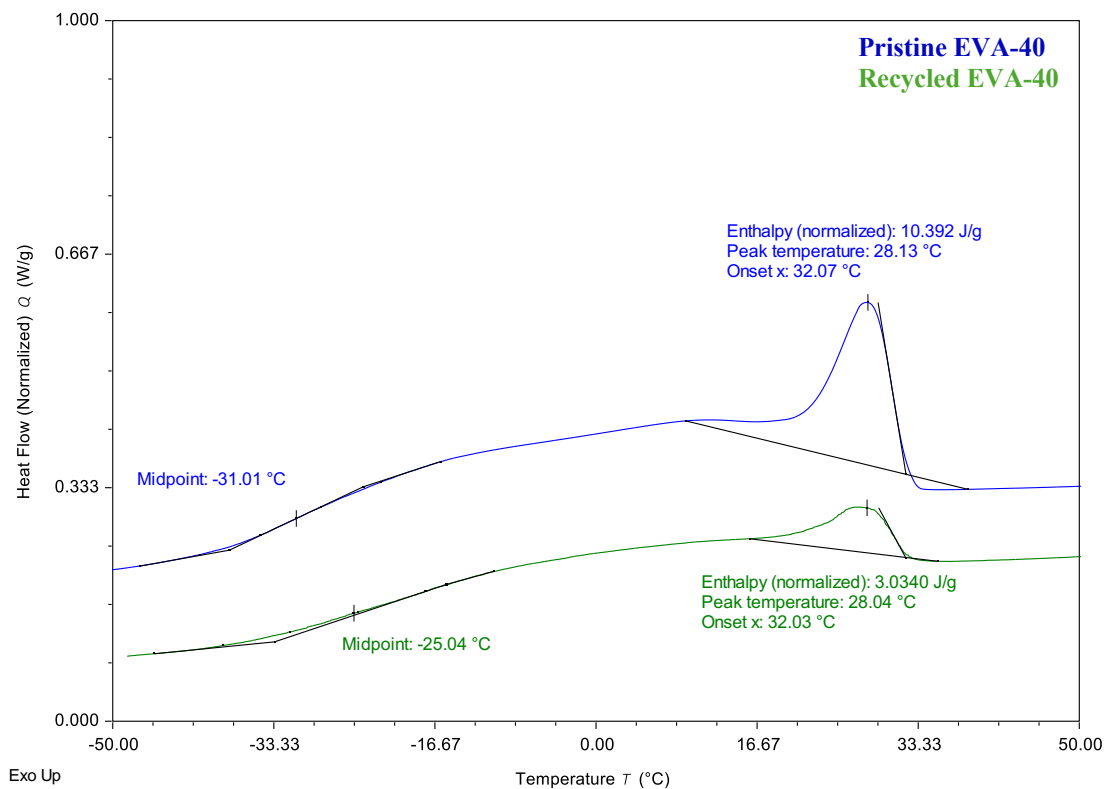


Figure S3-1. GC Chromatogram for Table 3-1 Entry 1.

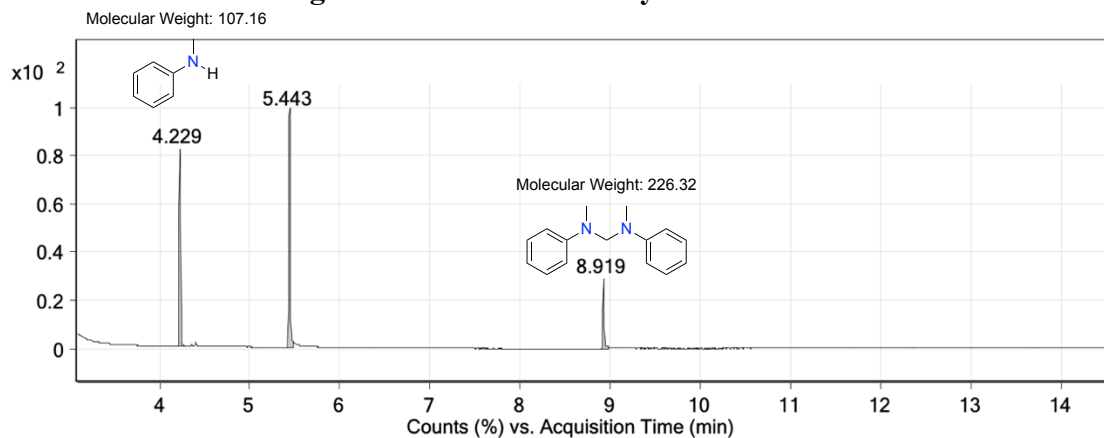


Figure S3-2. ^1H NMR spectrum of Table 3-1 Entry 1 (400 MHz, CDCl_3 , 298K).

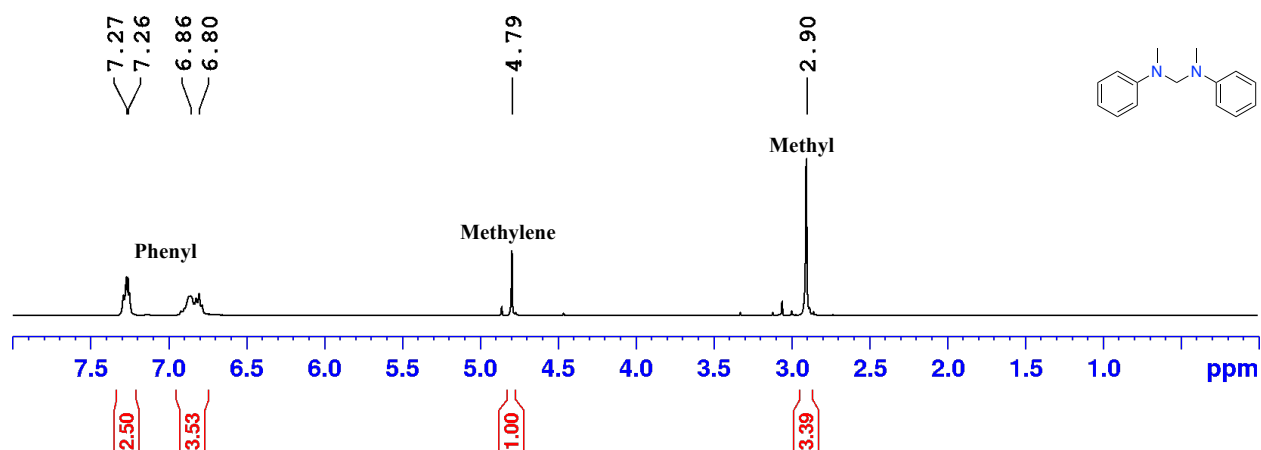


Figure S3-3. HSQC NMR $\{^1\text{H}, ^{13}\text{C}\}$ spectrum of Table 3-1 Entry 1 (400 MHz, CDCl_3 , 298K).

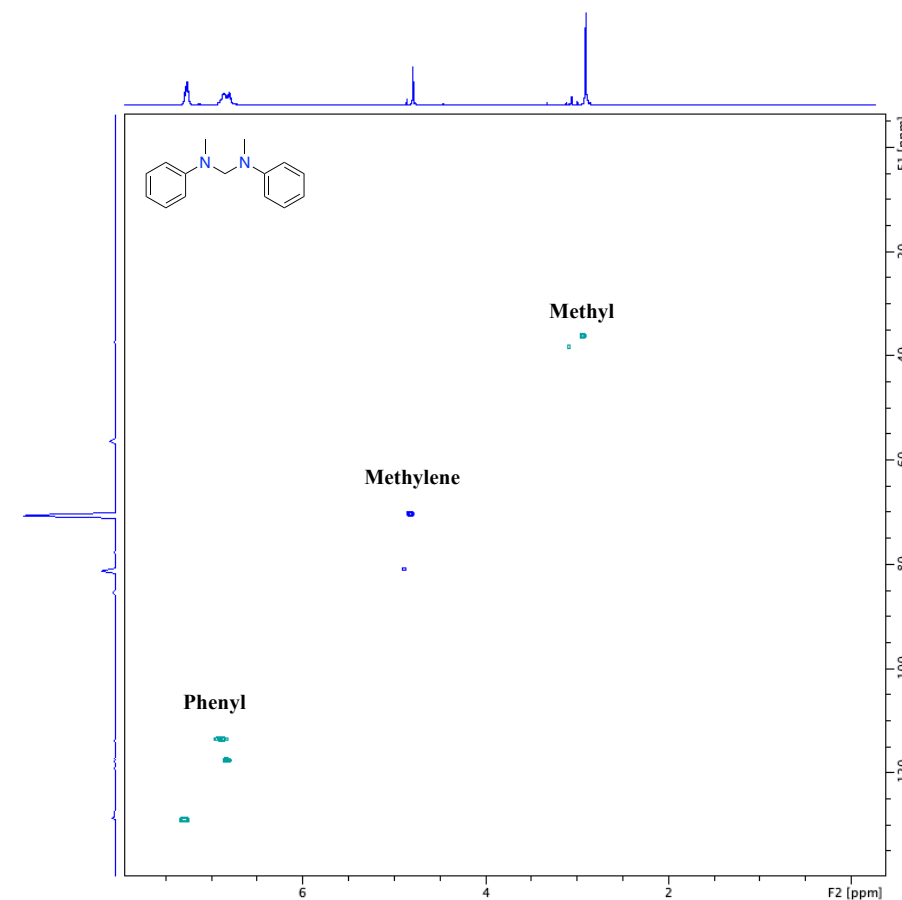
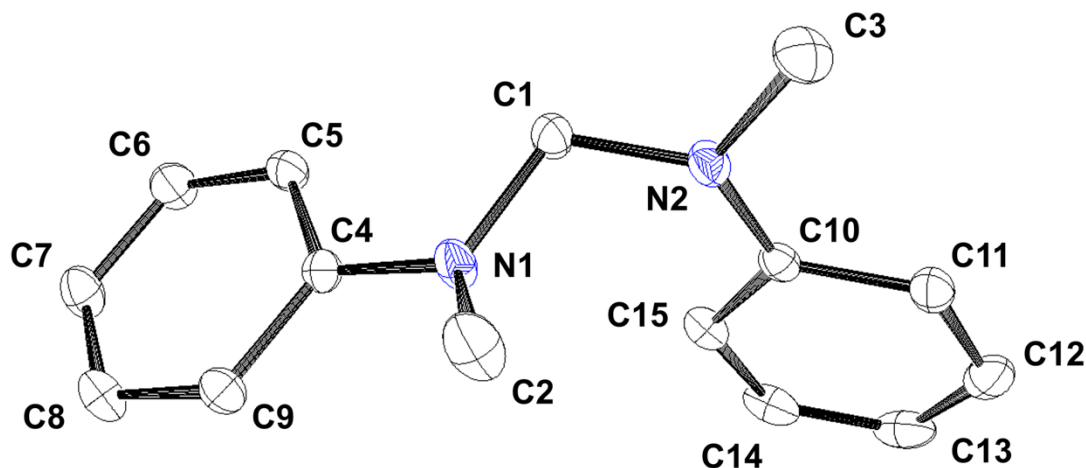


Figure S3-4. Crystal structure of **a1**.



ORTEP representation of the solid-state molecular structure of **a1** plotted with 50% probability ellipsoids for non-hydrogen atoms. Hydrogen atoms omitted for clarity. Selected bond lengths (Å) and angles (deg): N1–C1, 1.4432(12); N1–C2, 1.4508(12); N1–C4, 1.3830(12); N2–C1, 1.4670(11); N2–C3, 1.4536(12); N2–C10, 1.3897(11); N1–C1–N2, 111.75(7); C1–N1–C2, 118.91(8); C2–N1–C4, 119.83(8); C1–N1–C4, 120.39(8); C1–N2–C3, 116.85(8); C1–N2–C10, 118.63(7); C3–N2–C10, 118.65(7).

Table S1. Crystallographic parameters for <i>N,N'</i> -dimethyl- <i>N,N'</i> -diphenylmethylenediamine (a1).			
Crystal data		Data collection	
Empirical formula	C ₁₅ H ₁₈ N ₂	Diffractometer	Bruker Apex DUO
Formula weight	226.31	Abs corr (T _{min} , T _{max})	0.6993, 0.7461
Crystal System, space group	monoclinic, C2/c	2 θ range for data collection	4.076° ≤ 2 θ ≤ 61.19°
Temperature (K)	100	Index Ranges	-26 ≤ h ≤ 26, -9 ≤ k ≤ 9, -28 ≤ l ≤ 28
a, b, c (Å)	18.3166(7), 6.9280(3), 20.0908(8)	No. of measured reflections	31007
α, β, γ (°)	90, 95.8730(10), 90	No. of independent reflections	3873 [R _{int} = 0.0298, R _{sigma} = 0.0182]
Volume (Å ³)	2536.09(18)		
ρ_{calc} (g/cm ³)	1.185		
Z	8	Refinement	
Radiation type	MoK α (λ = 0.71073)	Data/restraints/parameters	3873/0/156
μ (mm ⁻¹)	0.07	Goodness-of-fit on F ²	1.029
F(000)	976	Final R indexes [I > 2 σ (I)]	R1 = 0.0401, wR2 = 0.1027
Crystal size (mm ³)	0.2 × 0.18 × 0.1	Final R indexes [all data]	R1 = 0.0528, wR2 = 0.1114
Crystal shape	Plate	Largest diff. peak/hole / e Å ⁻³	0.35/-0.22
Computer programs: APEX3, SAINT, SHELXL2018, OLEX2, ORTEP-3 for windows			
^a R ₁ = $\sum F_o - F_c / \sum F_o $. ^b wR ₂ = $[\sum (w(F_o^2 - F_c^2)^2) / \sum (w(F_o^2)^2)]^{1/2}$.			

Figure S3-5. Representative ^1H NMR spectrum of Table 3-1 Entry 2 and 3 (400 MHz, CDCl_3 , 298K).

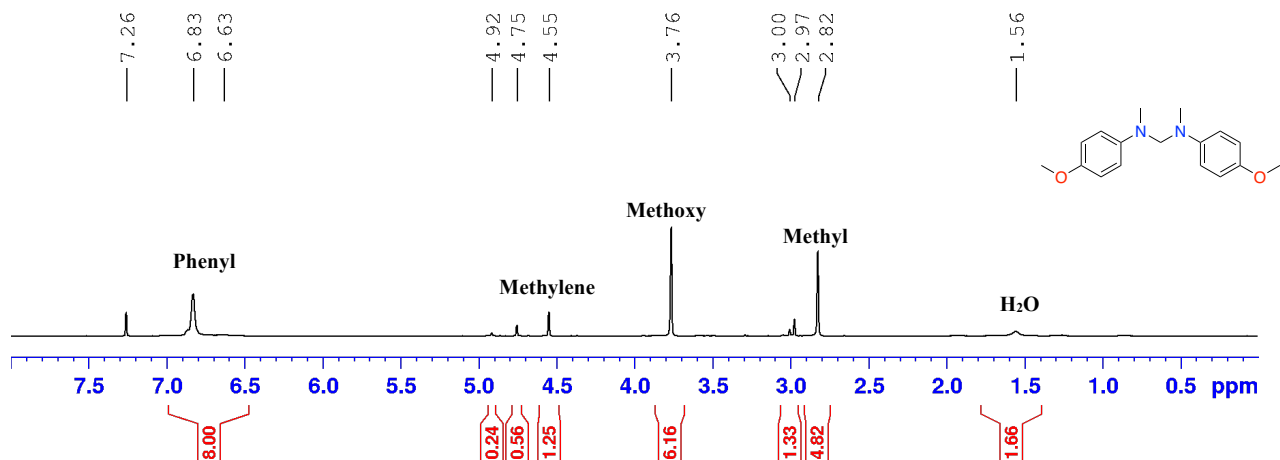


Figure S3-6. GC Chromatogram of Table 3-1 Entry 2.

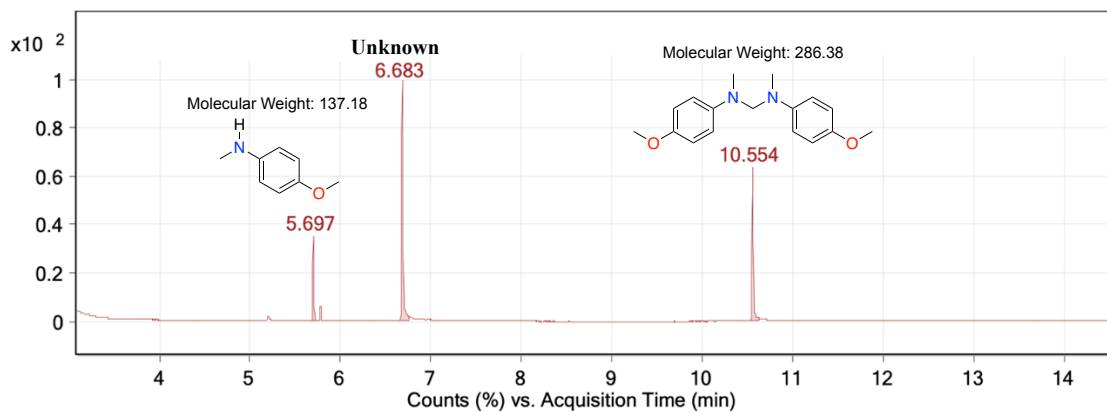


Figure S3-7. GC Chromatogram for Table 3-1 Entry 3.

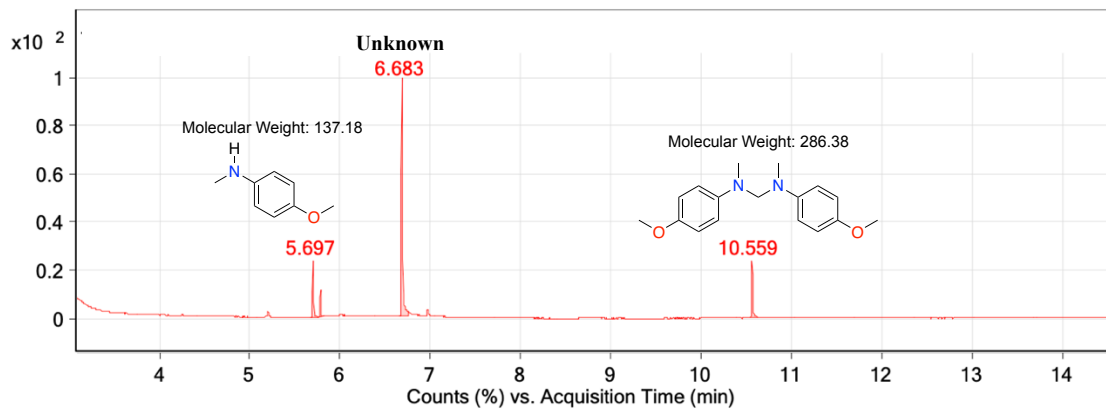


Figure S3-8. Representative GC chromatogram for Table 3-1 Entry 4 to 8.

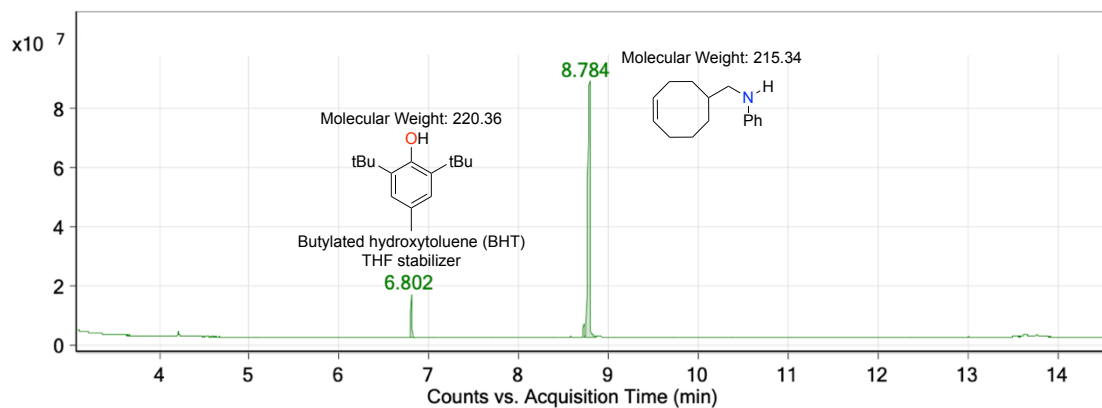


Figure S3-9. Representative ^1H NMR spectrum of Table 3-1 Entry 4 to 8 (400 MHz, CDCl_3 , 298K).

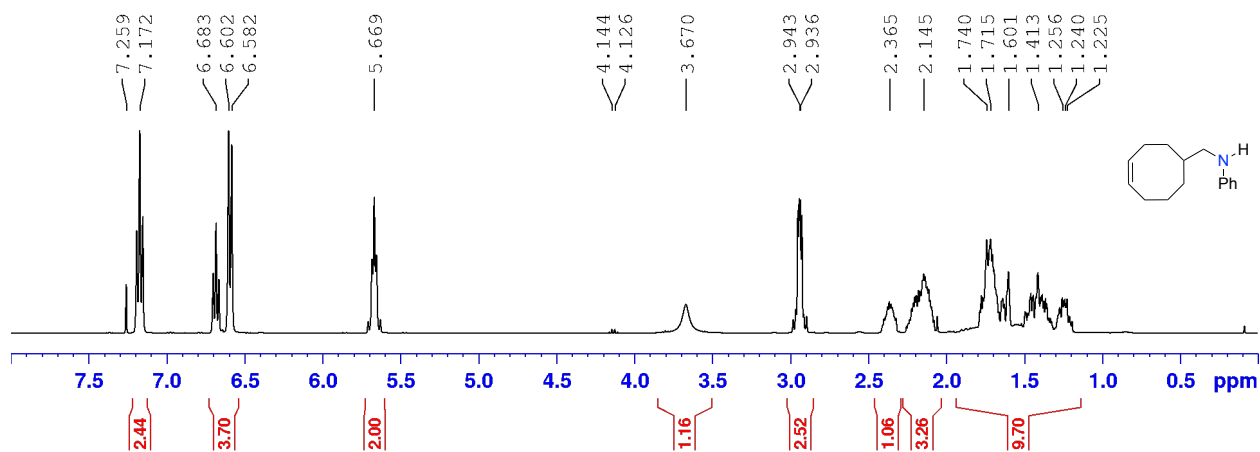
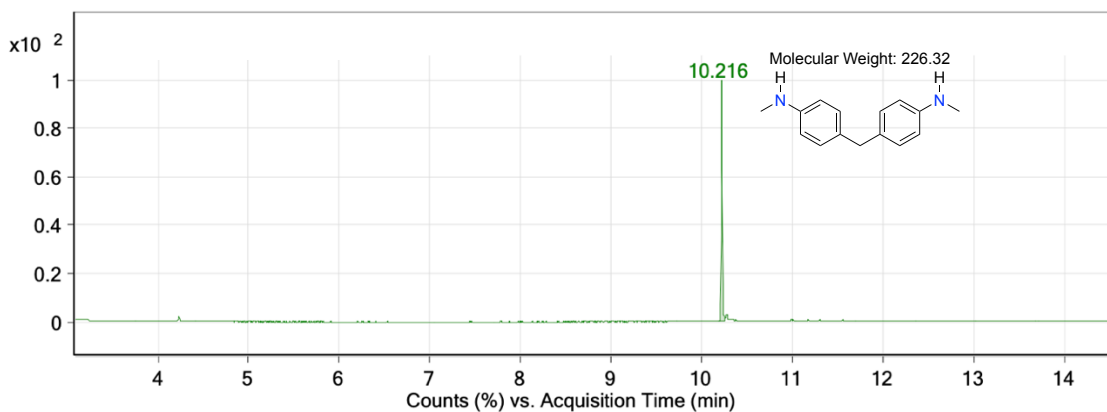


Figure S3-10. GC Chromatogram and mass spectrum for Table 3-2 Entry 1.



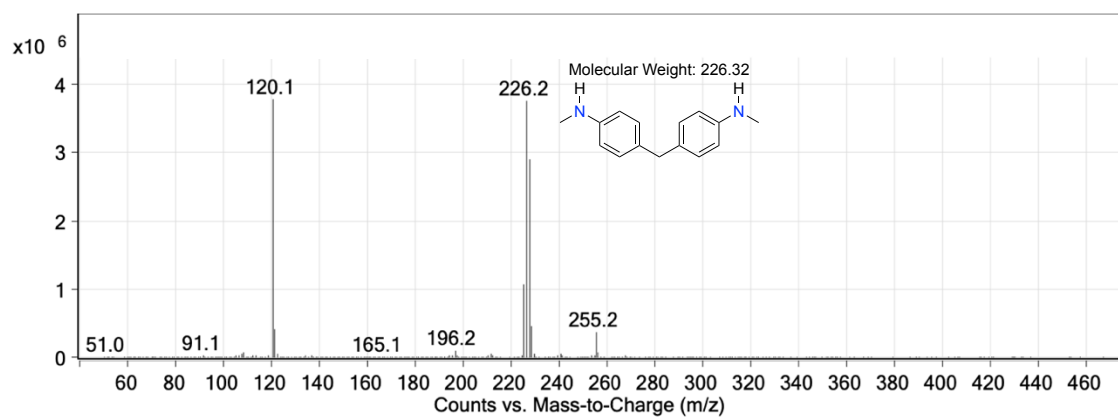


Figure S3-11. ¹H NMR spectrum of Table 3-2 Entry 1 (400 MHz, CDCl₃, 298K).

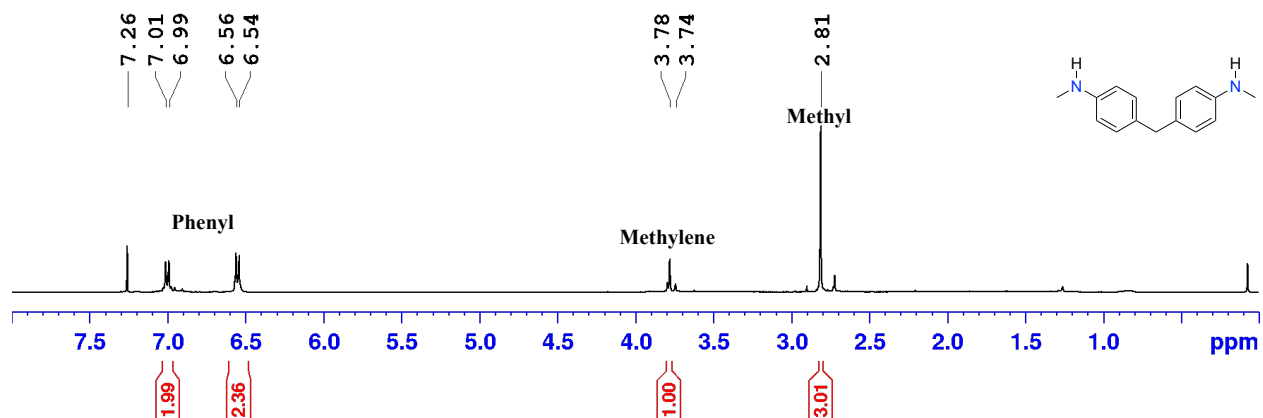


Figure S3-12. HSQC NMR $\{^1\text{H}, ^{13}\text{C}\}$ spectrum for Table 3-2 Entry 1 (400 MHz, CDCl_3 , 298K).

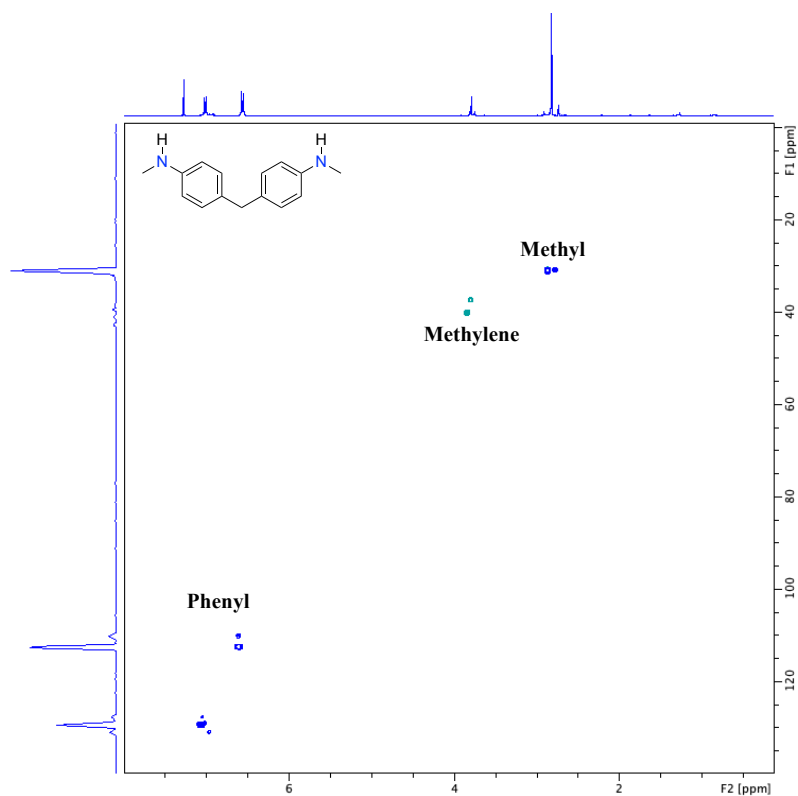


Figure S3-13. GC Chromatogram and mass spectrum for Table 3-2 Entry 2.

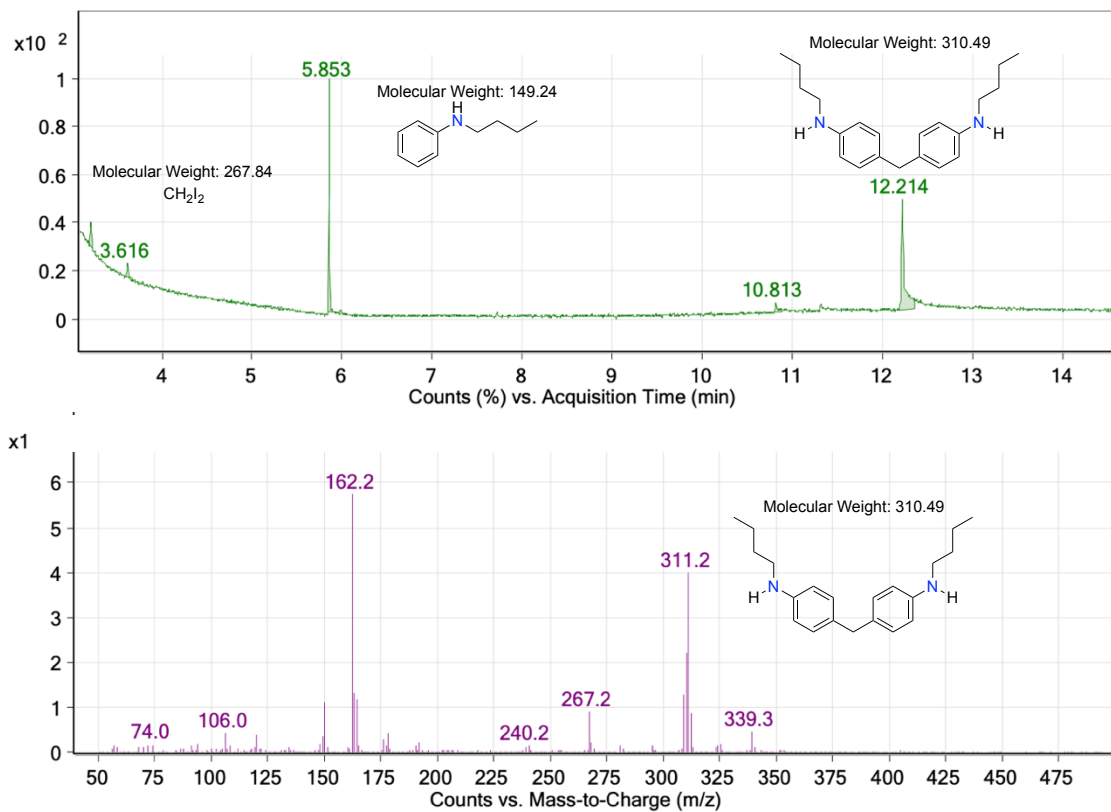


Figure S3-14. ^1H NMR spectrum of Table 3-2 Entry 2 (400 MHz, CDCl_3 , 298K).

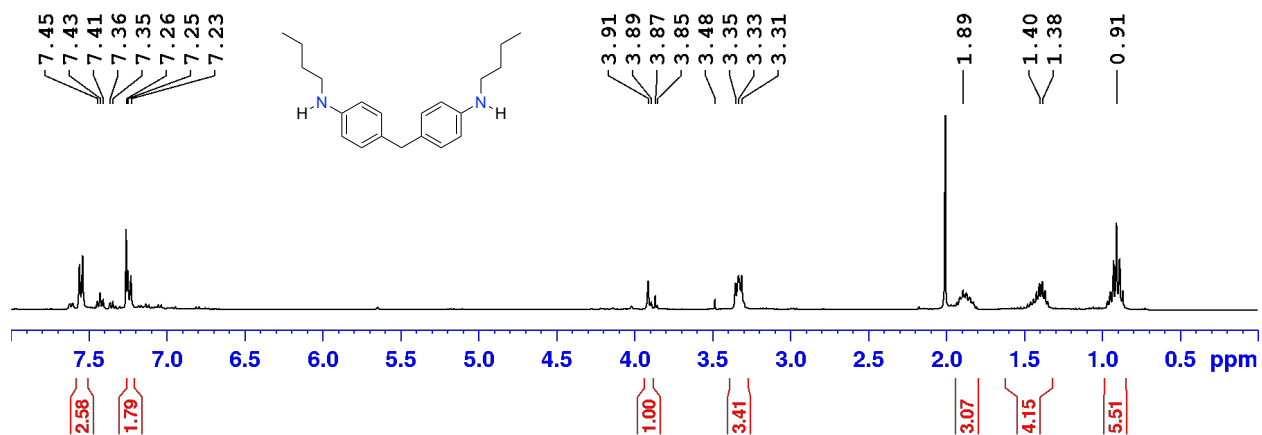


Figure S3-15. ^1H NMR spectrum of a1 (400 MHz, CD_3CN , 298K).

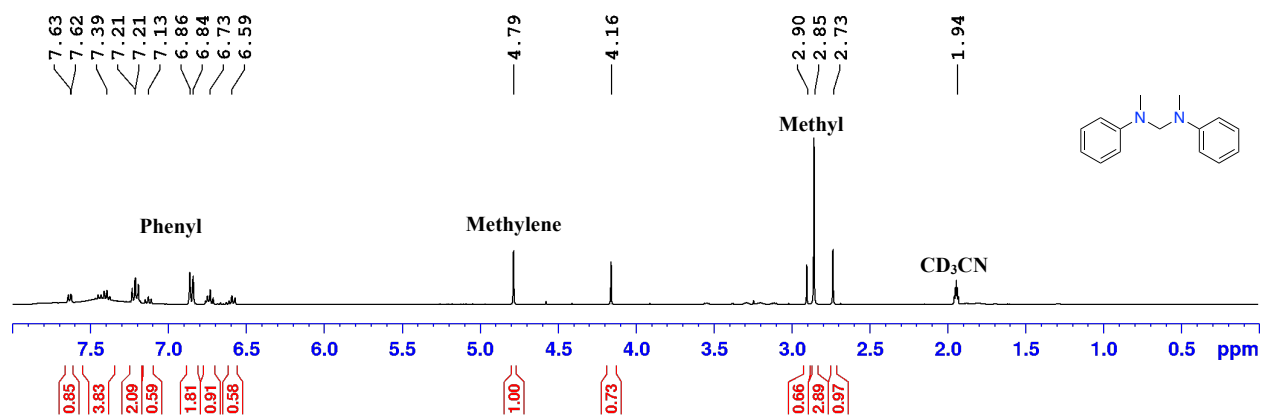


Figure S3-16. HSQC NMR $\{^1\text{H}, ^{13}\text{C}\}$ spectrum of a1 (400 MHz, CD_3CN , 298K).

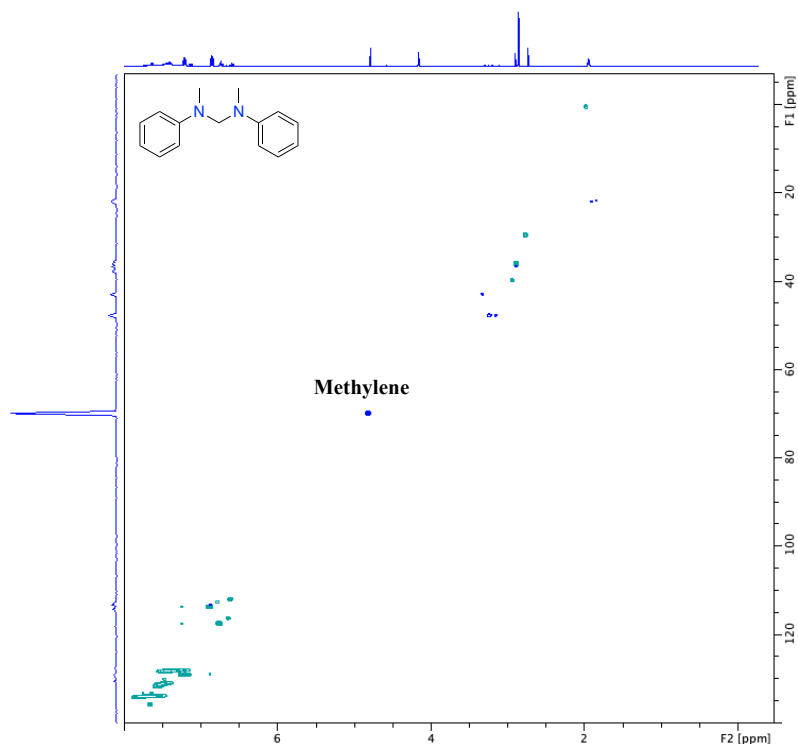


Figure S3-17. HSQC NMR $\{^1\text{H}, ^{13}\text{C}\}$ spectrum of c2 (400 MHz, CD_3CN , 298K).

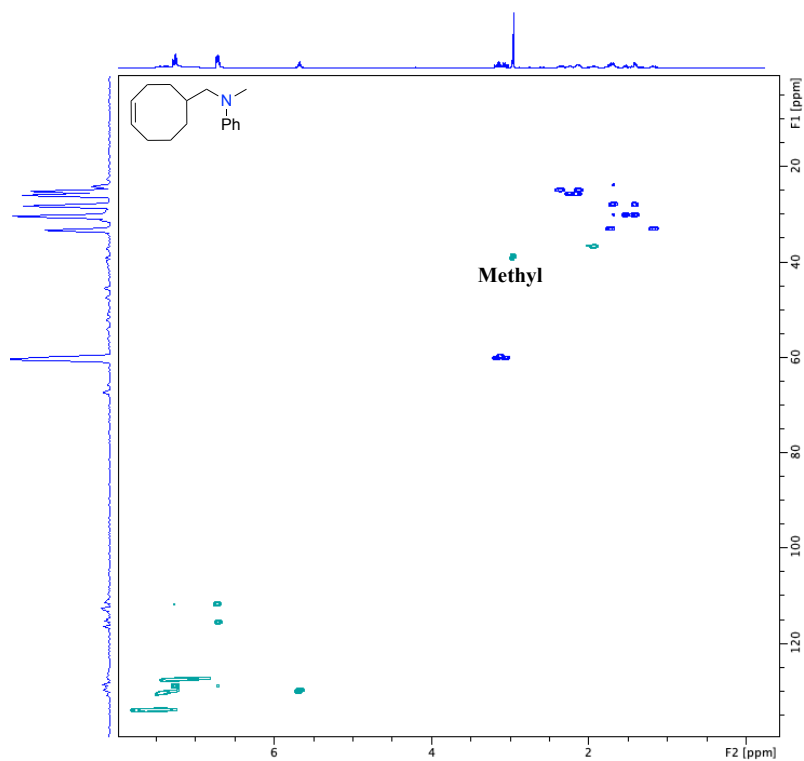


Figure S3-18. GC Chromatogram and mass spectrum for c1.

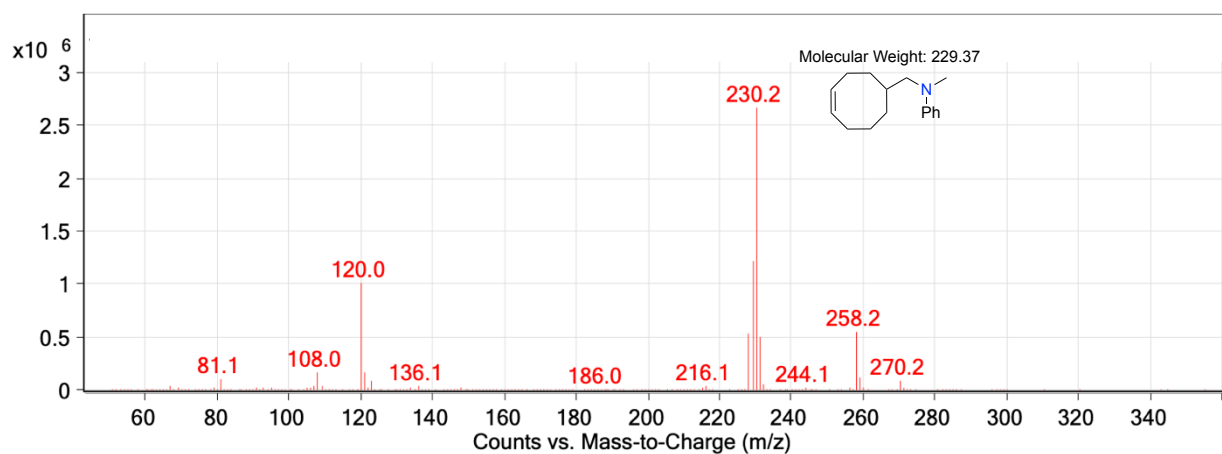


Figure S3-19. Representative HSQC NMR $\{^1\text{H}, ^{13}\text{C}\}$ spectrum of Table 3-3 Entry 3 to 5 (400 MHz, CD_3CN , 298K).

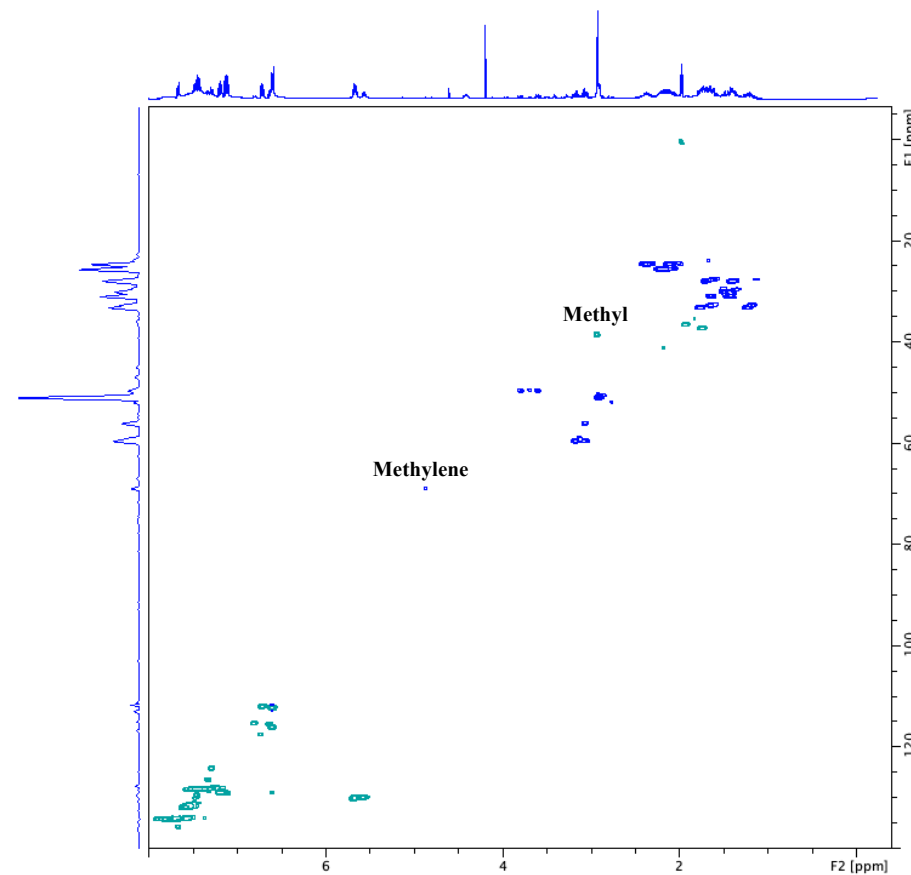


Figure S3-20. ^1H NMR spectrum of a3 (400 MHz, CD_3CN , 298K).

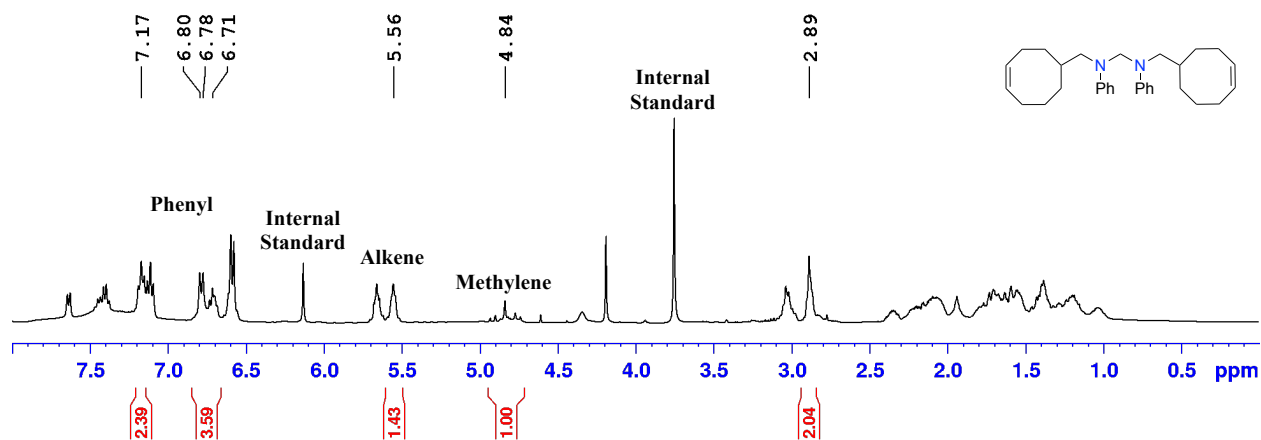


Figure S3-21. HSQC NMR $\{^1\text{H}, ^{13}\text{C}\}$ spectrum of a3 (400 MHz, CD_3CN , 298K).

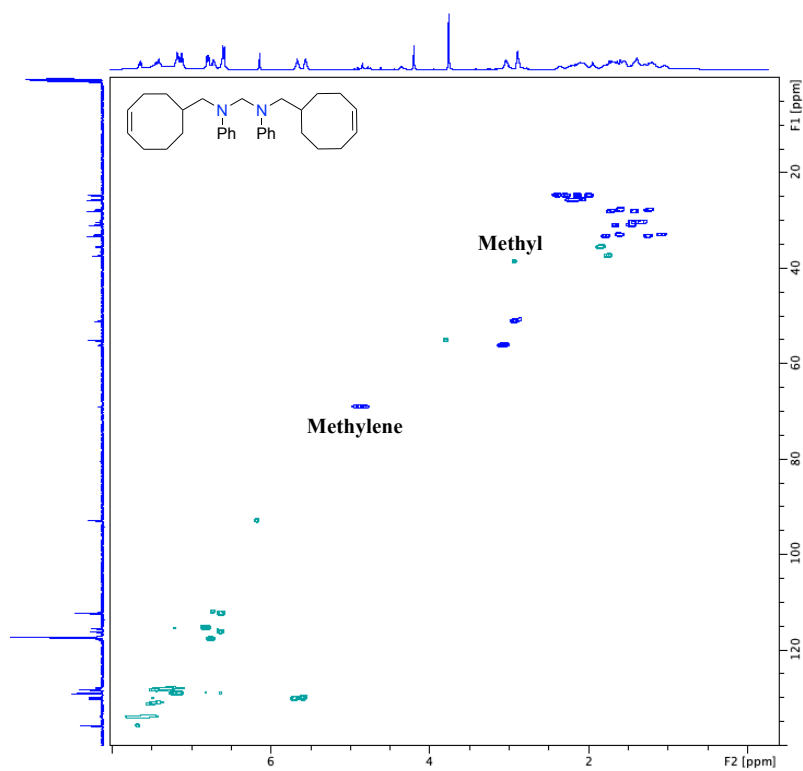


Figure S3-22. COSY NMR $\{^1\text{H}, ^1\text{H}\}$ spectrum of a3 (400 MHz, CD_3CN , 298K).

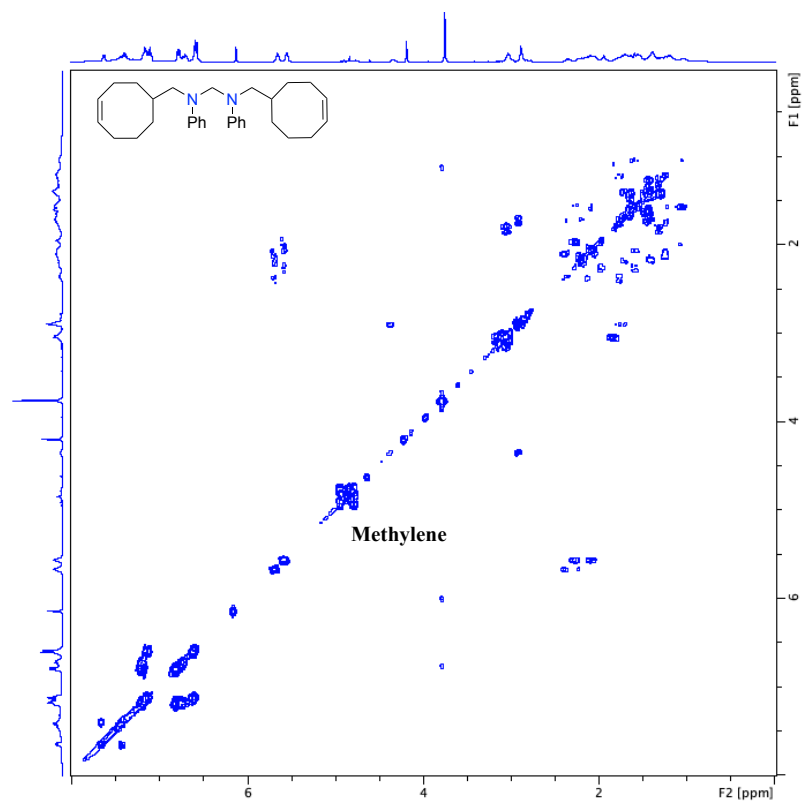


Figure S3-23. GC-MS mass spectrum of a5.

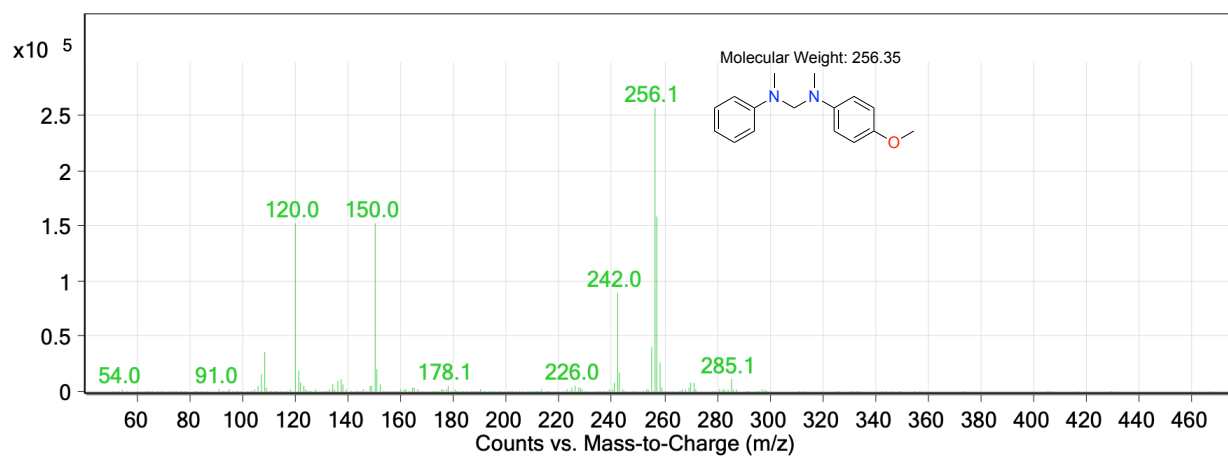


Figure S3-24. DSC thermogram of 40 mol% aminal-linked APO.

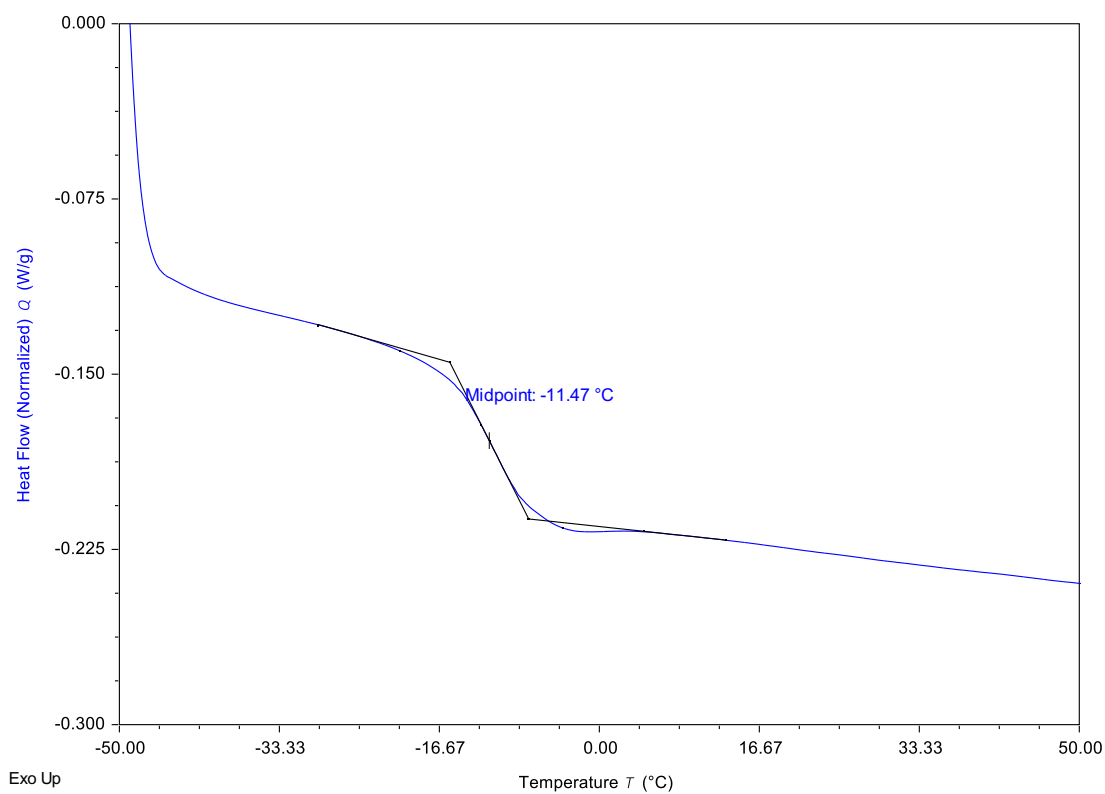


Figure S3-25. ^1H NMR spectrum of 5 mol% aminal-linked APO (400 MHz, CD_3CN , 298K).

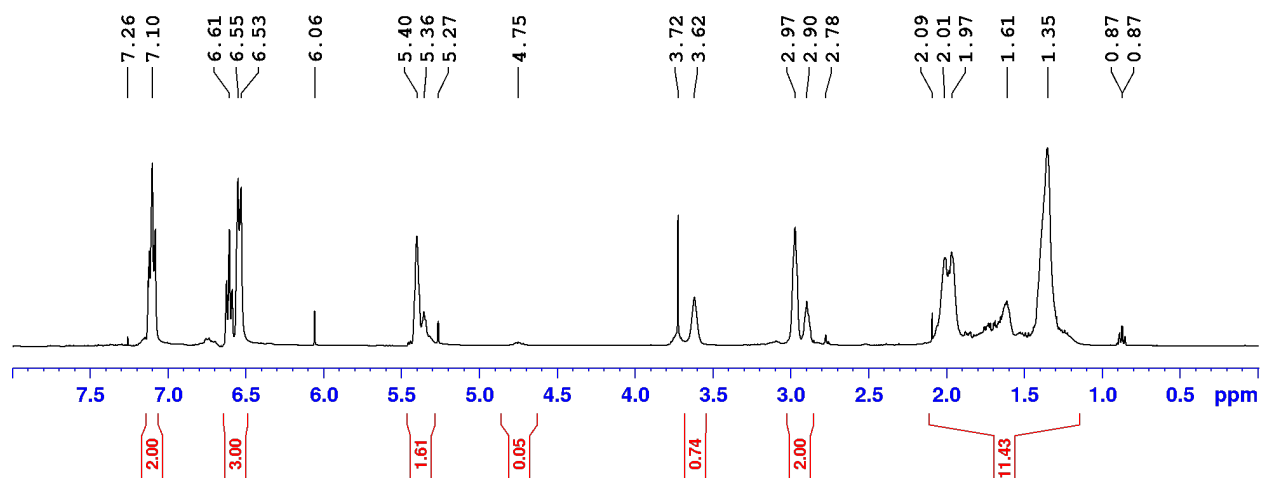


Figure S3-26. Master curves of 5 mol% aminal-linked APO from frequency sweep experiments at $T_{\text{ref}} = 30^\circ\text{C}$.

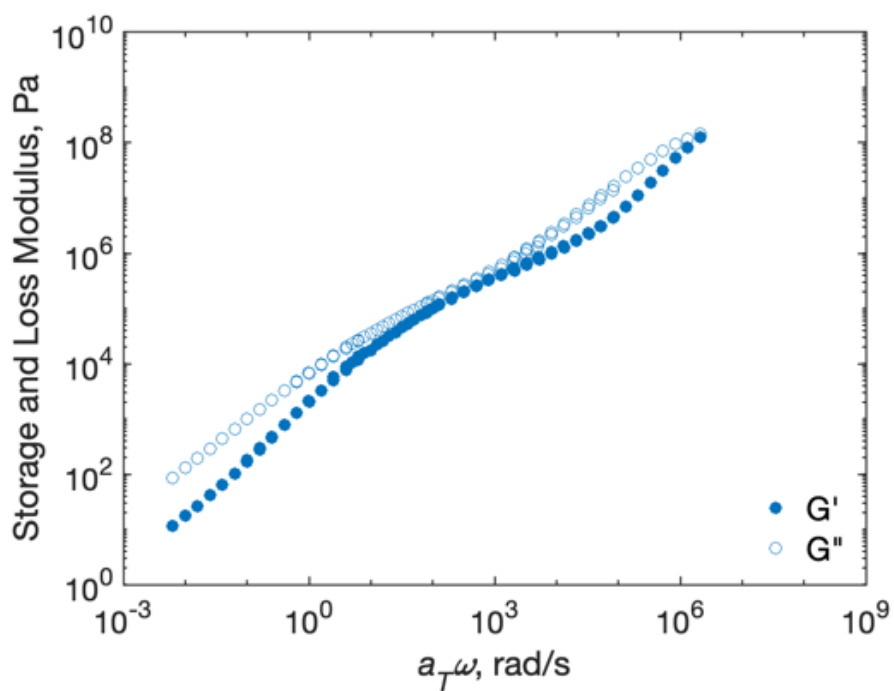


Figure S3-27. Stacked DSC thermograms of crosslinked APOs using different mol% of CH_2I_2 in DCM at 40 °C.

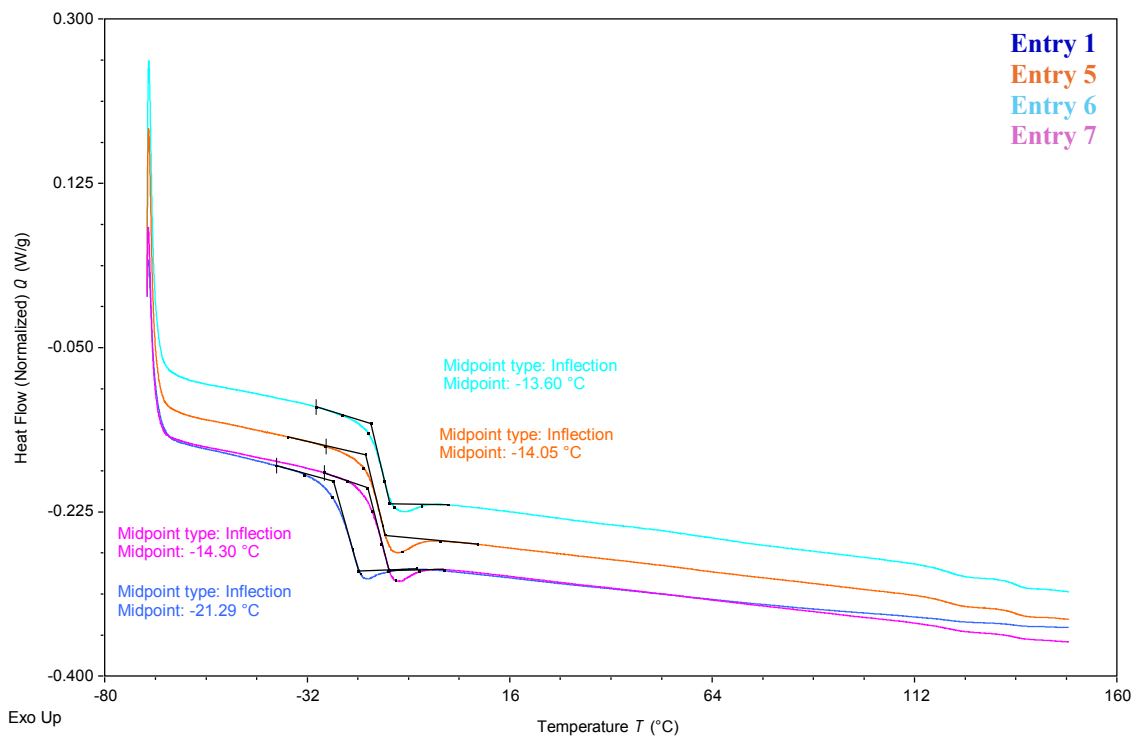


Figure S3-28. Stacked DSC thermograms of crosslinked APOs using different mol% of CH_2I_2 in THF at 60 °C.

